

SAP Fiori / C_FIORD

Demonstration Evidence Log

A living record of demonstrations, code, exam points, and gotchas
All Sessions · UX100 → UX410 · C_FIORD Exam Preparation

How to Use This Document

This document tracks real demonstrations across every training session. Each session section has the same structure:

- Exam Points Raised — the specific C_FIORD topics this session's code demonstrates
- Code Examples — working snippets that illustrate the concept correctly
- Watch Out For — real gotchas and compiler restrictions encountered while building the demo
- Your Demonstration Notes — a blank box for what you actually showed in the live session
- Your Code — a blank box to paste your own code as you build it out

Fill in the blank boxes as you go — this becomes a running record you can point to as evidence of practical coverage, and a handy revision aid since it pairs real code with the exact exam point it proves.

Master Quick Reference — CDS Syntax Pitfalls

Three real compiler errors were hit and fixed while building the live Flight/Booking demo (see the Live CDS Debugging Log near the end of this document for the full story). These are exam-relevant rules, not just one-off bugs:

Rule	Wrong	Right
No "/" operator	<code>price / qty</code>	<code>division(price, qty, 0)</code>
No functions in CASE WHEN	<code>when division(a,b,0)>=80</code>	<code>when a*100 >= 80*b (cross-multiply)</code>
No arithmetic in CASE WHEN	<code>when seatsocc*100/max>=80</code>	<code>precompute in a base view first</code>

⚠ Watch Out For

CASE WHEN conditions in ABAP CDS only support plain field-to-literal comparisons (field >= literal). No arithmetic operators and no function calls are allowed inside the condition. Any calculation must happen in the SELECT list — either as a column in the same view, or precomputed in a separate base/interface view, then consumed by reference in a higher-level view.

SESSION 1

Course Introduction & Setup

Orientation · Tools · Environment Setup

○ NOT YET DOCUMENTED

Exam Points Raised

Add the topics covered once this session is delivered — e.g. SAP Learning Hub navigation, BAS/Eclipse setup, system landscape overview.

Code Examples

No code typically demonstrated in an orientation session. Add notes here if any setup scripts or configuration steps were shown.

Your Demonstration Notes

What was actually demonstrated in this session:

Initial Introduction – Navigate from the C_FIORID certification page to the learning journeys. Additionally point out where you can book a learning preparation system to carry out exercises.

Your Code

Paste your code here

None

SESSION 2

Fiori Overview & Launchpad

UX100 · Fiori Overview & Launchpad — 11–20% of C_FIORD

• MATERIALS BUILT (PPTX + Study Guide)

Exam Points Raised

- 5 Fiori Principles: Role-based, Adaptive, Coherent, Simple, Delightful
- SAP Fiori Horizon UI anatomy: Spaces → Pages → Sections → Tiles
- Spaces & Pages replaced the deprecated Groups & Catalogs model
- Fiori Landscape three-tier architecture: FLP → Web Dispatcher (optional) → FES → BES
- OData V2 for Smart Controls vs OData V4 for Fiori Elements
- Catalog → Role → Tile chain for app visibility troubleshooting

Code Examples

Session 2 is largely conceptual (Launchpad, architecture, principles) — there is little CDS/UI5 code at this stage. The one piece of annotation syntax worth knowing early is the CDS header info pattern, since it reappears in every later session:

```
// Minimal @UI.headerInfo — drives the Object Page title bar
@UI.headerInfo.typeName:      'Flight'
@UI.headerInfo.typeNamePlural: 'Flights'
@UI.headerInfo.title.value:    'carrid'
@UI.headerInfo.description.value: 'carname'
```

⚠ Watch Out For

Spaces & Pages replaced Groups & Catalogs as the primary FLP UI in Fiori Horizon. Tiles in a Catalog can appear on multiple Pages — they're reused, not duplicated. OData V2 ≠ V4: Smart Controls require V2, Fiori Elements requires V4.

Your Demonstration Notes

What was actually demonstrated in this session:

Fiori Reference Library Demonstration – Reference Library itself, Implementation information (Installation, Configuration, Extensibility [Adaption projects]), PFCG role (adding catalogue)

Fiori Launchpad Dashboard – Adding apps through catalogues to your space/page, Variants and value defaults, Personalization features.

Custom Fiori Application Back-end – Eclipse IDE, CDS modelling, Service Definition & Binding, Gateway Service exploration & testing, Service Nodes, BSP application from front

end app deployed, Created new catalogue (tile & target mapping), adding catalogue to user and test the Fiori app through the Fiori Launchpad Dashboard.

Custom Fiori Application Front-end – BAS IDE, Fiori generator template (List Report), Test web app, Deployment yaml files, manifest.json discussion, xml annotations.

<https://github.com/AngeliqueSabTec/ZUXFioriDemo>

Your Code

Paste your code here

```
@AbapCatalog.sqlViewName: 'ZUX410FLIGHTUI004'
@AbapCatalog.compiler.compareFilter: true
@AccessControl.authorizationCheck: #CHECK
@endUserText.label: 'UX410 Flight UI'
@OData.publish: true
@OData.entitySet.name: 'FlightSet00'
@OData.entityType.name: 'Flight00'

@UI.headerInfo.typeName: 'Flight'
@UI.headerInfo.typeNamePlural: 'Flights'
@UI.headerInfo.title.value: 'carrid'
@UI.headerInfo.description.value: 'carrname'

@Search.searchable: true

define view ZUX410_C_FLIGHTUI00_14 as select from sflight as sf
  association to scarr as sc on sf.carrid = sc.carrid
  association to sflights as sfs on sfs.carrid = sf.carrid and
    sfs.connid = sf.connid and sfs.fldate = sf.fldate
  association [*] to ZUX410_C_BOOKINGUI00 as _Bookings
    on $projection.carrid = _Bookings.carrid and $projection.fldate =
      _Bookings.fldate and
    $projection.connid = _Bookings.connid
{
@UI.facet: [
{ purpose:#HEADER, type: #FIELDGROUP_REFERENCE, targetQualifier: 'flightinfo',
position:10 },
{ purpose:#HEADER, type: #FIELDGROUP_REFERENCE, targetQualifier: 'seatinfo',
position:20 },
{ purpose:#STANDARD, type: #FIELDGROUP_REFERENCE, targetQualifier: 'seatdata',
position:10},
{ purpose:#STANDARD, type: #FIELDGROUP_REFERENCE, targetQualifier: 'seatdata2',
position:20},
{ purpose:#STANDARD, id: 'seat1', label:'Seatsinfo 1', type:#COLLECTION,
position:30},
{ purpose:#STANDARD, id: 'seat2', label:'Seatsinfo 2', type:#COLLECTION,
position:40}
]
@UI.lineItem.position:10
  key sf.carrid,
@UI.lineItem.position:20
  key sf.connid,

@UI.identification.position:20
@UI.lineItem.position:30
  key sf.fldate,

@UI.identification.position:10
```

```

@Search.defaultSearchElement: true
@Search.fuzzinessThreshold: 0.7
@UI.selectionField.position:10
@UI.lineItem.position:40
    sc.carrname,

@UI.identification.position:30
@UI.fieldGroup : [
    {
        qualifier:'flightinfo', position:10
    }
]
@UI.selectionField.position:20
@UI.lineItem.position:70
    sfs.cityfrom,

@UI.identification.position:40
@UI.fieldGroup : [
    {
        qualifier:'flightinfo', position:20
    }
]
@UI.selectionField.position:30
@UI.lineItem.position:80
    sfs.cityto,

@UI.identification.position:50
@UI.lineItem.position:50
@UI.lineItem.importance: #HIGH
    sf.planetype,
@UI.lineItem.importance: #LOW
@UI.lineItem.position:60
    sf.price,
@Semantics.currencyCode: true
    sf.currency,
@Semantics.amount.currencyCode: 'currency'
@UI.lineItem.importance: #MEDIUM
@UI.lineItem.position:90
    sf.price * sf.seatsocc as Total,

@UI.fieldGroup : [
    {
        qualifier:'seatinfo', position:10
    }
]
    sf.seatsmax,
@UI.fieldGroup : [
    {
        qualifier:'seatinfo', position:20
    }
]
    sf.seatsocc,

@UI.fieldGroup : [
    {
        qualifier:'flightinfo', position:30
    }
]
    sc.url,
    sf.paymentsum,
@UI.fieldGroup : [
    {qualifier:'seatdata', position:10}
]
    sf.seatsmax_b,
@UI.fieldGroup : [
    {qualifier:'seatdata2', position:10}
]

```

```

]
    sf.seatsocc_b,
@UI.fieldGroup : [
    {qualifier:'seatdata', position:20}
]
    sf.seatsmax_f,
@UI.fieldGroup : [
    {qualifier:'seatdata2', position:10}
]
    sf.seatsocc_f,
    _Bookings
}

@AbapCatalog.sqlViewName: 'ZUX410BOOKING00'
@AbapCatalog.compiler.compareFilter: true
@AccessControl.authorizationCheck: #CHECK
@EndUserText.label: 'bookings'
define view ZUX410_C_BOOKINGUI00 as select from sbook {
    //sbook
    key mandt,
    key carrid,
    key connid,
    @UI.lineItem.position: 20
    key fldate,
    key bookid,
    customid,
    custtype,
    @UI.lineItem.position: 60
    smoker,
    luggweight,
    wunit,
    invoice,
    @UI.lineItem.position: 50
    class,
    forcuram,
    forcurkey,
    @UI.lineItem.position: 40
    loccuram,
    loccurkey,
    @UI.lineItem.position: 30
    order_date,
    counter,
    agencynum,
    cancelled,
    reserved,
    @UI.lineItem.position: 10
    passname,
    passform,

    passbirth
}

@EndUserText.label: 'ZUXFIORI_DEMO'
define service ZUXFIORI_DEMO {
    expose ZUX410_C_FLIGHTUI00_14;
}

```

Demonstration Notes

Screenshot Evidence — Backend: Activating the OData Service

Service Binding: ZUXFIORI_DEMO_BIND

General Information

This section describes general information about this service binding

Binding Type: OData V2 - UI

Service Versions

Define service versions associated with the service binding

type filter text

Version	Service Definition
1.0.0	ZUXFIORI_DEMO

Add...
Remove

Service Version Details

View information on selected service version

Local Service Endpoint: PublishedUnpublish

Service Information

Service URL: /sap/opu/odata/sap/ZUXFIORI_DEMO_BIND

type filter text

Entity Set and Association

FlightSet00

Preview...

Service Binding ZUXFIORI_DEMO_BIND — OData V2 UI binding published, exposing the FlightSet00 entity set.

[Main Menu](#)
[Edit](#)
[Goto](#)
[System](#)
[Help](#)
SAP
Activate and Maintain Services

[S4H \(1\) 300](#)

[More](#)

[Exit](#)

Service Catalog

[Type](#)
[Technical Service Name](#)
[Service Description](#)
[External Service Name](#)
[Namespace](#)
[OAuth](#)
[Soft State Status](#)
[Processing Mode](#)

[Add Service](#)
[Service Details](#)
[Load Metadata](#)
[Error Log](#)
[Request Statistics](#)
[Refresh Catalog](#)
[OAuth](#)
[Soft State](#)
[Processing Mode](#)
[Add to Transport](#)

Type	Technical Service Name	Service Description	External Service Name	Namespace	OAuth	Soft State Status	Processing Mode
REP	ZUSER_MENU	1 User Menu service	USER_MENU	AB2P	☐		Routing-based
REP	AWFNDSG_USER_SERVICE	1 Information Worker - User Service	USERSERVICE	AWFNDSV	☐		Co-deployed only
REP	ZUX_TRAVEL_SRV	1 OData Service for UX Travel	UX_TRAVEL_SRV		☐		Co-deployed only
REP	ZUX_UI_CUSTOMER_O2	1 Flight Customer - OData V2 UI Service	UX_UI_CUSTOMER_O2		☐		Co-deployed only
REP	ZVBLAPPL_DEF2_SRV	1 Service for VBL Application Definition	VBLAPPL_DEF2_SRV		☐		Co-deployed only
REP	ZVDM_CDSVIEW_BROWSERSES	1 VDM CDSView Browser Gateway Project	VDM_CDSVIEW_BROWSERSES		☐		Co-deployed only
REP	ZLS_BUCHEHT	1 Business display	ZLS_BUCHEHT		☐		Co-deployed only
REP	ZUX410_C_FLIGHTU00_11_CDS	1 UX410 Flight UI	ZUX410_C_FLIGHTU00_11_CDS		☐	Not Supported	Routing-based
REP	ZUX410_C_FLIGHTU00_12_CDS	1 UX410 Flight UI	ZUX410_C_FLIGHTU00_12_CDS		☐	Not Supported	Routing-based
REP	ZUX410_C_FLIGHTU00_13_CDS	1 UX410 Flight UI	ZUX410_C_FLIGHTU00_13_CDS		☐	Not Supported	Routing-based
REP	ZUX410_C_FLIGHTU00_14_CDS	1 UX410 Flight UI	ZUX410_C_FLIGHTU00_14_CDS		☐	Not Supported	Routing-based
REP	ZUXFORL_DEMO_BND	1 ZUXFORL_DEMO_BND	ZUXFORL_DEMO_BND		☒		Co-deployed only

ICF Nodes

[ICF Node](#)
[Call Browser](#)
[SAP Gateway Client](#)

[Status](#)
[ICF Node](#)
[Session Time-out](#)
[Soft State](#)
[Description](#)

Status	ICF Node	Session Time-out	Soft State	Description
OK	ODATA	00:00:00		Standard Mode

System Aliases

[Add System Alias](#)
[Remove System Alias](#)
[Customizing](#)
[Implementation](#)

[SAP System Alias](#)
[Description](#)
[Default System](#)
[Metadata](#)
[User Role](#)
[Host Name](#)

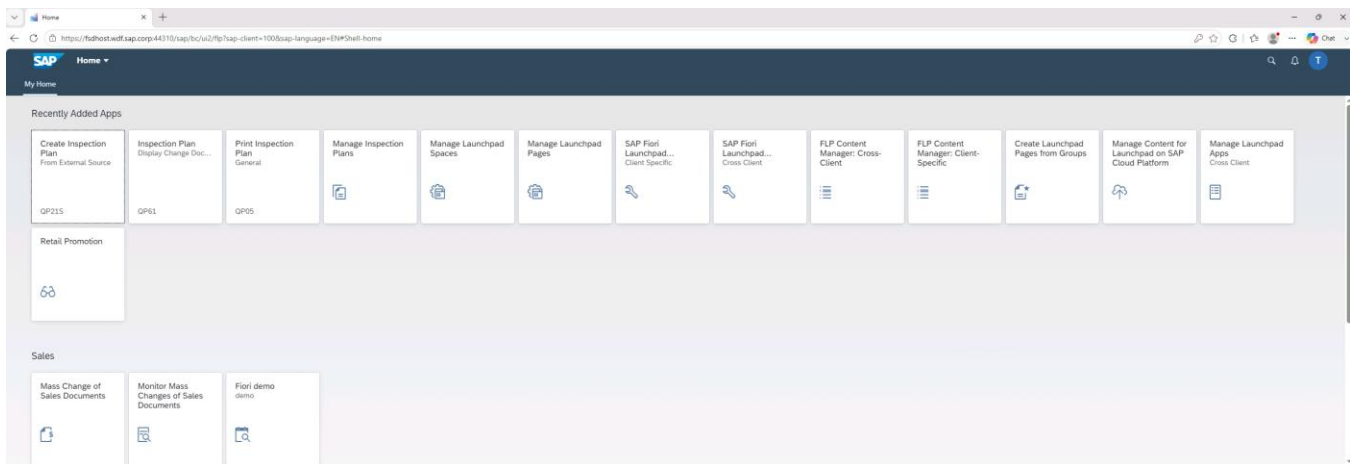
SAP System Alias	Description	Default System	Metadata	User Role	Host Name
------------------	-------------	----------------	----------	-----------	-----------

/IWFND/MAINT_SERVICE — Gateway Service Catalog showing the CDS-generated services (ZUX410 C FLIGHTUI00 11–14 CDS) and the active ZUXFIORI DEMO BIND service.

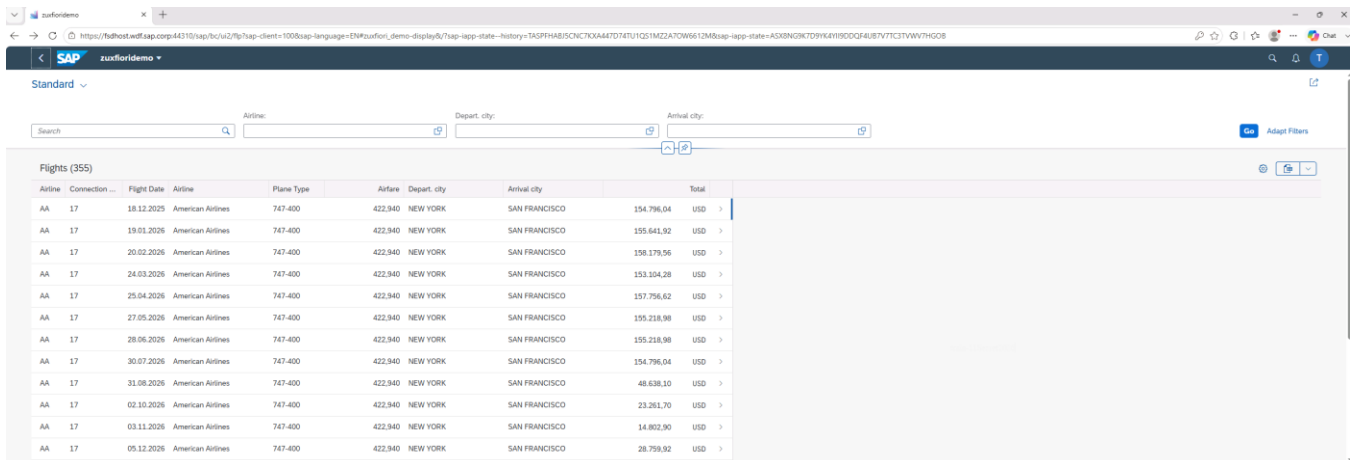
[illegible]

SAP Gateway Client — testing a GET request to FlightSet00. HTTP 200 OK confirms the service responds with valid OData XML.

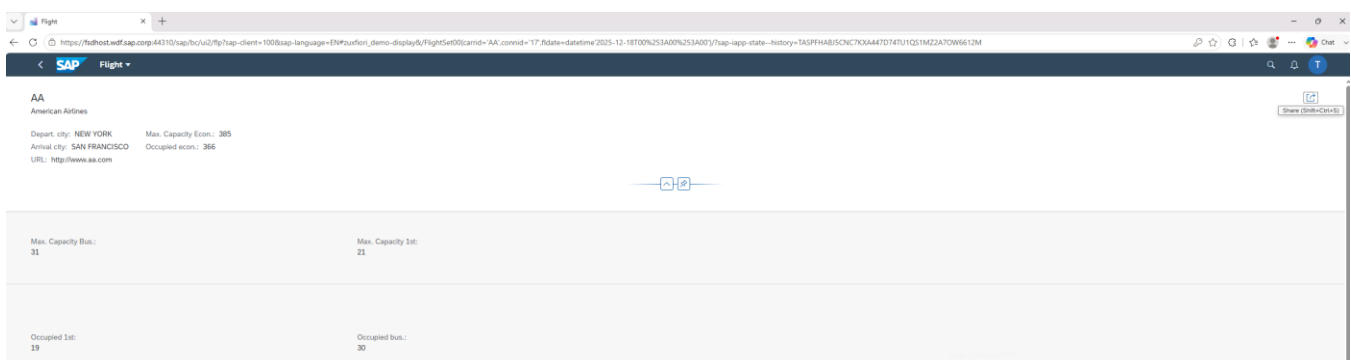
Screenshot Evidence — Front-End: Running the App in the Fiori Launchpad



Fiori Launchpad — My Home page with the 'Fiori demo' tile live under the Sales section, alongside the FLP/Spaces admin tiles used to configure it.



List Report — the running app generated by SAP Fiori Elements, showing the filter bar (Airline / Depart / Arrival city) and the Flights table with 355 results.



Object Page — flight detail view showing the header (carrier, route) and FieldGroup sections (Max Capacity / Occupied seats by class).

Screenshot Evidence — Catalog & Target Mapping Configuration

The screenshot shows the SAP Fiori catalog configuration interface. The top navigation bar includes 'Catalogs' and 'Groups'. The left sidebar shows a 'Catalog Collection' with a 'Drag to add' button and a list of catalogs: 'Education - Fiori Launchpad ...', 'Fiori_demo', and 'Manage Position - Fiori Applic...'. The main area displays the 'Fiori_demo' catalog configuration. The 'Target Mapping' tab is active, showing a table with columns: Semantic Object, Action, Navigation Type, Information, Desktop, Tablet, Phone, and Out. The table contains one row: 'zuxfiori_demo' for the 'display' action, linked to the 'SAPUI5 Fiori App'. A 'Configure Columns' button is visible. Below the table, there are links: 'Show similar target mappings', 'Create Target Mapping', 'Configure', 'Create Reference', 'Delete', 'Where Used', and 'Original'.

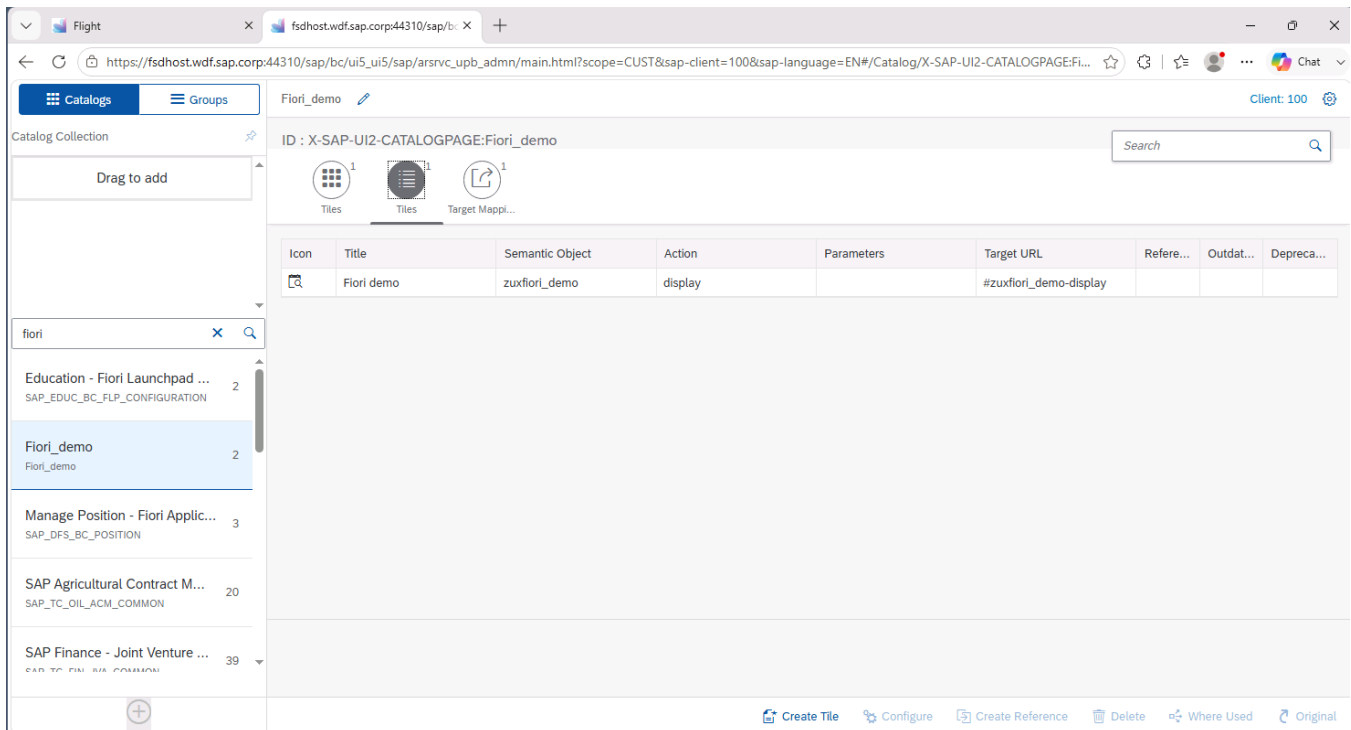
Semantic Object	Action	Navigation Type	Information	Desktop	Tablet	Phone	Out
zuxfiori_demo	display	SAPUI5 Fiori App		✓	✓	✓	

Manage Launchpad Pages — the Fiori_demo catalog’s Target Mapping tab, linking semantic object zuxfiori_demo + action display to the SAPUI5 app.

The screenshot shows the 'Configure: Fiori Demo' configuration page. The 'Intent' section has 'Semantic Object' set to 'zuxfiori_demo' and 'Action' set to 'display'. The 'Target' section has 'Application Type' set to 'SAPUI5 Fiori App', 'Title' set to 'Fiori Demo', 'URL' set to '/sap/bc/bsp/sap/zuxfiori_demo', and 'ID' set to 'zuxfioridemo.zuxfioridemo'. The 'General' section has 'Information' set to an empty field. Under 'Device Types', 'Desktop', 'Tablet', and 'Phone' are all checked. A 'Parameters' table is shown with columns: Name, Mandatory, Value, Is Regular Expression, Default Value, and Target N... The table is empty. There is a checkbox for 'Allow additional parameters' which is checked. At the bottom, there are 'Add' and 'Delete' buttons. The page footer says 'This is a reference tile' and 'Save Cancel'.

Name	Mandatory	Value	Is Regular Expression	Default Value	Target N...
Enter a name	☐		☐		

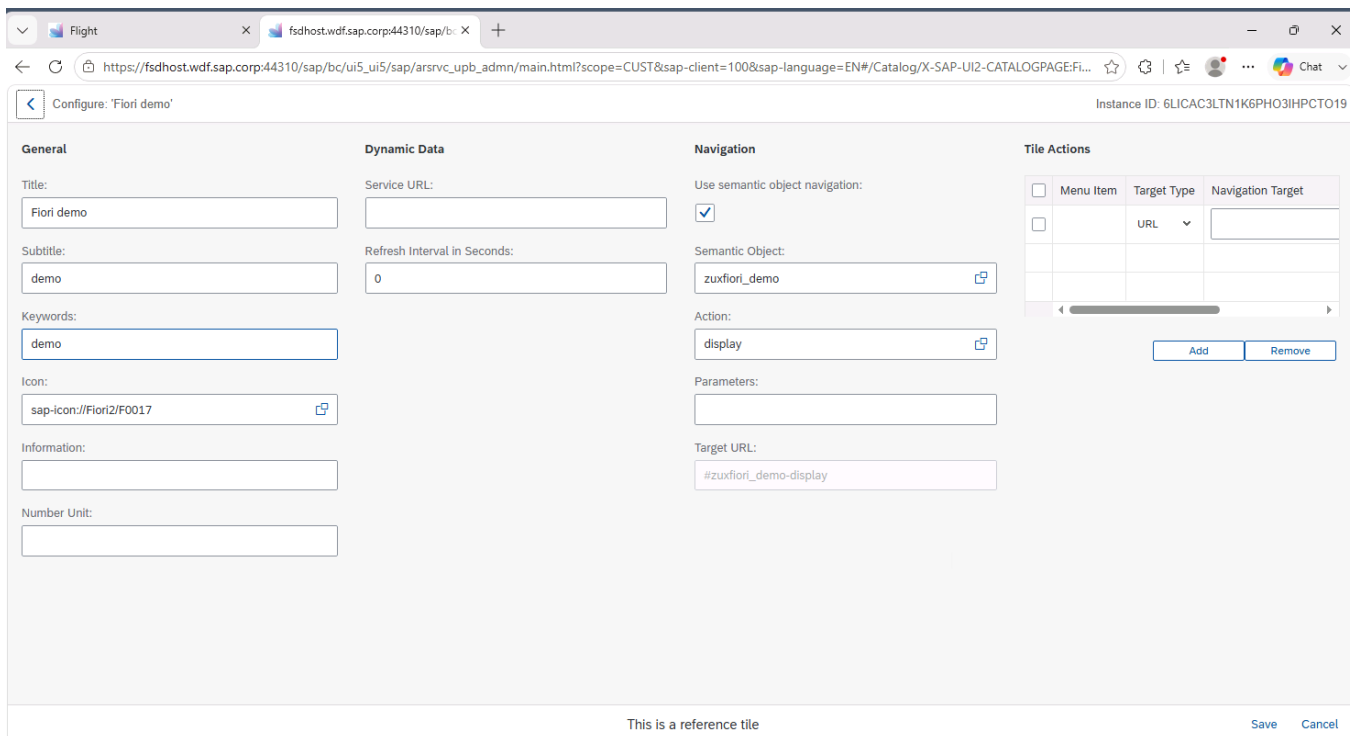
Target Mapping detail — Intent (Semantic Object / Action) and Target (Application Type, Title, URL) configuration for the demo app.



The screenshot shows the SAP Fiori catalog interface. On the left, a 'Catalog Collection' sidebar lists various tiles, with 'Fiori_demo' selected. The main area displays the 'Fiori_demo' tile entry in a list view. The tile's ID is 'X-SAP-UI2-CATALOGPAGE:Fiori_demo'. Below the ID, there is a table with columns: Icon, Title, Semantic Object, Action, Parameters, Target URL, Reference, Outdated, and Deprecation. The 'Fiori_demo' tile is listed with the Semantic Object 'zuxfiori_demo' and Action 'display'. The Target URL is '#zuxfiori_demo-display'.

Icon	Title	Semantic Object	Action	Parameters	Target URL	Reference	Outdated	Deprecation
	Fiori demo	zuxfiori_demo	display		#zuxfiori_demo-display			

Tiles (list view) — the Fiori demo tile entry with its resolved Target URL (#zuxfiori_demo-display).



The screenshot shows the 'Configure: Fiori demo' configuration page. The page is divided into four main sections: General, Dynamic Data, Navigation, and Tile Actions. The 'General' section contains fields for Title, Subtitle, Keywords, Icon, Information, and Number Unit. The 'Dynamic Data' section contains fields for Service URL, Refresh Interval in Seconds, and Semantic Object. The 'Navigation' section contains fields for Action, Parameters, and Target URL. The 'Tile Actions' section contains a table for configuring menu items and navigation targets.

General

Title: Fiori demo

Subtitle: demo

Keywords: demo

Icon: sap-icon://Fiori2/F0017

Information:

Number Unit:

Dynamic Data

Service URL:

Refresh Interval in Seconds: 0

Navigation

Use semantic object navigation: ☒

Semantic Object: zuxfiori_demo

Action: display

Parameters:

Target URL: #zuxfiori_demo-display

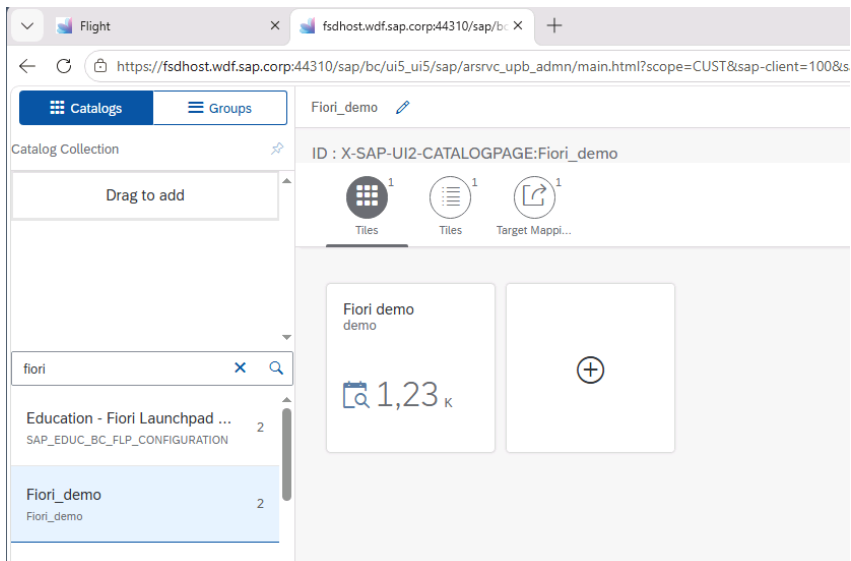
Tile Actions

Menu Item	Target Type	Navigation Target
☐	URL	

Buttons: Add, Remove

Footer: This is a reference tile. Save Cancel

Tile configuration detail — General (title, subtitle, icon), Navigation (semantic object/action), and Tile Actions settings.

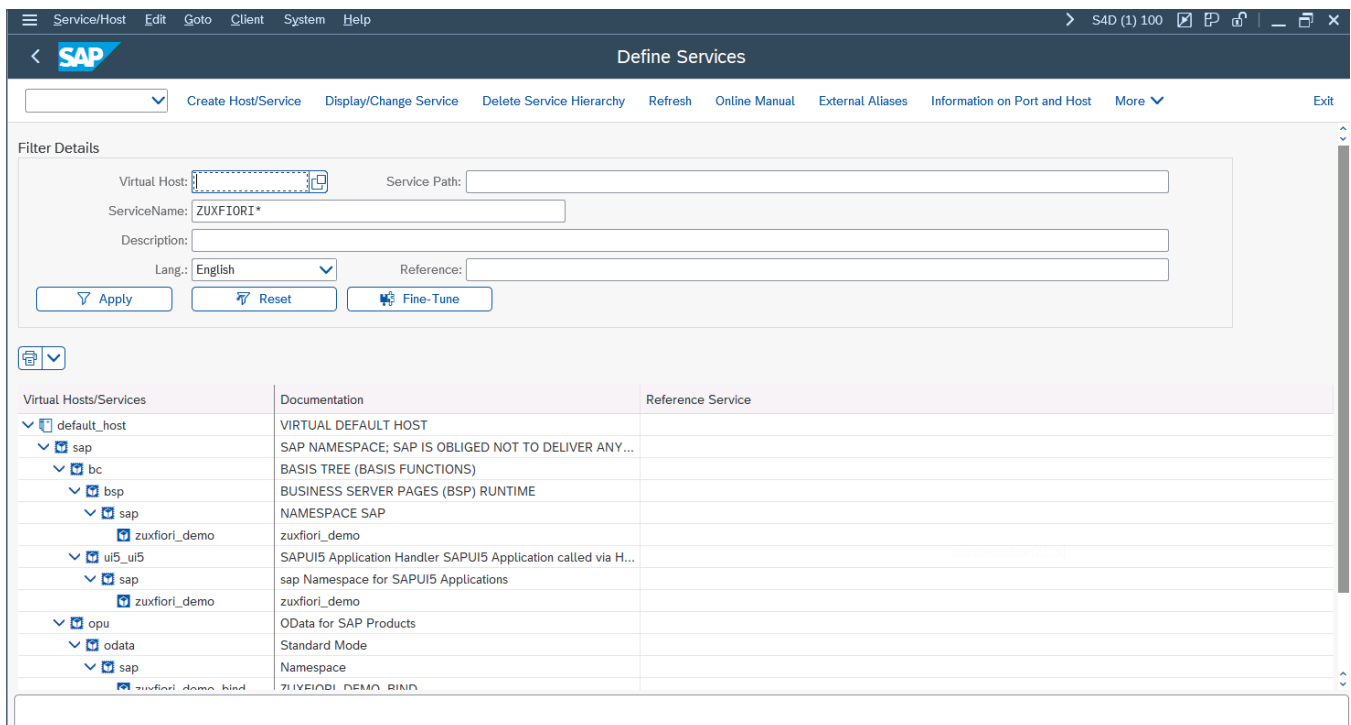


Tiles (grid view) — visual preview of the Fiori demo tile as it appears to end users, including its dynamic count.

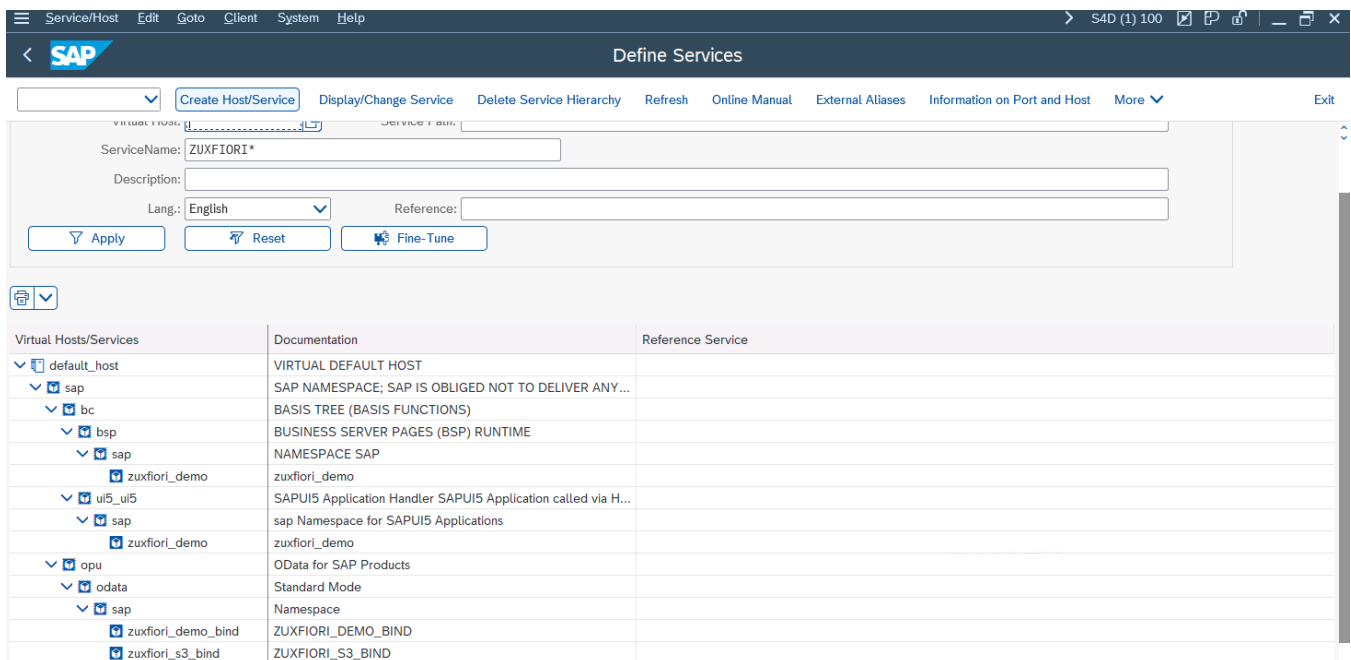
Screenshot Evidence — Semantic Object & Service Path Registration

SAP		
Display View "Semantic Objects - Customer View": Overview		
▼ Select All Select Block Deselect All More ▼		
Semantic Objects - Customer View		
Semantic Object	Semantic Object Name	Semantic Object Description
☐ CustomerBooking	Customer Booking	Semantic object for training class S4DEV
☐ STARTProcess	STARTProcess	START Test Processes
☐ STARTTestcase	STARTTestcase	START Testcases for fiori
☐ zuxfiori_demo	zuxfiori_demo	zuxfiori_demo

SM30 — Semantic Objects (Customer View) table showing zuxfiori_demo registered as a semantic object.

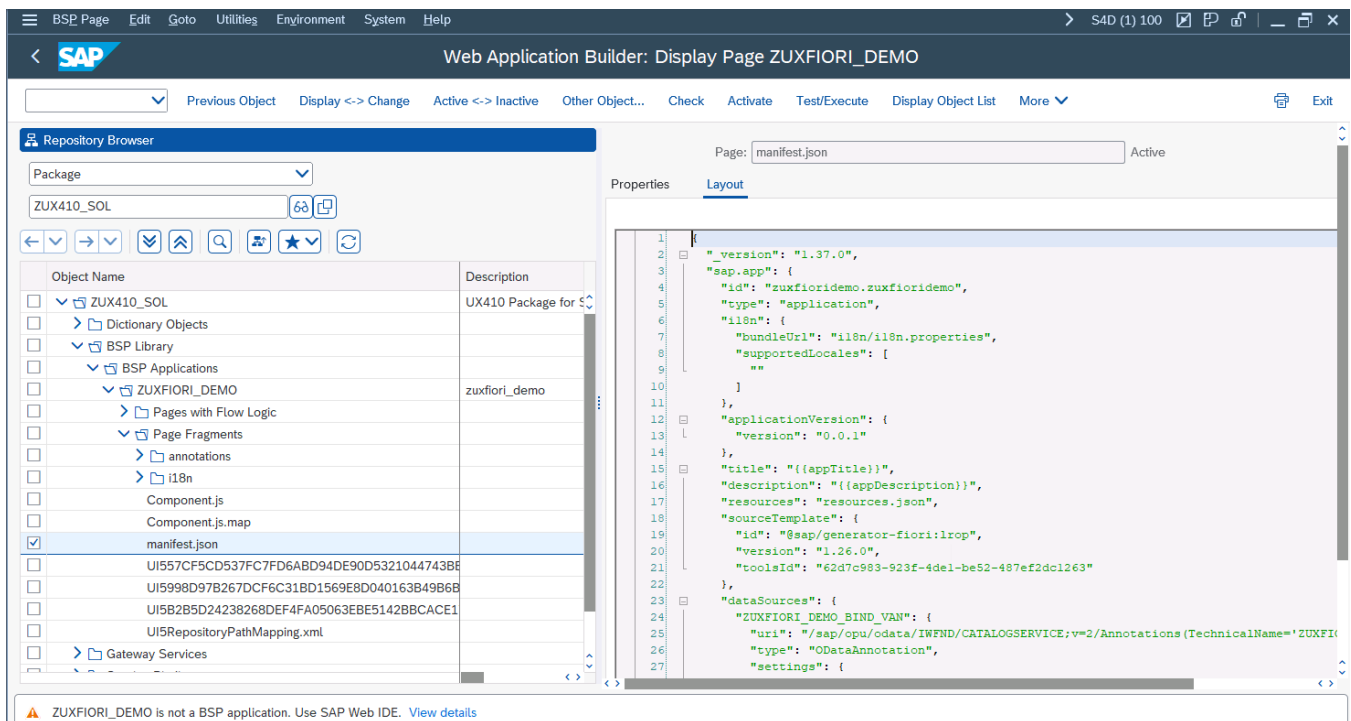


SICF (Define Services) — the BSP/UI5 application node tree for zuxfiiori_demo, plus the OData node ZUXFIORI_DEMO_BIND.

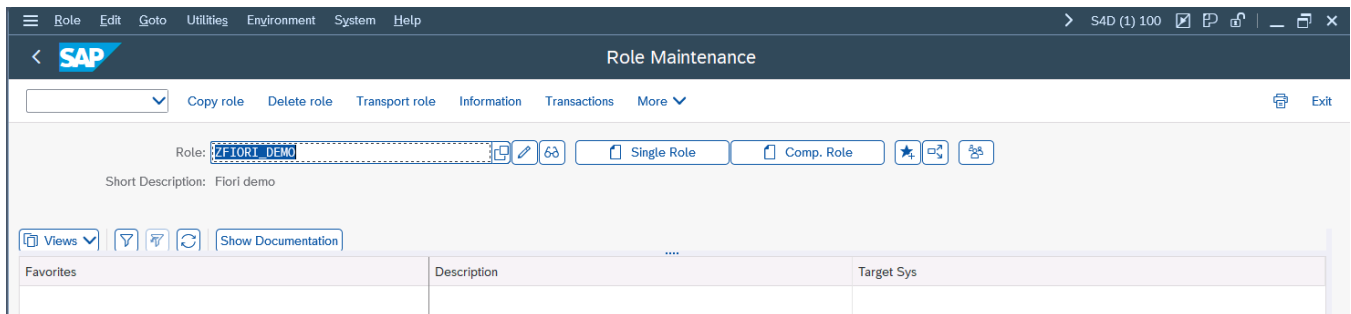


SICF tree expanded further — showing both zuxfiiori_demo_bind and zuxfiiori_s3_bind OData service nodes registered side by side.

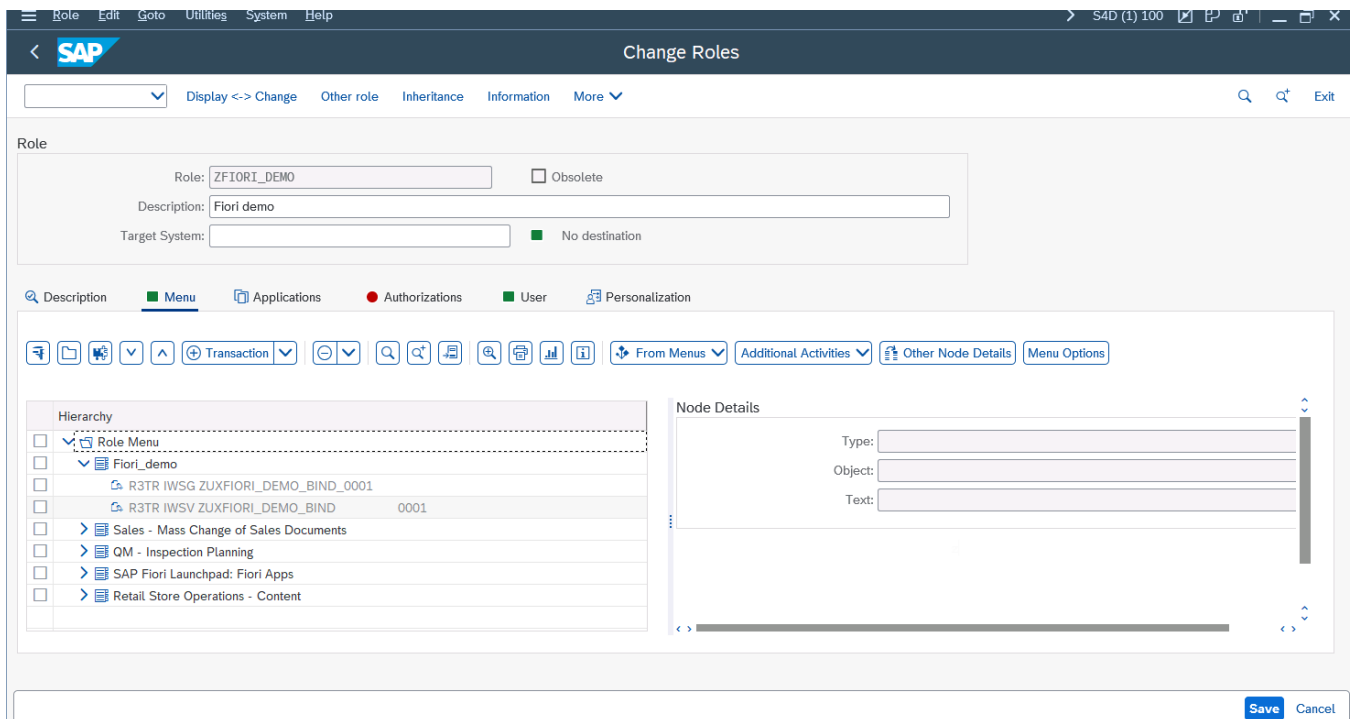
Screenshot Evidence — BSP Deployment & PFCG Role Setup



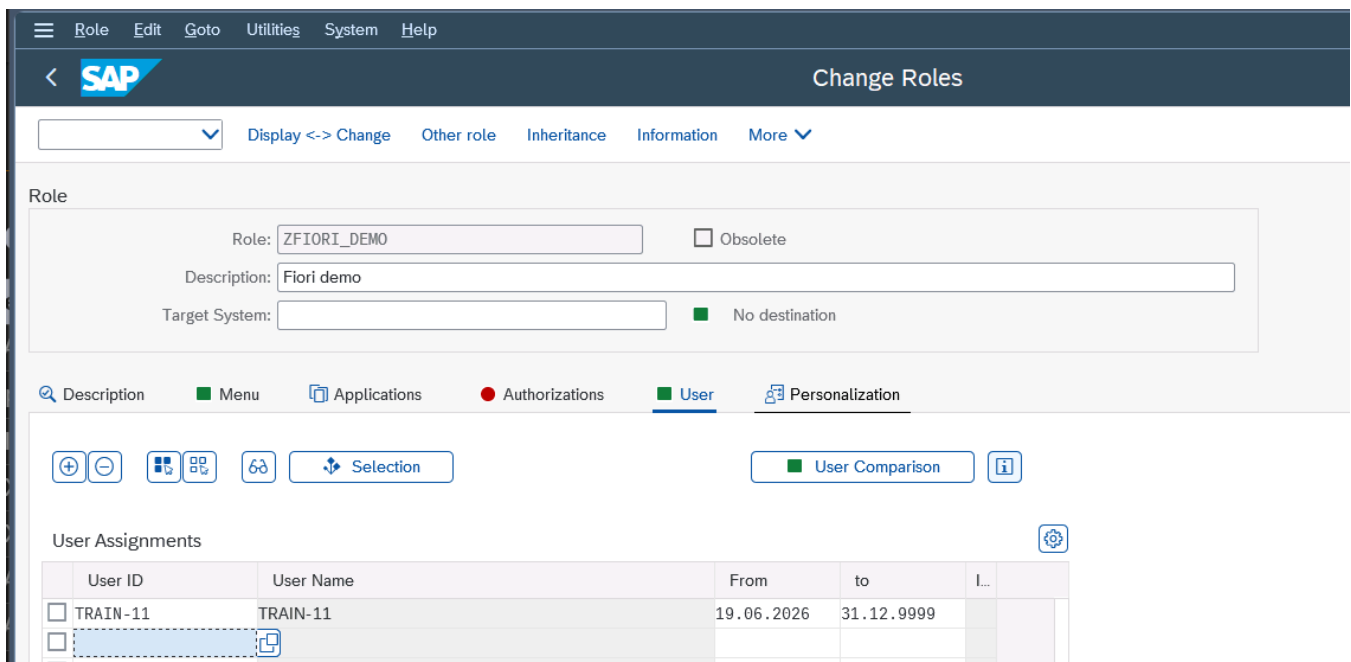
SE80 Web Application Builder — the deployed BSP application ZUXFIORI_DEMO, with manifest.json open. Note the warning that classic SE80 isn't the recommended tool for SAPUI5 apps — BAS/Fiori Tools is.



PFCG — Role Maintenance initial screen for role ZFIORI_DEMO.

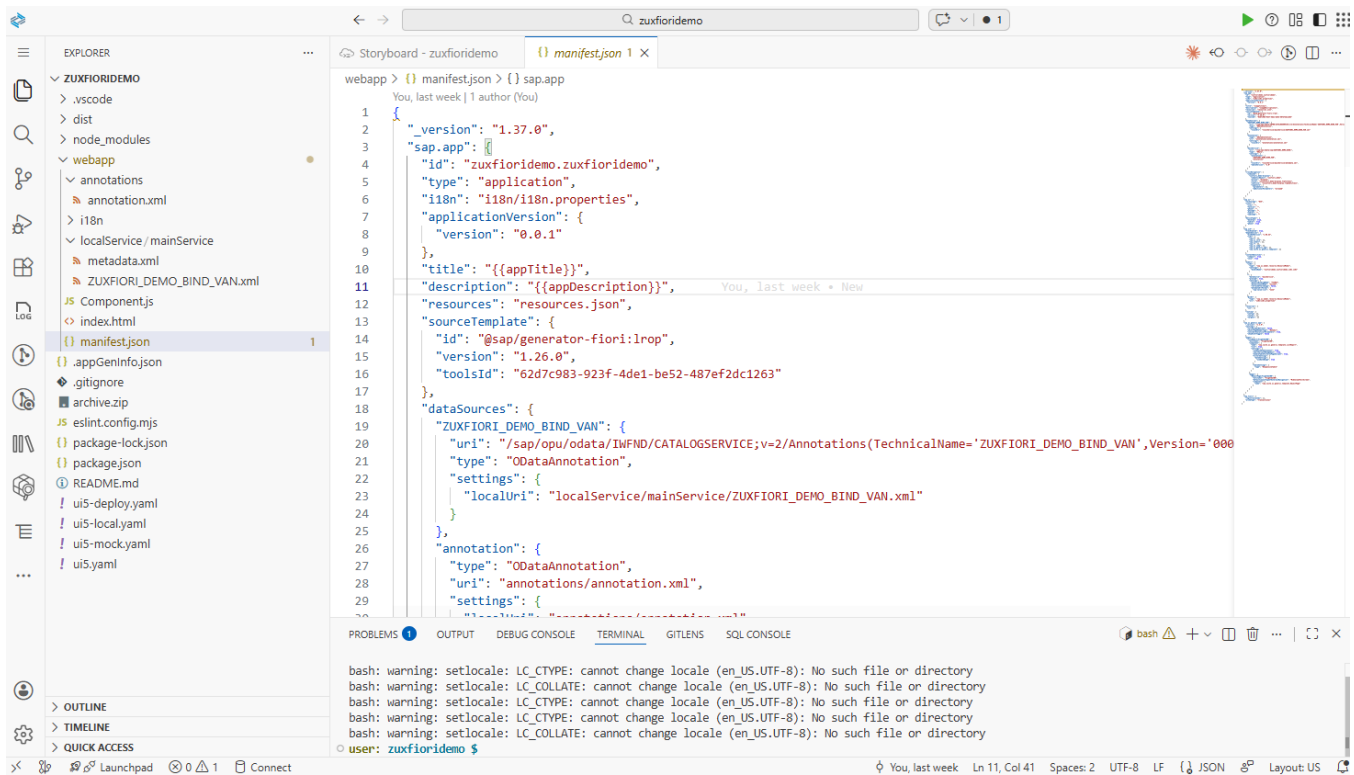


PFCG — Change Roles, Menu tab, showing the role hierarchy with the Fiori_demo catalog and target mapping node nested under it.

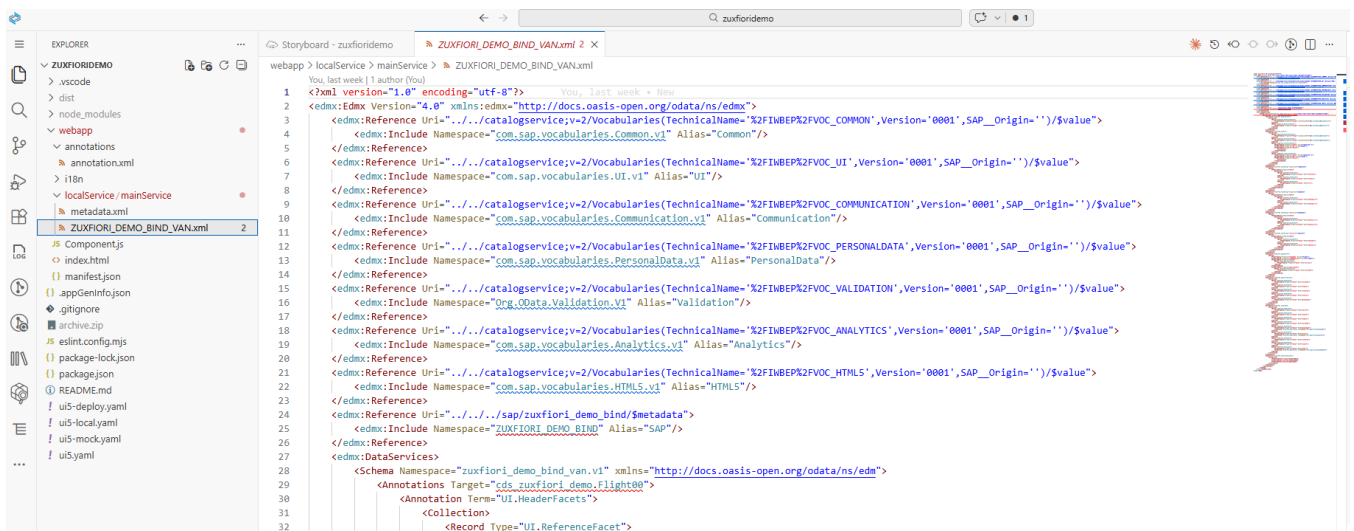


PFCG — Change Roles, User tab, showing TRAIN-11 assigned to the role with a validity period.

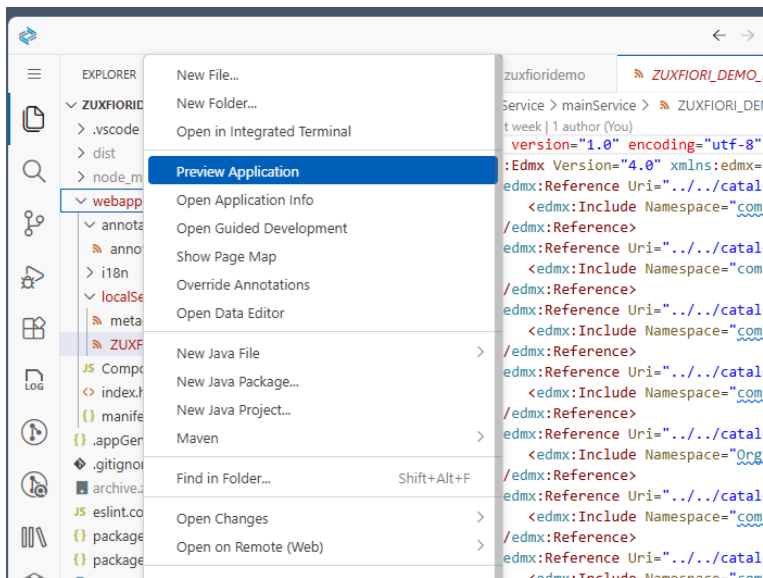
Screenshot Evidence — Front-End Development in SAP Business Application Studio



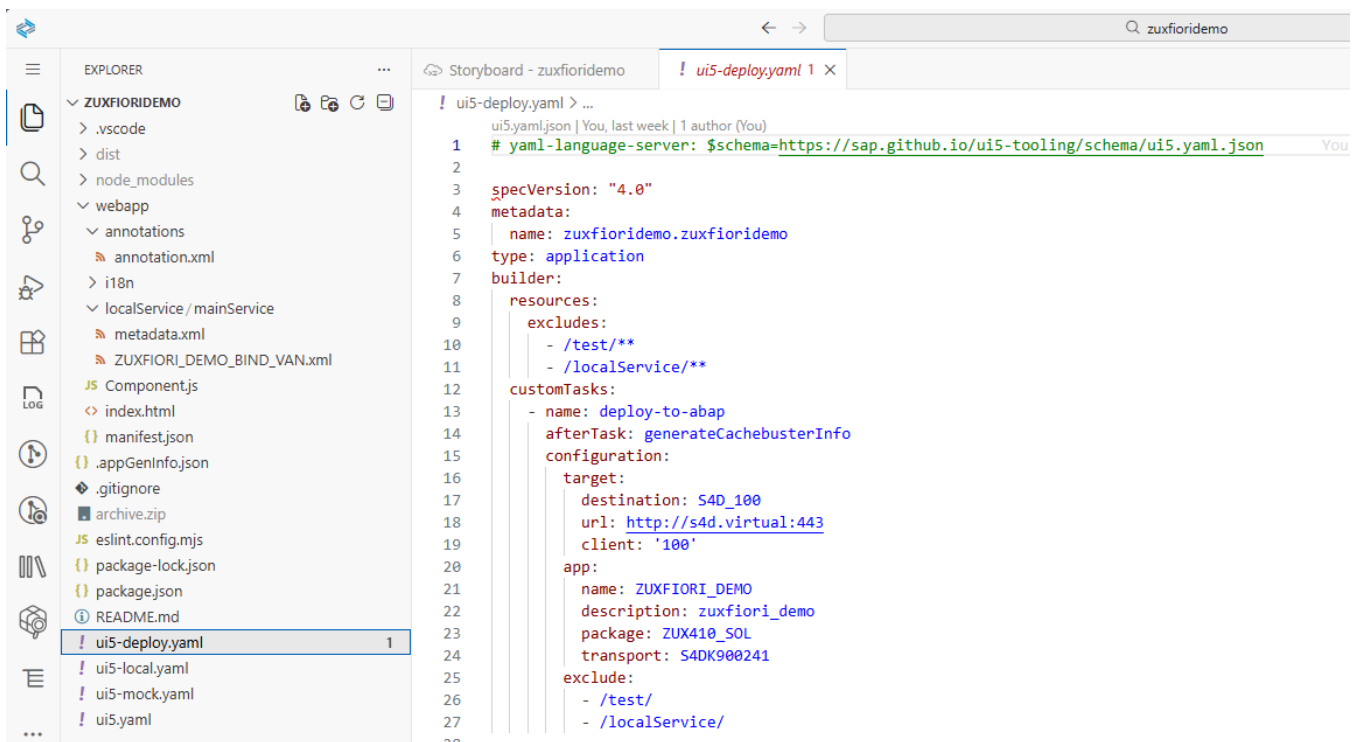
BAS Explorer — the zuxfioridemo project structure, with manifest.json open showing the dataSources block (OData annotation + ZUXFIORI_DEMO_BIND_VAN service).



Generated OData annotation file (ZUXFIORI_DEMO_BIND_VAN.xml) — the EDMX vocabulary references and the start of the UI.HeaderFacets annotation, generated from the CDS @UI annotations.



Fiori Tools context menu — Preview Application, Open Guided Development, Show Page Map, and Override Annotations: the core BAS commands for working with a Fiori Elements project.

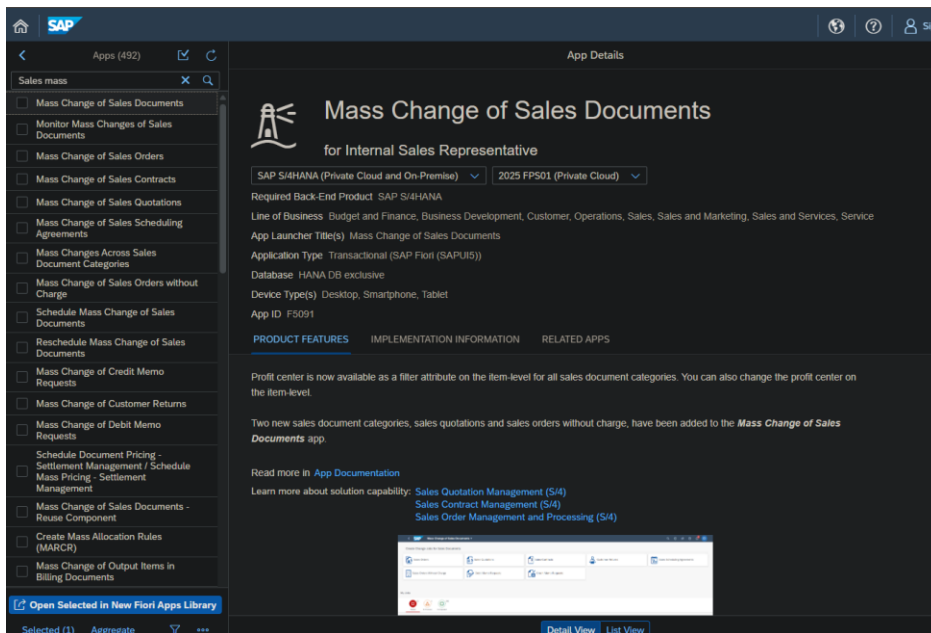


ui5-deploy.yaml — the deploy-to-abap configuration: target system (S4D_100), ABAP package (ZUX410_SOL), and transport request (S4DK900241).

Fiori Reference Library Demonstration:

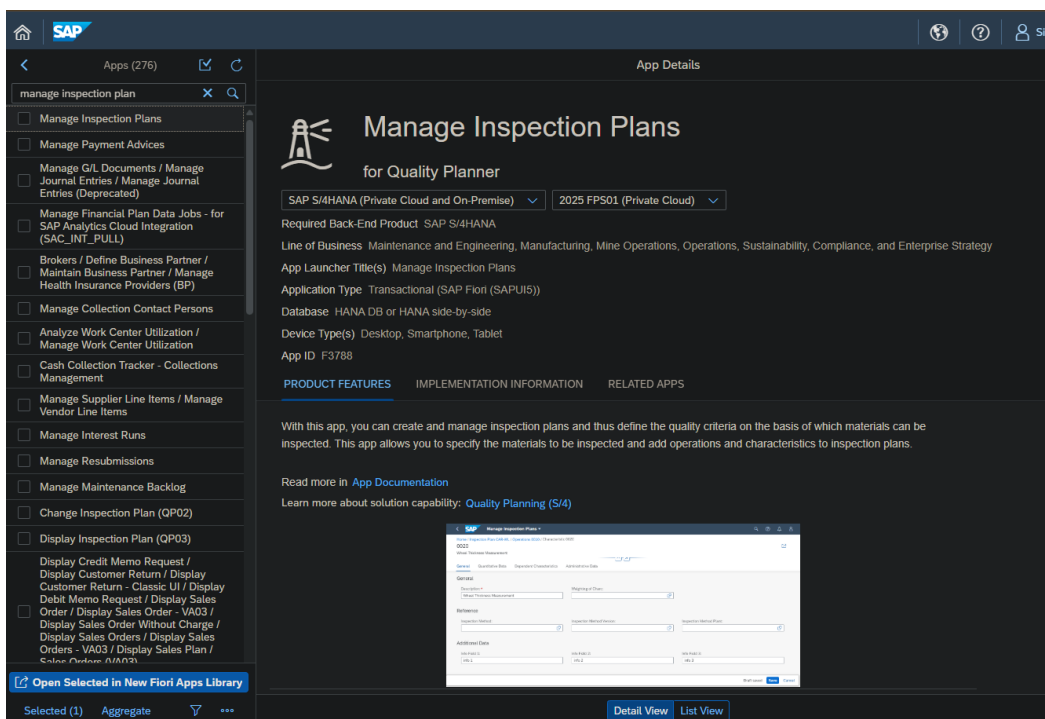
[https://fioriappslibrary.hana.ondemand.com/sap/fix/externalViewer/#/detail/Apps\('F5091'\)/S32PCE](https://fioriappslibrary.hana.ondemand.com/sap/fix/externalViewer/#/detail/Apps('F5091')/S32PCE)

Screenshot Evidence — Fiori Apps Reference Library: Example Lookups



Fiori Apps Library — 'Mass Change of Sales Documents' (F5091): Transactional app, HANA DB exclusive.

[https://fioriappslibrary.hana.ondemand.com/sap/fix/externalViewer/#!/detail/Apps\('F3788'\)/S32PCE](https://fioriappslibrary.hana.ondemand.com/sap/fix/externalViewer/#!/detail/Apps('F3788')/S32PCE)



Fiori Apps Library — 'Manage Inspection Plans' (F3788): Transactional app, HANA DB or HANA side-by-side.

[https://fioriappslibrary.hana.ondemand.com/sap/fix/externalViewer/#!/detail/Apps\('F0336A'\)/S32PCE](https://fioriappslibrary.hana.ondemand.com/sap/fix/externalViewer/#!/detail/Apps('F0336A')/S32PCE)

Retail Promotion (S/4HANA)
for Promotion Specialist (Retail)

SAP S/4HANA (Private Cloud and On-Premise) | 2025 FPS01 (Private Cloud)

Required Back-End Product: SAP S/4HANA
Line of Business: Merchandising, Sales and Marketing, Supply Chain
App Launcher Title(s): Retail Promotion
Application Type: Transactional, Fact sheet (SAP Fiori elements)
Database: HANA DB exclusive
Device Type(s): Desktop, Smartphone, Tablet
App ID: F0336A

PRODUCT FEATURES | IMPLEMENTATION INFORMATION | RELATED APPS

With this app, you can access promotion data and context information easily and efficiently. The app displays all relevant information about a promotion in one place and enables you to navigate directly to related business objects.

Read more in [App Documentation](#)
Learn more about solution capability: [Promotion Management \(S/4\)](#)
[Promotional Procurement \(S/4\)](#)

Selected (1) | Aggregate | [Detail View](#) | [List View](#)

Fiori Apps Library — ‘Retail Promotion’ (F0336A): a combined Transactional + Fact Sheet app type — useful for showing that one app can implement more than one floorplan.

[https://fioriappslibrary.hana.ondemand.com/sap/fix/externalViewer/#!/detail/Apps\('F4512'\)/S32PCE](https://fioriappslibrary.hana.ondemand.com/sap/fix/externalViewer/#!/detail/Apps('F4512')/S32PCE)

Manage Launchpad Pages
for Administrator

SAP S/4HANA (Private Cloud and On-Premise) | 2025 FPS01 (Private Cloud)

Required Back-End Product: SAP S/4HANA
App Launcher Title(s): Manage Launchpad Pages
Application Type: Transactional (SAP Fiori (SAPUI5))
Database: HANA DB or HANA side-by-side
Device Type(s): Desktop
App ID: F4512

PRODUCT FEATURES | IMPLEMENTATION INFORMATION | RELATED APPS

With this app you can manage the launchpad pages, define the layout of pages and edit, copy or delete existing pages.

Read more in [App Documentation](#)

Selected (1) | Aggregate | [Detail View](#) | [List View](#)

Fiori Apps Library — ‘Manage Launchpad Pages’ (F4512): the admin app used earlier in this log to configure Spaces and Pages.

Screenshot Evidence — Role Authorization Check

SAP Change Roles

Role: ZFIORT_DEMO ☐ Obsolete

Description: Fiori demo

Target System: ☐ No destination

Menu

Hierarchy

- ☐ Role Menu
- ☐ Fiori_demo
- ☒ Sales - Mass Change of Sales Documents
- ☐ QM - Inspection Planning
- ☐ SAP Fiori Launchpad: Fiori Apps
- ☐ Retail Store Operations - Content

Node Details

Type:

Object:

Text:

Save Cancel

PFCG — Change Roles, Authorizations tab entry point, with the Fiori_demo catalog node selected in the Menu hierarchy.

SESSION 3

SAPUI5 Development

UX400 + UX402 · SAPUI5 Foundations — 21–30% of C_FIORD

• MATERIALS BUILT (PPTX + Study Guide)

Exam Points Raised

Add the topics covered once this session is delivered. Typical UX400/UX402 exam points to capture:

- MVC architecture: Controller (JS/TS) holds logic, View (XML) holds no logic, Model holds data
- sap.m control library — Table, List, Dialog, MessageBox, data binding
- OData V2 vs V4 model creation and CRUD operations in a controller
- Routing & navigation: manifest.json routes/targets, router.navTo()
- Fragments and custom dialogs

Code Examples

Add SAPUI5 controller/view snippets here as they're demonstrated. Placeholder pattern for an OData V2 model read with a filter:

```
// Example pattern – replace with your actual demo code
onInit: function () {
    var oModel = new sap.ui.model.odata.v2.ODataModel(
        '/sap/opu/odata/sap/ZUXFIORI_DEMO_CDS/'
    );
    this.getView().setModel(oModel);
}
```

Your Demonstration Notes

What was actually demonstrated in this session:

Custom Fiori Application Back-end – Eclipse IDE, Expanding CDS modelling, Service Definition & Binding, Gateway Service exploration & testing.

Custom Fiori Application Front-end – BAS IDE, Fiori generator template (List Report), Test web app, Deployment yaml files, deep focus on manifest.json, deep focus on xml annotations. Running through some git version control commands. Additionally, demonstrating Smart controls (OData v2). Show casing some i18n internalization files. Show casing the local service (mock data) options for Unit Testing. Routing and Navigation demonstrated as well.

<https://github.com/AngeliqueSabTec/ZUXFioriDemoS3>

Your Code

Paste your code here

```
@AbapCatalog.sqlViewName: 'ZUX410BOOKING00'
@AbapCatalog.compiler.compareFilter: true
@AccessControl.authorizationCheck: #CHECK
@EndUserText.label: 'bookings'
define view ZUX410_C BOOKINGUI00 as select from sbook {
  //sbook
  key mandt,
  key carrid,
  key connid,
  @UI.lineItem.position: 20
  key fldate,
  key bookid,
  customid,
  custtype,
  @UI.lineItem.position: 60
  smoker,
  luggweight,
  wunit,
  invoice,
  @UI.lineItem.position: 50
  class,
  forcuram,
  forcurkey,
  @UI.lineItem.position: 40
  loccuram,
  loccurkey,
  @UI.lineItem.position: 30
  order_date,
  counter,
  agencynum,
  cancelled,
  reserved,
  @UI.lineItem.position: 10
  passname,
  passform,

  passbirth
}

@AbapCatalog.sqlViewName:      'ZUX410FLIBASE'
@AbapCatalog.dataMaintenance:   #RESTRICTED
@AccessControl.authorizationCheck: #CHECK
@EndUserText.label:             'UX410 Flight Base (Interface View)'

define view ZUX410 I FLIGHTUI00 14 BASE as select from sflight as sf

association to scarr as sc
  on sf.carrid = sc.carrid

association to spfli as sp
  on sf.carrid = sp.carrid
  and sf.connid = sp.connid

association [*] to ZUX410_C_BOOKINGUI00 as _Bookings
  on $projection.carrid = _Bookings.carrid
  and $projection.connid = _Bookings.connid
  and $projection.fldate = _Bookings.fldate

{
  key sf.carrid,
  key sf.connid,
  key sf.fldate,
```

```

sc.carrname,
sc.url,

sp.cityfrom,
sp.cityto,
sp.airpfrom,
sp.airpto,
sp.distance,

sf.planetype,
sf.price,
sf.currency,

sf.price * sf.seatsocc as TotalRevenue,

sf.paymentsum,
sf.seatsmax,
sf.seatsocc,
sf.seatsmax_b,
sf.seatsocc_b,
sf.seatsmax_f,
sf.seatsocc_f,

division( sf.seatsocc * 100, sf.seatsmax, 0 ) as OccupancyPct,
_Bookings
}

```

```

@AbapCatalog.sqlViewName:      'ZUX410FLIGHTUIENH'
@AbapCatalog.compiler.compareFilter: true
@AbapCatalog.dataMaintenance:  #RESTRICTED
@AccessControl.authorizationCheck: #CHECK
@endUserText.label:            'UX410 Flight UI - Enhanced'

@OData.entitySetName:         'FlightSet'
@OData.entityType.name:       'Flight'

@UI.headerInfo.typeName:      'Flight'
@UI.headerInfo.typeNamePlural: 'Flights'
@UI.headerInfo.title.value:    'carrid'
@UI.headerInfo.description.value: 'carrname'

@Search.searchable: true

define view ZUX410 C FLIGHTUI00 14 Enced as select from ZUX410_I_FLIGHTUI00_14_BASE as
base

{
  @UI.facet: [
    { purpose: #HEADER, type: #FIELDGROUP_REFERENCE,
      targetQualifier: 'FlightInfo', position: 10, label: 'Flight Info' },
    { purpose: #HEADER, type: #FIELDGROUP_REFERENCE,
      targetQualifier: 'SeatSummary', position: 20, label: 'Seat Summary' },

    { id: 'FlightSection', purpose: #STANDARD, type: #COLLECTION,
      label: 'Flight Details', position: 10 },
    { id: 'SeatSection', purpose: #STANDARD, type: #COLLECTION,
      label: 'Seat Availability', position: 20 },
    { id: 'BookingSection', purpose: #STANDARD, type: #LINEITEM_REFERENCE,
      label: 'Bookings', targetElement: '_Bookings', position: 30 },

    { id: 'GeneralFacet', parentId: 'FlightSection', purpose: #STANDARD,
      type: #FIELDGROUP_REFERENCE, targetQualifier: 'GeneralData',
      label: 'General Data', position: 10 },
    { id: 'RouteFacet', parentId: 'FlightSection', purpose: #STANDARD,

```

```

    type: #FIELDGROUP_REFERENCE, targetQualifier: 'RouteData',
    label: 'Route', position: 20 },

    { id: 'EconomyFacet', parentId: 'SeatSection', purpose: #STANDARD,
      type: #FIELDGROUP_REFERENCE, targetQualifier: 'EconomySeats',
      label: 'Economy Class', position: 10 },
    { id: 'BusinessFacet', parentId: 'SeatSection', purpose: #STANDARD,
      type: #FIELDGROUP_REFERENCE, targetQualifier: 'BusinessSeats',
      label: 'Business Class', position: 20 },
    { id: 'FirstFacet', parentId: 'SeatSection', purpose: #STANDARD,
      type: #FIELDGROUP_REFERENCE, targetQualifier: 'FirstClassSeats',
      label: 'First Class', position: 30 }
  ]

@Common.label:      'Airline'
@UI.lineItem:       [{ position: 10, label: 'Airline' }]
@UI.selectionField: [{ position: 10 }]
key base.carriid,

@Common.label:      'Flight No.'
@UI.lineItem:       [{ position: 20, label: 'Flight No.' }]
@UI.selectionField: [{ position: 20 }]
key base.connid,

@Common.label:      'Flight Date'
@UI.lineItem:       [{ position: 30, label: 'Date' }]
@UI.identification: [{ position: 10 }]
@UI.selectionField: [{ position: 30 }]
key base.fldate,

@Common.label:      'Airline Name'
@Search.defaultSearchElement: true
@Search.fuzzinessThreshold: 0.7
@UI.lineItem:       [{ position: 40, label: 'Airline Name',
                      importance: #HIGH }]
@UI.identification: [{ position: 20 }]
@UI.selectionField: [{ position: 40 }]
@UI.fieldGroup:     [{ qualifier: 'FlightInfo', position: 10 },
                      { qualifier: 'GeneralData', position: 10 }]

base.carrname,

@Common.label:      'Carrier URL'
@UI.fieldGroup:     [{ qualifier: 'FlightInfo', position: 20 }]
base.url,

@Common.label:      'Departure City'
@UI.lineItem:       [{ position: 50, label: 'From' }]
@UI.fieldGroup:     [{ qualifier: 'RouteData', position: 10 }]
base.cityfrom,

@Common.label:      'Arrival City'
@UI.lineItem:       [{ position: 60, label: 'To' }]
@UI.fieldGroup:     [{ qualifier: 'RouteData', position: 20 }]
base.cityto,

@Common.label:      'Departure Airport'
@UI.fieldGroup:     [{ qualifier: 'RouteData', position: 30 }]
base.airpfrom,

@Common.label:      'Arrival Airport'
@UI.fieldGroup:     [{ qualifier: 'RouteData', position: 40 }]
base.airpto,

@Common.label:      'Distance (km)'
@UI.fieldGroup:     [{ qualifier: 'RouteData', position: 50 }]
base.distance,

```

```

@Common.label:      'Plane Type'
@UI.lineItem:       [{ position: 70, label: 'Aircraft', importance: #HIGH }]
@UI.identification: [{ position: 30 }]
@UI.fieldGroup:     [{ qualifier: 'GeneralData', position: 20 }]
base.planetype,

@Common.label:      'Base Price'
@Semantics.amount.currencyCode: 'currency'
@UI.lineItem:       [{ position: 80, label: 'Price',
                      importance: #MEDIUM,
                      criticality: 'PriceCriticality',
                      criticalityRepresentation: #WITH_ICON }]
@UI.identification: [{ position: 40 }]
@UI.fieldGroup:     [{ qualifier: 'GeneralData', position: 30 }]
base.price,

@Semantics.currencyCode: true
@UI.hidden: true
base.currency,

@Common.label:      'Total Revenue'
@Semantics.amount.currencyCode: 'currency'
@UI.lineItem:       [{ position: 90, label: 'Total Revenue',
                      importance: #LOW }]
@UI.identification: [{ position: 50 }]
@UI.fieldGroup:     [{ qualifier: 'GeneralData', position: 40 }]
base.TotalRevenue,

base.paymentsum,

@Common.label:      'Max Economy Seats'
@UI.fieldGroup:     [{ qualifier: 'SeatSummary', position: 10 },
                    { qualifier: 'EconomySeats', position: 10 }]
base.seatsmax,

@Common.label:      'Economy Occupied'
@UI.lineItem:       [{ position: 100, label: 'Eco. Occ.',
                      importance: #LOW }]
@UI.fieldGroup:     [{ qualifier: 'SeatSummary', position: 20 },
                    { qualifier: 'EconomySeats', position: 20 }]
base.seatsocc,

@Common.label:      'Business Max'
@UI.fieldGroup:     [{ qualifier: 'BusinessSeats', position: 10 }]
base.seatsmax_b,

@Common.label:      'Business Occupied'
@UI.fieldGroup:     [{ qualifier: 'BusinessSeats', position: 20 }]
base.seatsocc_b,

@Common.label:      'First Class Max'
@UI.fieldGroup:     [{ qualifier: 'FirstClassSeats', position: 10 }]
base.seatsmax_f,

@Common.label:      'First Class Occupied'
@UI.fieldGroup:     [{ qualifier: 'FirstClassSeats', position: 20 }]
base.seatsocc_f,

@UI.hidden: true
case
  when base.price <= 500 then 3
  when base.price <= 1000 then 2
  else 1
end as PriceCriticality,

```

```

@UI.hidden: true
case
  when base.OccupancyPct >= 80 then 3
  when base.OccupancyPct >= 50 then 2
  else 1
end as OccupancyCriticality,

@UI.hidden: true
base.OccupancyPct,

_bookings
}

```

```

@EndUserText.label: 'Session 3 Demo'
define service ZUXFIORI_DEMO_S3 {
  expose ZUX410_C_FLIGHTUI00_14_Enced as FlightSet;
  expose ZUX410_C_BOOKINGUI00 as BookingSet;
}

```

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta http-equiv="X-UA-Compatible" content="IE=edge">
  <title>zfioridemos3</title>
  <style>
    html, body, body > div, #container, #container-uiarea {
      height: 100%;
    }
  </style>
  <script>
    id="sap-ui-bootstrap"
    src="resources/sap-ui-core.js"
    data-sap-ui-theme="sap_fiori_3"
    data-sap-ui-resourceroots='{
      "zfioridemos3": "./"
    }'
    data-sap-ui-oninit="module:sap/ui/core/ComponentSupport"
    data-sap-ui-compatVersion="edge"
    data-sap-ui-async="true"
    data-sap-ui-frameOptions="trusted"
  ></script>
</head>
<body class="sapUiBody sapUiSizeCompact" id="content">
  <div>
    data-sap-ui-component
    data-name="zfioridemos3"
    data-id="container"
    data-settings='{ "id" : "zfioridemos3" }'
    data-handle-validation="true"
  ></div>
</body>
</html>

```

```

{
  "_version": "1.37.0",
  "sap.app": {
    "id": "zfioridemos3",
    "type": "application",
    "i18n": "i18n/i18n.properties",
    "applicationVersion": {
      "version": "0.0.1"
    },
  },

```

```

"title": "{{appTitle}}",
"description": "{{appDescription}}",
"resources": "resources.json",
"sourceTemplate": {
  "id": "@sap/generator-fiori:drop",
  "version": "1.27.0",
  "toolsId": "025c90d1-d346-4baa-bfac-624e7c7aa1a9"
},
"dataSources": {
  "ZUXFIORI_S3_BIND_VAN": {
    "uri":
"/sap/opu/odata/IWFND/CATALOGSERVICE;v=2/Annotations(TechnicalName='ZUXFIORI_S3_BIND_VAN',
Version='0001')/$value/",
    "type": "ODataAnnotation",
    "settings": {
      "localUri": "localService/mainService/ZUXFIORI_S3_BIND_VAN.xml"
    }
  },
  "annotation": {
    "type": "ODataAnnotation",
    "uri": "annotations/annotation.xml",
    "settings": {
      "localUri": "annotations/annotation.xml"
    }
  },
  "mainService": {
    "uri": "/sap/opu/odata/sap/ZUXFIORI_S3_BIND/",
    "type": "OData",
    "settings": {
      "annotations": [
        "ZUXFIORI_S3_BIND_VAN",
        "annotation"
      ],
      "localUri": "localService/mainService/metadata.xml",
      "odataVersion": "2.0"
    }
  }
},
"crossNavigation": {
  "inbounds": {
    "zfioridemos3-display": {
      "semanticObject": "zfioridemos3",
      "action": "display",
      "title": "{{zfioridemos3-display.flpTitle}}",
      "subTitle": "{{zfioridemos3-display.flpSubtitle}}",
      "signature": {
        "parameters": {},
        "additionalParameters": "allowed"
      }
    }
  }
},
"sap.ui": {
  "technology": "UI5",
  "icons": {
    "icon": "",
    "favIcon": "",
    "phone": "",
    "phone@2": "",
    "tablet": "",
    "tablet@2": ""
  },
  "deviceTypes": {
    "desktop": true,
    "tablet": true,

```

```

    "phone": true
  }
},
"sap.ui5": {
  "flexEnabled": true,
  "dependencies": {
    "minUI5Version": "1.96.33",
    "libs": {
      "sap.m": {},
      "sap.ui.core": {},
      "sap.ushell": {},
      "sap.f": {},
      "sap.ui.comp": {},
      "sap.ui.generic.app": {},
      "sap.suite.ui.generic.template": {}
    }
  },
  "contentDensities": {
    "compact": true,
    "cozy": true
  },
  "models": {
    "i18n": {
      "type": "sap.ui.model.resource.ResourceModel",
      "settings": {
        "bundleName": "zfioridemos3.i18n.i18n"
      }
    },
    "": {
      "dataSource": "mainService",
      "preload": true,
      "settings": {
        "defaultBindingMode": "TwoWay",
        "defaultCountMode": "Inline",
        "refreshAfterChange": false,
        "metadataUrlParams": {
          "sap-value-list": "none"
        }
      }
    },
    "@i18n": {
      "type": "sap.ui.model.resource.ResourceModel",
      "uri": "i18n/i18n.properties"
    }
  },
  "resources": {
    "css": []
  },
  "routing": {
    "config": {},
    "routes": [],
    "targets": {}
  }
},
"sap.ui.generic.app": {
  "_version": "1.3.0",
  "settings": {
    "forceGlobalRefresh": false,
    "objectPageHeaderType": "Dynamic",
    "considerAnalyticalParameters": true,
    "showDraftToggle": false
  },
  "pages": {
    "ListReport|FlightSet": {
      "entitySet": "FlightSet",
      "component": {

```

```

"name": "sap.suite.ui.generic.template.ListReport",
"list": true,
"settings": {
  "condensedTableLayout": true,
  "smartVariantManagement": true,
  "enableTableFilterInPageVariant": true,
  "filterSettings": {
    "dateSettings": {
      "useDateRange": true
    }
  },
  "tableSettings": {
    "type": "ResponsiveTable"
  }
},
"pages": {
  "ObjectPage|FlightSet": {
    "entitySet": "FlightSet",
    "defaultLayoutTypeIfExternalNavigation": "MidColumnFullScreen",
    "component": {
      "name": "sap.suite.ui.generic.template.ObjectPage"
    },
    "pages": {
      "ObjectPage|to_Bookings": {
        "navigationProperty": "to_Bookings",
        "entitySet": "BookingSet",
        "defaultLayoutTypeIfExternalNavigation": "MidColumnFullScreen",
        "component": {
          "name": "sap.suite.ui.generic.template.ObjectPage"
        }
      }
    }
  }
}
},
"sap.fiori": {
  "registrationIds": [],
  "archetype": "transactional"
}
}

```

```

sap.ui.define(
  ["sap/suite/ui/generic/template/lib/AppComponent"],
  function (Component) {
    "use strict";

    return Component.extend("zfioridemos3.Component", {
      metadata: {
        manifest: "json"
      }
    });
  }
);

```

```

<?xml version="1.0" encoding="utf-8"?>
<edmx:Edmx Version="4.0" xmlns:edmx="http://docs.oasis-open.org/odata/ns/edmx">
  <edmx:Reference
Uri="../../catalogservice;v=2/Vocabularies(TechnicalName='%2FIWBEP%2FVOC_COMMON',Version='0001',SAP__Origin='')/$value">
    <edmx:Include Namespace="com.sap.vocabularies.Common.v1" Alias="Common"/>
  </edmx:Reference>

```

```

    <edmx:Reference
Uri=".../catalogservice;v=2/Vocabularies(TechnicalName='%2FIWBEP%2FVOC_UI',Version='0001',SAP__Origin='')/$value">
    <edmx:Include Namespace="com.sap.vocabularies.UI.v1" Alias="UI"/>
    </edmx:Reference>
    <edmx:Reference
Uri=".../catalogservice;v=2/Vocabularies(TechnicalName='%2FIWBEP%2FVOC_COMMUNICATION',Version='0001',SAP__Origin='')/$value">
    <edmx:Include Namespace="com.sap.vocabularies.Communication.v1"
Alias="Communication"/>
    </edmx:Reference>
    <edmx:Reference
Uri=".../catalogservice;v=2/Vocabularies(TechnicalName='%2FIWBEP%2FVOC_PERSONALDATA',Version='0001',SAP__Origin='')/$value">
    <edmx:Include Namespace="com.sap.vocabularies.PersonalData.v1"
Alias="PersonalData"/>
    </edmx:Reference>
    <edmx:Reference
Uri=".../catalogservice;v=2/Vocabularies(TechnicalName='%2FIWBEP%2FVOC_VALIDATION',Version='0001',SAP__Origin='')/$value">
    <edmx:Include Namespace="Org.odata.Validation.V1" Alias="Validation"/>
    </edmx:Reference>
    <edmx:Reference
Uri=".../catalogservice;v=2/Vocabularies(TechnicalName='%2FIWBEP%2FVOC_ANALYTICS',Version='0001',SAP__Origin='')/$value">
    <edmx:Include Namespace="com.sap.vocabularies.Analytics.v1" Alias="Analytics"/>
    </edmx:Reference>
    <edmx:Reference
Uri=".../catalogservice;v=2/Vocabularies(TechnicalName='%2FIWBEP%2FVOC_HTML5',Version='0001',SAP__Origin='')/$value">
    <edmx:Include Namespace="com.sap.vocabularies.HTML5.v1" Alias="HTML5"/>
    </edmx:Reference>
    <edmx:Reference Uri=".../..../sap/zuxfiori_s3_bind/$metadata">
    <edmx:Include Namespace="ZUXFIORI_S3_BIND" Alias="SAP"/>
    </edmx:Reference>
    <edmx:DataServices>
    <Schema Namespace="zuxfiori_s3_bind_van.v1" xmlns="http://docs.oasis-open.org/odata/ns/edm">
    <Annotations Target="cds_zuxfiori_demo_s3.BookingSetType">
    <Annotation Term="UI.LineItem">
    <Collection>
    <Record Type="UI.DataField">
    <PropertyValue Property="Value" Path="passname"/>
    </Record>
    <Record Type="UI.DataField">
    <PropertyValue Property="Value" Path="fldate"/>
    </Record>
    <Record Type="UI.DataField">
    <PropertyValue Property="Value" Path="order_date"/>
    </Record>
    <Record Type="UI.DataField">
    <PropertyValue Property="Value" Path="loccuram"/>
    </Record>
    <Record Type="UI.DataField">
    <PropertyValue Property="Value" Path="class"/>
    </Record>
    <Record Type="UI.DataField">
    <PropertyValue Property="Value" Path="smoker"/>
    </Record>
    </Collection>
    </Annotation>
    </Annotations>
    <Annotations Target="cds_zuxfiori_demo_s3.Flight/currency">
    <Annotation Term="UI.Hidden"/>
    </Annotations>
    <Annotations Target="cds_zuxfiori_demo_s3.Flight/PriceCriticality">

```

```

        <Annotation Term="UI.Hidden"/>
    </Annotations>
    <Annotations Target="cds_zuxfiori_demo_s3.Flight/OccupancyCriticality">
        <Annotation Term="UI.Hidden"/>
    </Annotations>
    <Annotations Target="cds_zuxfiori_demo_s3.Flight/OccupancyPct">
        <Annotation Term="UI.Hidden"/>
    </Annotations>
    <Annotations Target="cds_zuxfiori_demo_s3.Flight">
        <Annotation Term="UI.Facets">
            <Collection>
                <Record Type="UI.CollectionFacet">
                    <PropertyValue Property="Label" String="Flight Details"/>
                    <PropertyValue Property="ID" String="FlightSection"/>
                    <PropertyValue Property="Facets">
                        <Collection>
                            <Record Type="UI.ReferenceFacet">
                                <PropertyValue Property="Label" String="General
Data"/>
                                <PropertyValue Property="ID"
String="GeneralFacet"/>
                                <PropertyValue Property="Target"
AnnotationPath="@UI.FieldGroup#GeneralData"/>
                            </Record>
                            <Record Type="UI.ReferenceFacet">
                                <PropertyValue Property="Label" String="Route"/>
                                <PropertyValue Property="ID" String="RouteFacet"/>
                                <PropertyValue Property="Target"
AnnotationPath="@UI.FieldGroup#RouteData"/>
                            </Record>
                        </Collection>
                    </PropertyValue>
                </Record>
                <Record Type="UI.CollectionFacet">
                    <PropertyValue Property="Label" String="Seat Availability"/>
                    <PropertyValue Property="ID" String="SeatSection"/>
                    <PropertyValue Property="Facets">
                        <Collection>
                            <Record Type="UI.ReferenceFacet">
                                <PropertyValue Property="Label" String="Economy
Class"/>
                                <PropertyValue Property="ID"
String="EconomyFacet"/>
                                <PropertyValue Property="Target"
AnnotationPath="@UI.FieldGroup#EconomySeats"/>
                            </Record>
                            <Record Type="UI.ReferenceFacet">
                                <PropertyValue Property="Label" String="Business
Class"/>
                                <PropertyValue Property="ID"
String="BusinessFacet"/>
                                <PropertyValue Property="Target"
AnnotationPath="@UI.FieldGroup#BusinessSeats"/>
                            </Record>
                            <Record Type="UI.ReferenceFacet">
                                <PropertyValue Property="Label" String="First
Class"/>
                                <PropertyValue Property="ID" String="FirstFacet"/>
                                <PropertyValue Property="Target"
AnnotationPath="@UI.FieldGroup#FirstClassSeats"/>
                            </Record>
                        </Collection>
                    </PropertyValue>
                </Record>
            </Collection>
        </Annotation>
    </Annotations>
    <Record Type="UI.ReferenceFacet">
        <PropertyValue Property="Label" String="Bookings"/>
    </Record>

```

```

        <PropertyValue Property="ID" String="BookingSection"/>
        <PropertyValue Property="Target"
AnnotationPath="to_Bookings/@UI.LineItem"/>
    </Record>
</Collection>
</Annotation>
<Annotation Term="UI.HeaderFacets">
    <Collection>
        <Record Type="UI.ReferenceFacet">
            <PropertyValue Property="Label" String="Flight Info"/>
            <PropertyValue Property="Target"
AnnotationPath="@UI.FieldGroup#FlightInfo"/>
        </Record>
        <Record Type="UI.ReferenceFacet">
            <PropertyValue Property="Label" String="Seat Summary"/>
            <PropertyValue Property="Target"
AnnotationPath="@UI.FieldGroup#SeatSummary"/>
        </Record>
    </Collection>
</Annotation>
<Annotation Term="UI.FieldGroup" Qualifier="BusinessSeats">
    <Record>
        <PropertyValue Property="Data">
            <Collection>
                <Record Type="UI.DataField">
                    <PropertyValue Property="Value" Path="seatsmax_b"/>
                </Record>
                <Record Type="UI.DataField">
                    <PropertyValue Property="Value" Path="seatsocc_b"/>
                </Record>
            </Collection>
        </PropertyValue>
    </Record>
</Annotation>
<Annotation Term="UI.FieldGroup" Qualifier="EconomySeats">
    <Record>
        <PropertyValue Property="Data">
            <Collection>
                <Record Type="UI.DataField">
                    <PropertyValue Property="Value" Path="seatsmax"/>
                </Record>
                <Record Type="UI.DataField">
                    <PropertyValue Property="Value" Path="seatsocc"/>
                </Record>
            </Collection>
        </PropertyValue>
    </Record>
</Annotation>
<Annotation Term="UI.FieldGroup" Qualifier="FirstClassSeats">
    <Record>
        <PropertyValue Property="Data">
            <Collection>
                <Record Type="UI.DataField">
                    <PropertyValue Property="Value" Path="seatsmax_f"/>
                </Record>
                <Record Type="UI.DataField">
                    <PropertyValue Property="Value" Path="seatsocc_f"/>
                </Record>
            </Collection>
        </PropertyValue>
    </Record>
</Annotation>
<Annotation Term="UI.FieldGroup" Qualifier="FlightInfo">
    <Record>
        <PropertyValue Property="Data">
            <Collection>

```

```

        <Record Type="UI.DataField">
            <PropertyValue Property="Value" Path="carrname"/>
        </Record>
        <Record Type="UI.DataField">
            <PropertyValue Property="Value" Path="url"/>
        </Record>
    </Collection>
</PropertyValue>
</Record>
</Annotation>
<Annotation Term="UI.FieldGroup" Qualifier="GeneralData">
    <Record>
        <PropertyValue Property="Data">
            <Collection>
                <Record Type="UI.DataField">
                    <PropertyValue Property="Value" Path="carrname"/>
                </Record>
                <Record Type="UI.DataField">
                    <PropertyValue Property="Value" Path="planetype"/>
                </Record>
                <Record Type="UI.DataField">
                    <PropertyValue Property="Value" Path="price"/>
                </Record>
                <Record Type="UI.DataField">
                    <PropertyValue Property="Value" Path="TotalRevenue"/>
                </Record>
            </Collection>
        </PropertyValue>
    </Record>
</Annotation>
<Annotation Term="UI.FieldGroup" Qualifier="RouteData">
    <Record>
        <PropertyValue Property="Data">
            <Collection>
                <Record Type="UI.DataField">
                    <PropertyValue Property="Value" Path="cityfrom"/>
                </Record>
                <Record Type="UI.DataField">
                    <PropertyValue Property="Value" Path="cityto"/>
                </Record>
                <Record Type="UI.DataField">
                    <PropertyValue Property="Value" Path="airpfrom"/>
                </Record>
                <Record Type="UI.DataField">
                    <PropertyValue Property="Value" Path="airpto"/>
                </Record>
                <Record Type="UI.DataField">
                    <PropertyValue Property="Value" Path="distance"/>
                </Record>
            </Collection>
        </PropertyValue>
    </Record>
</Annotation>
<Annotation Term="UI.FieldGroup" Qualifier="SeatSummary">
    <Record>
        <PropertyValue Property="Data">
            <Collection>
                <Record Type="UI.DataField">
                    <PropertyValue Property="Value" Path="seatsmax"/>
                </Record>
                <Record Type="UI.DataField">
                    <PropertyValue Property="Value" Path="seatsocc"/>
                </Record>
            </Collection>
        </PropertyValue>
    </Record>

```

```

</Annotation>
<Annotation Term="UI.HeaderInfo">
  <Record>
    <PropertyValue Property="TypeName" String="Flight"/>
    <PropertyValue Property="TypeNamePlural" String="Flights"/>
    <PropertyValue Property="Title">
      <Record Type="UI.DataField">
        <PropertyValue Property="Value" Path="carrid"/>
      </Record>
    </PropertyValue>
    <PropertyValue Property="Description">
      <Record Type="UI.DataField">
        <PropertyValue Property="Value" Path="carrname"/>
      </Record>
    </PropertyValue>
  </Record>
</Annotation>
<Annotation Term="UI.Identification">
  <Collection>
    <Record Type="UI.DataField">
      <PropertyValue Property="Value" Path="fldate"/>
    </Record>
    <Record Type="UI.DataField">
      <PropertyValue Property="Value" Path="carrname"/>
    </Record>
    <Record Type="UI.DataField">
      <PropertyValue Property="Value" Path="planetype"/>
    </Record>
    <Record Type="UI.DataField">
      <PropertyValue Property="Value" Path="price"/>
    </Record>
    <Record Type="UI.DataField">
      <PropertyValue Property="Value" Path="TotalRevenue"/>
    </Record>
  </Collection>
</Annotation>
<Annotation Term="UI.LineItem">
  <Collection>
    <Record Type="UI.DataField">
      <PropertyValue Property="Label" String="Airline"/>
      <PropertyValue Property="Value" Path="carrid"/>
    </Record>
    <Record Type="UI.DataField">
      <PropertyValue Property="Label" String="Flight No."/>
      <PropertyValue Property="Value" Path="connid"/>
    </Record>
    <Record Type="UI.DataField">
      <PropertyValue Property="Label" String="Date"/>
      <PropertyValue Property="Value" Path="fldate"/>
    </Record>
    <Record Type="UI.DataField">
      <PropertyValue Property="Label" String="Airline Name"/>
      <PropertyValue Property="Value" Path="carrname"/>
      <Annotation Term="UI.Importance"
EnumMember="UI.ImportanceType/High"/>
    </Record>
    <Record Type="UI.DataField">
      <PropertyValue Property="Label" String="From"/>
      <PropertyValue Property="Value" Path="cityfrom"/>
    </Record>
    <Record Type="UI.DataField">
      <PropertyValue Property="Label" String="To"/>
      <PropertyValue Property="Value" Path="cityto"/>
    </Record>
    <Record Type="UI.DataField">
      <PropertyValue Property="Label" String="Aircraft"/>

```

```

        <PropertyValue Property="Value" Path="planetype"/>
        <Annotation Term="UI.Importance"
EnumMember="UI.ImportanceType/High"/>
    </Record>
    <Record Type="UI.DataField">
        <PropertyValue Property="Label" String="Price"/>
        <PropertyValue Property="Criticality"
Path="PriceCriticality"/>
        <PropertyValue Property="CriticalityRepresentation"
EnumMember="UI.CriticalityRepresentationType/WithIcon"/>
        <PropertyValue Property="Value" Path="price"/>
        <Annotation Term="UI.Importance"
EnumMember="UI.ImportanceType/Medium"/>
    </Record>
    <Record Type="UI.DataField">
        <PropertyValue Property="Label" String="Total Revenue"/>
        <PropertyValue Property="Value" Path="TotalRevenue"/>
        <Annotation Term="UI.Importance"
EnumMember="UI.ImportanceType/Low"/>
    </Record>
    <Record Type="UI.DataField">
        <PropertyValue Property="Label" String="Eco. Occ."/>
        <PropertyValue Property="Value" Path="seatsocc"/>
        <Annotation Term="UI.Importance"
EnumMember="UI.ImportanceType/Low"/>
    </Record>
</Collection>
</Annotation>
<Annotation Term="UI.SelectionFields">
    <Collection>
        <PropertyPath>carrid</PropertyPath>
        <PropertyPath>connid</PropertyPath>
        <PropertyPath>fldate</PropertyPath>
        <PropertyPath>carrname</PropertyPath>
    </Collection>
</Annotation>
</Annotations>
</Schema>
</edmx:DataServices>
</edmx:Edmx>

```

```

# This is the resource bundle for zfiordemos3

#Texts for manifest.json

#XTIT: Application name
appTitle=zfiordemos3

#YDES: Application description
appDescription=zfiordemos3

#XFLD,36
zfiordemos3-display.flpTitle=zfiordemos3

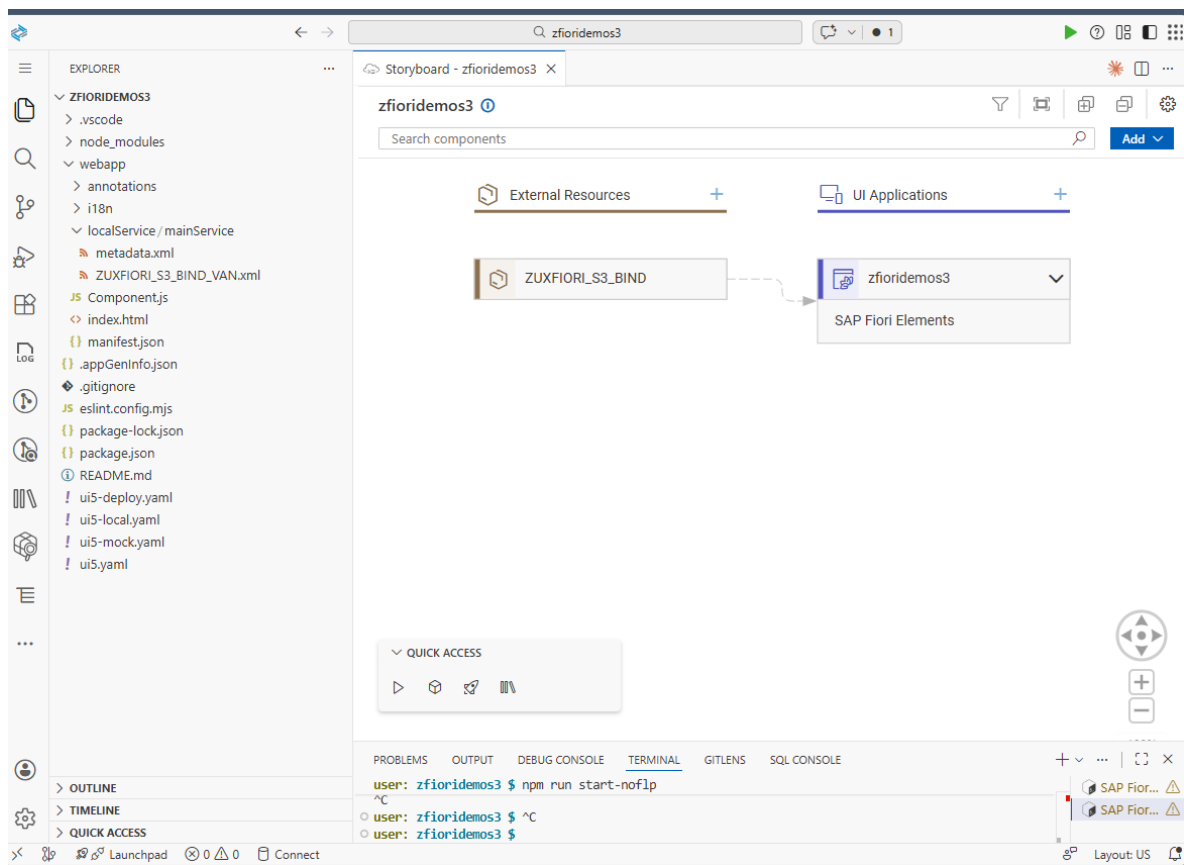
#XFLD,51
zfiordemos3-display.flpSubtitle=zuxfioris3_semobj

```

Front-end Fiori Application:

<https://github.com/AngeliqueSabTec/ZUXFioriDemoS3/blob/main/README.md>

Screenshot Evidence — SAPUI5 Project Structure: Session 3 Build (zfiordemos3)



BAS Storyboard — a second, dedicated project (zfioridemos3) wiring the ZUXFIORI_S3_BIND external resource to a new SAP Fiori Elements UI application.

z Fiori Demo 3

EXPLORER

- z FIORIDEMOS3
 - .vscode
 - node_modules
 - webapp
 - annotations
 - i18n
 - localService/mainService
 - metadata.xml
 - ZUXFIORI_S3_BIND_VAN.xml
 - Component.js
 - index.html
 - manifest.json
 - appGenInfo.json
 - .gitignore
 - eslint.config.mjs
 - package-lock.json
 - package.json
 - README.md
 - ui5-deploy.yaml
 - ui5-local.yaml
 - ui5-mock.yaml
 - ui5.yaml

Storyboard - z Fioridemos3

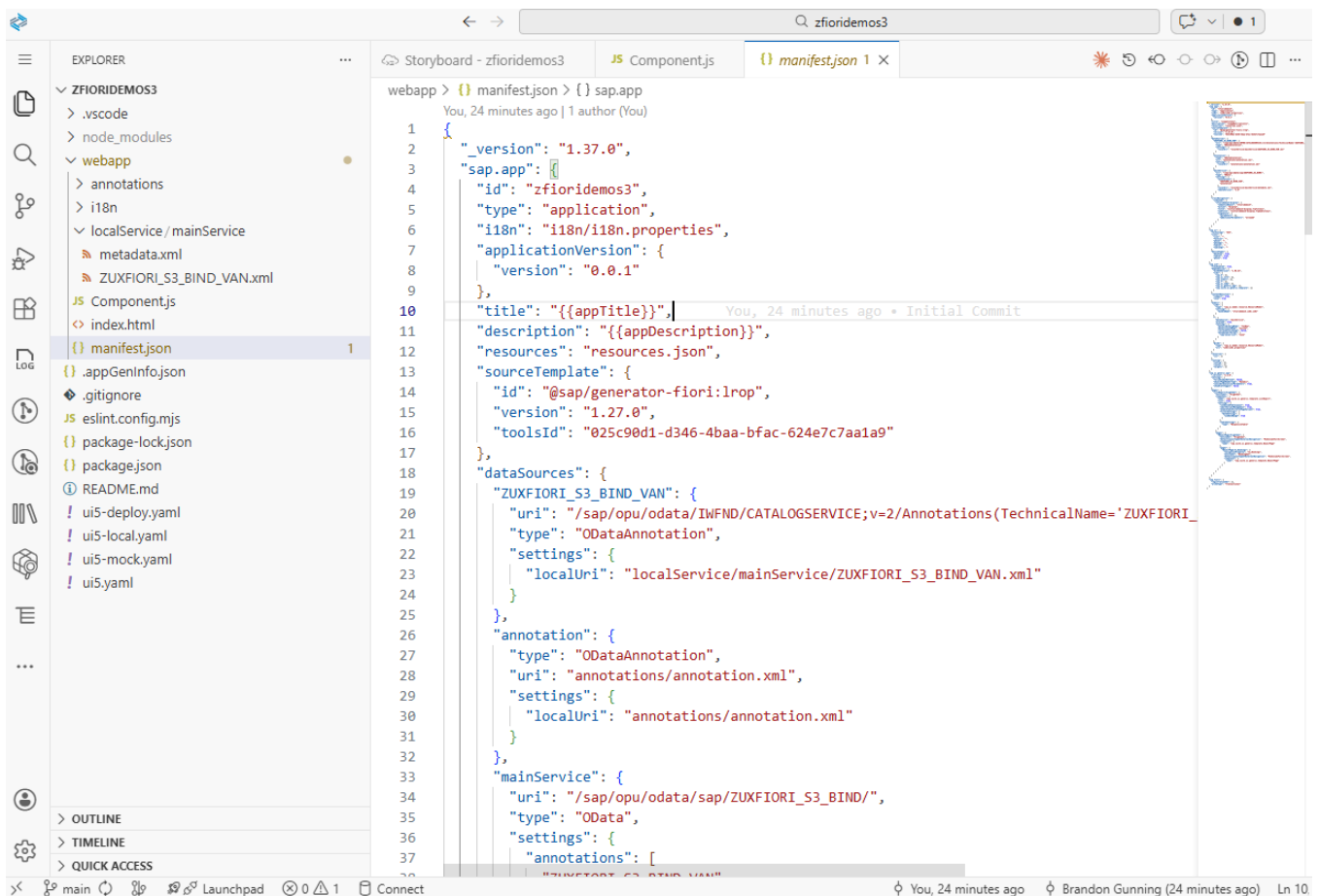
index.html

```

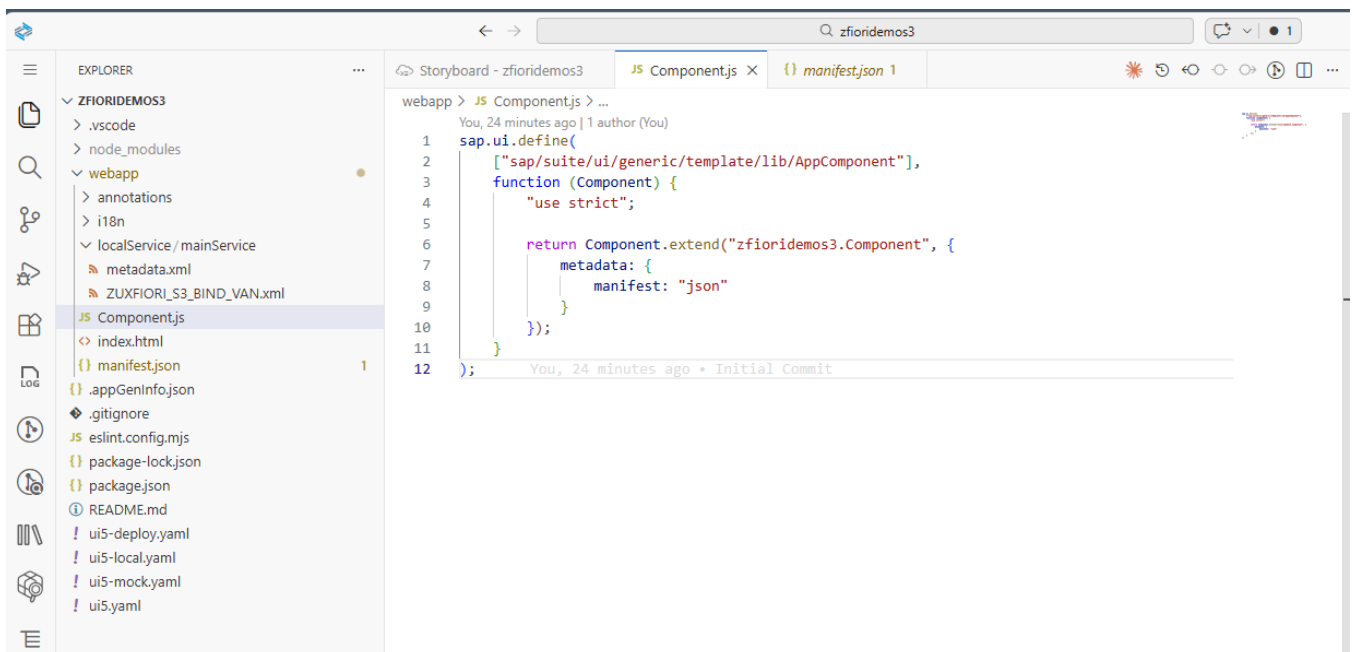
webapp > < index.html > html > head > meta
You, 23 minutes ago | 1 author (You)
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <meta http-equiv="X-UA-Compatible" content="IE=edge">
7   <title>z Fioridemos3</title>
8   <style>
9     html, body, body > div, #container, #container-uiarea {
10       height: 100%;
11     }
12   </style>
13   <script
14     id="sap-ui-bootstrap"
15     src="resources/sap-ui-core.js"
16     data-sap-ui-theme="sap_fiori_3"
17     data-sap-ui-resourceroots='{
18       "z Fioridemos3": "."
19     }'
20     data-sap-ui-oninit="module:sap/ui/core/ComponentSupport"
21     data-sap-ui-compatVersion="edge"
22     data-sap-ui-async="true"
23     data-sap-ui-frameOptions="trusted"
24   ></script>
25 </head>
26 <body class="sapUiBody sapUiSizeCompact" id="content">
27   <div
28     data-sap-ui-component
29     data-name="z Fioridemos3"
30     data-id="container"
31     data-settings='{ "id" : "z Fioridemos3" }'
32     data-handle-validation="true"
33   ></div>
34 </body>
35 </html>

```

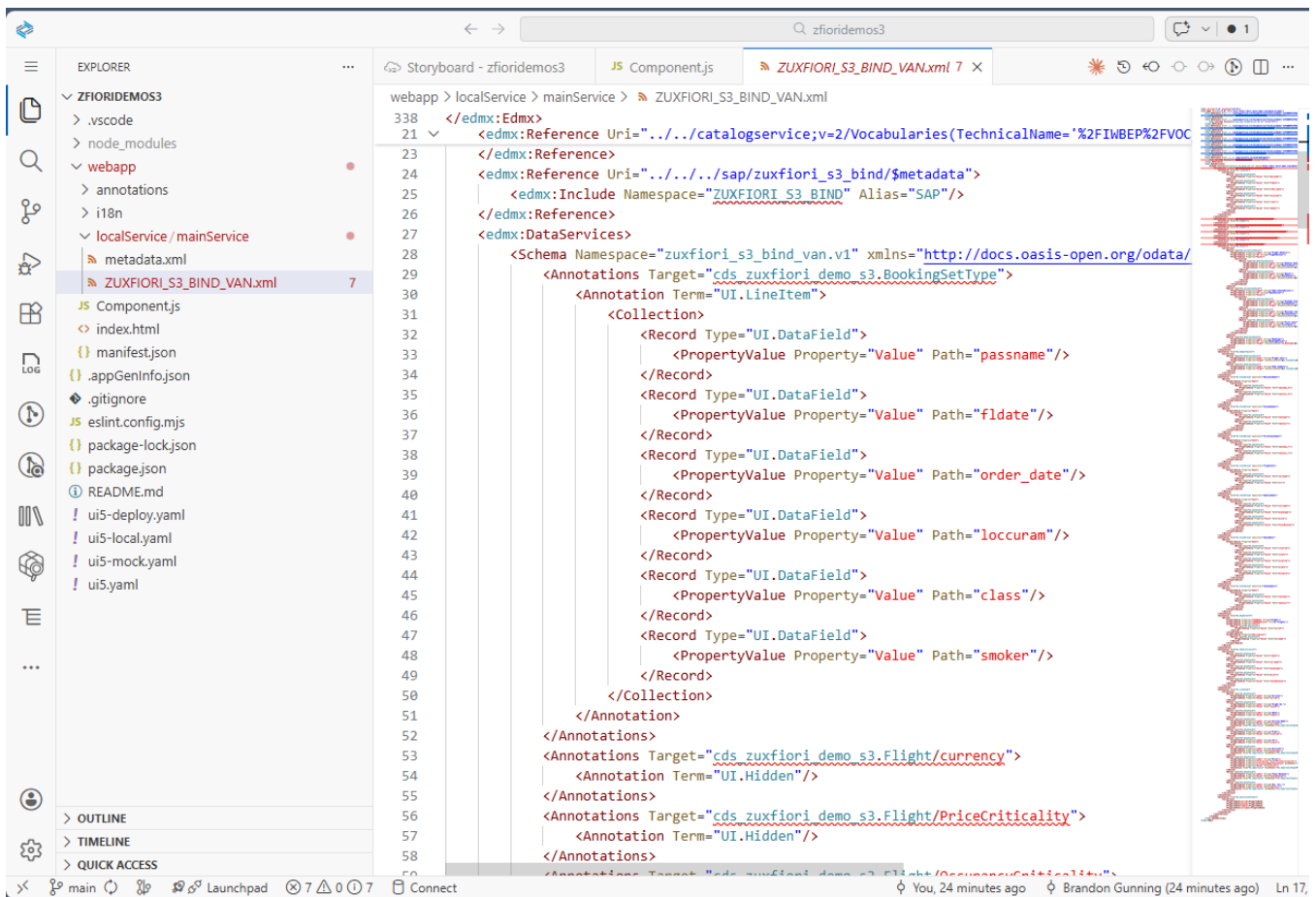
index.html — the SAPUI5 bootstrap for z Fioridemos3, loading the sap_fiori_3 theme and the z Fioridemos3 component.



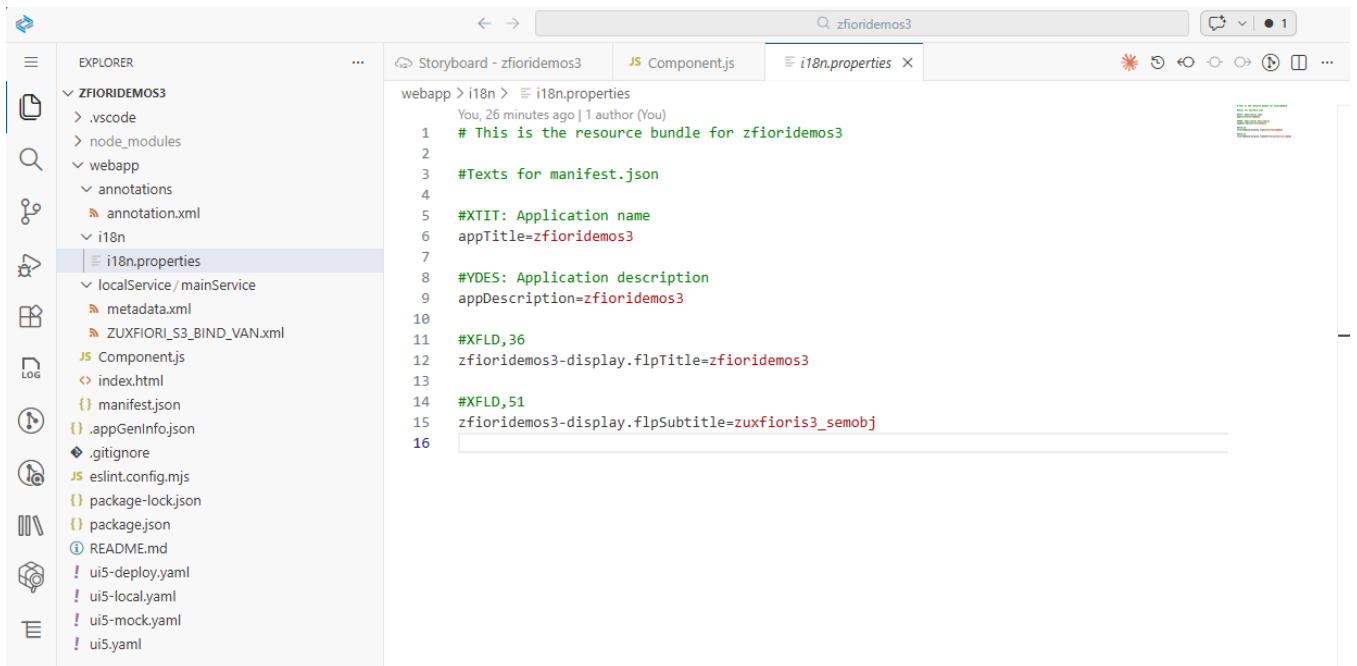
manifest.json — dataSources configuration pointing to the ZUXFIORI_S3_BIND OData service and its generated annotation file.



Component.js — the application's root UIComponent, extending sap.suite.ui.generic.template.lib.AppComponent and driven entirely by manifest.json.



Generated OData annotation file (ZUXFIORI_S3_BIND_VAN.xml) — the UI.LineItem collection for the Booking entity (passname, fldate, order_date, loccuram, class, smoker) plus Hidden annotations for currency and PriceCriticality.



i18n.properties — the resource bundle for zfiordemos3, including the app title and FLP tile subtitle texts.

SESSION 4

Fiori Elements with ABAP RAP (Live Build)

UX405 · Fiori Elements & FPM (21–30%) · Annotations (11–20%) · OData & BTP (11–20%)

i Note — This Was RAP, Not CAP

The live Flight app built in this session runs on ABAP RAP in Eclipse ADT (Data Definitions, Behavior Definitions, Service Definition/Binding) — not a CAP Node.js project. The Code Examples and Your Code sections below are already written in RAP's ABAP CDS syntax to match what was actually built. The Exam Points Raised bullets describe CAP concepts on purpose, since UX405's course material and the C_FIORD exam objectives are written around CAP — the underlying ideas (@UI.* annotations, actions, criticality, facets) are the same either way, only the backend and exact syntax differ. Session 5 will build the CAP counterpart of this same Flight scenario on the same OData V4 service.

• MATERIALS BUILT (PPTX + Study Guide)

Exam Points Raised

- Fiori Elements = annotation-driven, no hand-coded XML views
- CAP = Cloud Application Programming Model (Node.js/Java on BTP), always OData V4
- @UI.lineItem → table columns | @UI.selectionField → filter bar | @UI.facet → Object Page structure
- Bound action (needs row selection) vs Unbound action (independent)
- Criticality values are integers: 1=red, 2=amber, 3=green, 0=neutral
- @Common.SideEffects → annotation-driven auto-refresh, not JS event handlers
- ALP (Analytical List Page) requires SAP HANA; List Report works with Any DB

Code Examples

List Report filter bar

```
@UI.selectionField: [{ position: 10 }]
key sf.carriid,

@UI.selectionField: [{ position: 20 }]
key sf.connid,
```

Criticality colouring on a list column

```
@UI.lineItem: [{ position: 80, label: 'Price',
  criticality: 'PriceCriticality',
  criticalityRepresentation: #WITH_ICON }]
sf.price,

@UI.hidden: true
case
  when sf.price <= 500 then 3 // green
  when sf.price <= 1000 then 2 // amber
  else 1 // red
end as PriceCriticality,
```

Object Page facet structure with nested sections

```
@UI.facet: [  
  { id: 'FlightSection', purpose: #STANDARD, type: #COLLECTION,  
    label: 'Flight Details', position: 10 },  
  { id: 'GeneralFacet', parentId: 'FlightSection',  
    purpose: #STANDARD, type: #FIELDGROUP_REFERENCE,  
    targetQualifier: 'GeneralData', label: 'General Data',  
    position: 10 }  
]
```

⚠ Watch Out For

FE ≠ Freestyle SAPUI5. CAP ≠ ABAP Gateway. OData V4 only with CAP — never V2. ALP = SAP HANA only. Bound action needs row selection; unbound doesn't.

Watch Out For — Live RAP Debugging Log

Ten real errors were hit and resolved while building the live Flight app on this ABAP RAP / OData V4 system — far more than a textbook walkthrough would show, and genuinely more useful for exam prep because of it. Several trace back to this training system running an older on-premise release than the newest RAP features assume. Recognising "this is a release gap, not my syntax" is itself an exam-relevant skill.

What Happened	Root Cause	The Fix
RAP_SERVICES_ADT021 — "(Un-)Publishing of SRVB ... in Customizing Client not allowed"	SCC4 client role is set to Customizing, which blocks Service Binding publish via ADT.	Publish via transaction /IWFND/V4_ADMIN → Publish Service Groups instead (SAP Note 3101976). Nothing was wrong with the CDS objects.
"',' was expected, not '(' on strict(2);	This kernel doesn't recognise versioned strict mode — a newer-release addition.	Use bare strict; with no version number.
"'dependent master' was expected, not ':' on authorization: none;	authorization: none is only valid on an individual operation, never as the entity-level statement.	Omit the authorization keyword entirely at BO level — it's optional, not something you switch off with none.
"must be view entity" / "must be authorization master or dependent" under strict mode	strict; mode enforces modern view-entity syntax and a mandatory authorization declaration together.	Dropped strict mode for this training BO rather than build a full authorization implementation class.
"ZI_Flight_S4_OBW is not a root entity"	An abstract entity needs its own Behavior Definition only if declared as root.	Changed to define root abstract entity ...
"'with control' is only supported in ... 'abstract / with hierarchy' + release 756 gap	UI-enforced mandatory action-parameter fields need a software component release this system doesn't have.	Genuine platform limit — validated inside the action method instead (IF ... IS INITIAL + NEW_MESSAGE_WITH_TEXT).
"The operation requires a key (SORTED or HASHED) on table column(s)"	FILTER needs a declared secondary key; READ ENTITIES result tables only carry the primary %tky.	Used LOOP AT ... WHERE instead of FILTER — works on any standard table.
"Unexpected word '-' inside a CASE WHEN condition	Some CDS releases reject arithmetic inside a WHEN condition, even though it's fine as a plain computed column.	Rephrased as a direct comparison (seatsocc < seatsmax) instead of subtracting first.

sap.fe.macros.internal.Field: 'dataField' must be '\$Type' ... but is 'undefined'	<macros:Field> was pointed at a whole @UI.FieldGroup collection instead of one field entry.	Pointed metaPath at one specific field (.../Data/0), or dropped the Building Block where a plain control was enough.
"'save' was expected, not 'side'." (side effects block in Behavior Definition)	RAP side effects in the Behavior Definition only exist from release 2302 onward; this system predates it.	Designed the action to READ and return the updated record as its own result — Fiori Elements refreshes the row from that response, no dedicated side effect needed.

Take-away for the exam: several of these are genuine ABAP CDS / RAP language rules (arithmetic in CASE WHEN, FILTER needing a secondary key, authorization: none scope) rather than one-off typos — know them cold. The rest are release-availability gaps (strict mode versioning, with control, side effects in the Behavior Definition) — know that these features exist and what release introduced them, even on a system where they can't be tested directly.

Your Demonstration Notes

What was actually demonstrated in this session:
Created Backend ODATA V4 service (Interface View, Consumption View, Service Definition and Service Binding)
Service Publishing Debugging – Register the Flight Odata V4 as a service group with a system alias
Bound action (cancelFlight) and unbound/static action (openBookingWindow) — full RAP Behavior Definition, Behavior Projection, and Behavior Implementation class with EML READ/MODIFY ENTITIES
Table column criticality colouring on Price, driven by a calculated CASE WHEN field (PriceCriticality) on the interface view
Object Page header (@UI.headerInfo) and body structure — @UI.facet + @UI.fieldGroup building a 'Flight Details' section
Dynamic field control (@Common.FieldControl) on Price — Optional while seats are available, ReadOnly once fully booked — tied to the cancelFlight action's effect
Multiple List Report views — Open / Closed tabs via @UI.presentationVariant + @UI.selectionVariant + @UI.selectionPresentationVariant
Intent-based navigation on Connection Number (@UI.item type: #WITH_INTENT_BASED_NAVIGATION)
FPM custom section on the Object Page (BookingStatus fragment, built via Guided Development) with a conditional MessageStrip bound to SeatsAvailable
Investigated RAP Side Effects (Behavior Definition syntax) — confirmed not supported on this system's release; relied on the action's own result payload to refresh Price / Criticality / Field Control instead
https://github.com/AngeliqueSabTec/ZUXFioriDemoS4

Your Code

Paste your code here

```
@AbapCatalog.sqlViewName: 'ZIFLIGHTTR11'
@AbapCatalog.compiler.compareFilter: true
@AbapCatalog.preserveKey: true
@AccessControl.authorizationCheck: #NOT_REQUIRED
@EndUserText.label: 'Session 4 - Flight Interface View'
define root view ZI_Flight_S4
  as select from sflight
{
  key carrid          as Carrid,
  key connid          as Connid,
  key fldate          as Fldate,
  price              as Price,
  currency           as CurrencyCode,
  planetype          as Planetype,
  seatsmax           as SeatsMax,
  seatsocc           as SeatsOcc,
  seatsmax - seatsocc as SeatsAvailable,

  case
    when price <= 500 then 3 -- green / Positive
    when price <= 1000 then 2 -- amber / Critical
    else 1 -- red / Negative
  end
    as PriceCriticality,

  case
    when seatsocc < seatsmax then 3 -- seats still available
    else 1 -- fully booked, closed
  end
    as PriceFieldControl
}

@AccessControl.authorizationCheck: #NOT_REQUIRED
@EndUserText.label: 'Session 4 - Flight Consumption View'
@Metadata.ignorePropagatedAnnotations: true
@UI.headerInfo: {
  typeName: 'Flight',
  typeNamePlural: 'Flights',
  title: { value: 'Connid' },
  description: { value: 'Carrid' }
}
@UI.presentationVariant: [
  { qualifier: 'pOpen', visualizations: [{ type: #AS_LINEITEM }] },
  { qualifier: 'pClosed', visualizations: [{ type: #AS_LINEITEM }] }
]
@UI.selectionVariant: [
  { qualifier: 'sOpen', filter: 'SeatsAvailable GT 0' },
  { qualifier: 'sClosed', filter: 'SeatsAvailable EQ 0' }
]
@UI.selectionPresentationVariant: [
  { qualifier: 'Open', text: 'Open', presentationVariantQualifier:
    'pOpen', selectionVariantQualifier: 'sOpen' },
  { qualifier: 'Closed', text: 'Closed', presentationVariantQualifier: 'pClosed',
    selectionVariantQualifier: 'sClosed' }
]

define root view entity ZC_Flight_S4 as projection on ZI_Flight_S4
{
  @UI.facet: [
    { id: 'FlightDetails', purpose: #STANDARD, type: #FIELDGROUP_REFERENCE, label:
      'Flight Details', position: 10, targetQualifier: 'FlightDetails' }
  ]
}
```

```

        @UI.selectionField: [{ position: 10 }]
        @UI.lineItem: [{ position: 90, type: #FOR_ACTION, dataAction:
'cancelFlight', label: 'Cancel Flight' }]
key Carrid,
        @UI.selectionField: [{ position: 20 }]
        @UI.lineItem: [{ position: 30, type: #WITH_INTENT_BASED_NAVIGATION,
semanticObject: 'Flight', semanticObjectAction: 'displayFactSheet' }]
key Connid,
        @UI.lineItem: [{ position: 100, type: #FOR_ACTION, dataAction:
'openBookingWindow', label: 'Open Booking Window' }]
        @UI.fieldGroup: [{ qualifier: 'FlightDetails', position: 10 }]
key Fldate,

        @UI.lineItem: [{ position: 10, label: 'Price', criticality: 'PriceCriticality'
}]
        @Semantics.amount.currencyCode: 'CurrencyCode'
        @UI.fieldGroup: [{ qualifier: 'FlightDetails', position: 20 }]
        Price,
        CurrencyCode,
        @UI.fieldGroup: [{ qualifier: 'FlightDetails', position: 30 }]
        Planetype,
        @UI.fieldGroup: [{ qualifier: 'FlightDetails', position: 40 }]
        SeatsMax,
        @UI.fieldGroup: [{ qualifier: 'FlightDetails', position: 50 }]
        SeatsOcc,
        @UI.fieldGroup: [{ qualifier: 'FlightDetails', position: 60 }]
        SeatsAvailable,
        PriceCriticality,
        PriceFieldControl
}

```

```

@EndUserText.label: 'Session 4 - Open Booking Window Input'
define root abstract entity ZI_Flight_S4_OBW
{
@EndUserText.label: 'Flight Date'
    fldate : abap.dats;
}

```

```

projection;
//strict; //Comment this line in to enable strict mode. The strict mode is
prerequisite to be future proof regarding syntax and to be able to release your BO.

```

```

define behavior for ZC Flight S4 alias Flight
{
    use action cancelFlight;
    use action openBookingWindow;
}

```

```

managed implementation in class zbp_i_flight_s4 unique;

```

```

define behavior for ZI_Flight_S4 alias Flight
persistent table SFLIGHT
lock master
{
    field ( readonly ) Carrid, Connid, Fldate, PriceCriticality, PriceFieldControl;

    action cancelFlight result [1] $self;
    static action openBookingWindow parameter ZI_Flight_S4_OBW result [1] $self;

    mapping for sflight
    {
        Carrid          = carrid;

```

```

Connid      = connid;
Fldate      = fldate;
Price       = price;
CurrencyCode = currency;
Planetype   = planetype;
SeatsMax    = seatsmax;
SeatsOcc    = seatsocc;
}
}

```

```

CLASS lhc_Flight DEFINITION INHERITING FROM cl_abap_behavior_handler.
PRIVATE SECTION.

```

```

METHODS cancelFlight FOR MODIFY
IMPORTING keys FOR ACTION Flight~cancelFlight RESULT result.

```

```

METHODS openBookingWindow FOR MODIFY
IMPORTING keys FOR ACTION Flight~openBookingWindow RESULT result.

```

```

ENDCLASS.

```

```

CLASS lhc_Flight IMPLEMENTATION.

```

```

METHOD cancelFlight.

```

```

" BOUND - acts only on the selected instance(s)
READ ENTITIES OF ZI_Flight_S4 IN LOCAL MODE
ENTITY Flight
  FIELDS ( SeatsOcc ) WITH CORRESPONDING #( keys )
RESULT DATA(flights).

```

```

MODIFY ENTITIES OF ZI_Flight_S4 IN LOCAL MODE

```

```

ENTITY Flight
  UPDATE FIELDS ( SeatsMax ) WITH VALUE #(
    FOR flight IN flights
      ( %tky      = flight-%tky
        SeatsMax = flight-SeatsOcc ) )
  FAILED DATA(update_failed)
  REPORTED DATA(update_reported).

```

```

READ ENTITIES OF ZI_Flight_S4 IN LOCAL MODE

```

```

ENTITY Flight
  ALL FIELDS WITH CORRESPONDING #( keys )
RESULT DATA(updated_flights).

```

```

result = VALUE #( FOR flight IN updated_flights
  ( %tky = flight-%tky %param = flight ) ).

```

```

ENDMETHOD.

```

```

METHOD openBookingWindow.

```

```

DATA(booking_date) = keys[ 1 ]-%param-fldate.

```

```

IF booking_date IS INITIAL.

```

```

  APPEND VALUE #( %msg = NEW MESSAGE WITH TEXT(
    severity = if_abap_behv_message=>severity-error
    text      = 'Please enter a valid flight date' ) )
  TO reported-flight.

```

```

  RETURN.

```

```

ENDIF.

```

```

DATA update_tab TYPE TABLE FOR UPDATE ZI_Flight_S4\@==@@Flight.

```

```

READ ENTITIES OF ZI_Flight_S4 IN LOCAL MODE

```

```

ENTITY Flight
  FIELDS ( Fldate ) WITH VALUE #( ( %control-Fldate = if_abap_behv=>mk-on ) )
RESULT DATA(all_flights).

```

```

LOOP AT all_flights INTO DATA(flight) WHERE Fldate = booking_date.
  update_tab = VALUE #( BASE update_tab
    ( %tky = flight-%tky SeatsOcc = 0 ) ).
ENDLOOP.

MODIFY ENTITIES OF ZI_Flight_S4 IN LOCAL MODE
  ENTITY Flight
    UPDATE FIELDS ( SeatsOcc ) WITH update_tab
  FAILED DATA(update_failed)
  REPORTED DATA(update_reported).

result = VALUE #( ( %param = VALUE #( ) ) ).
ENDMETHOD.

ENDCLASS.

```

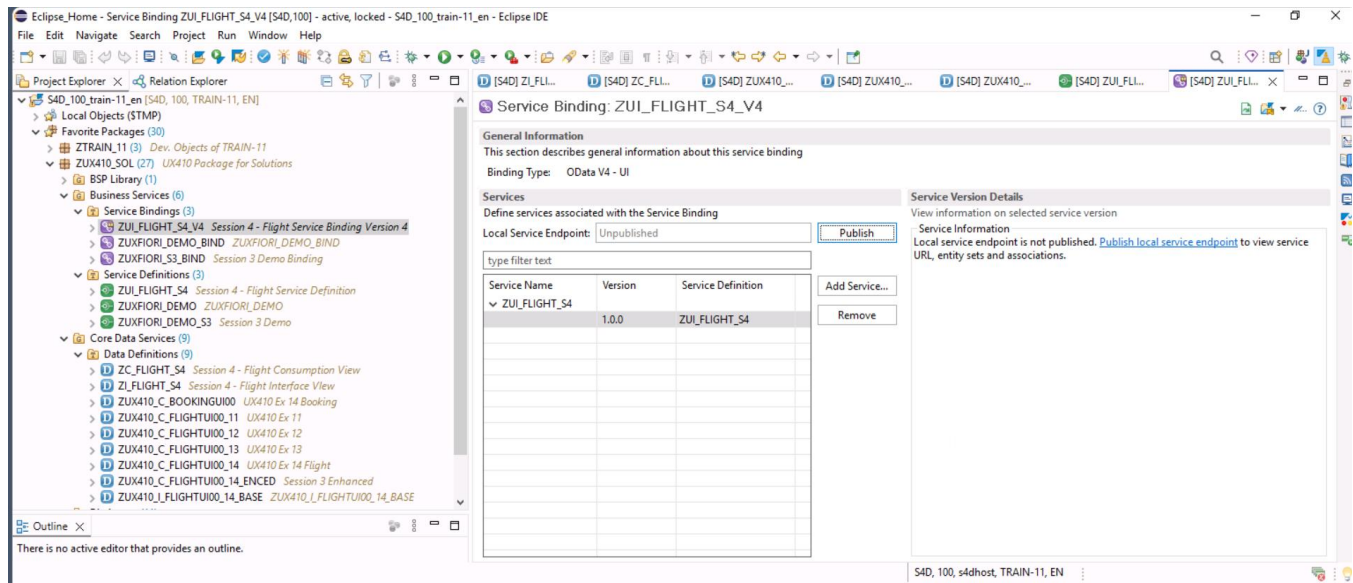
```

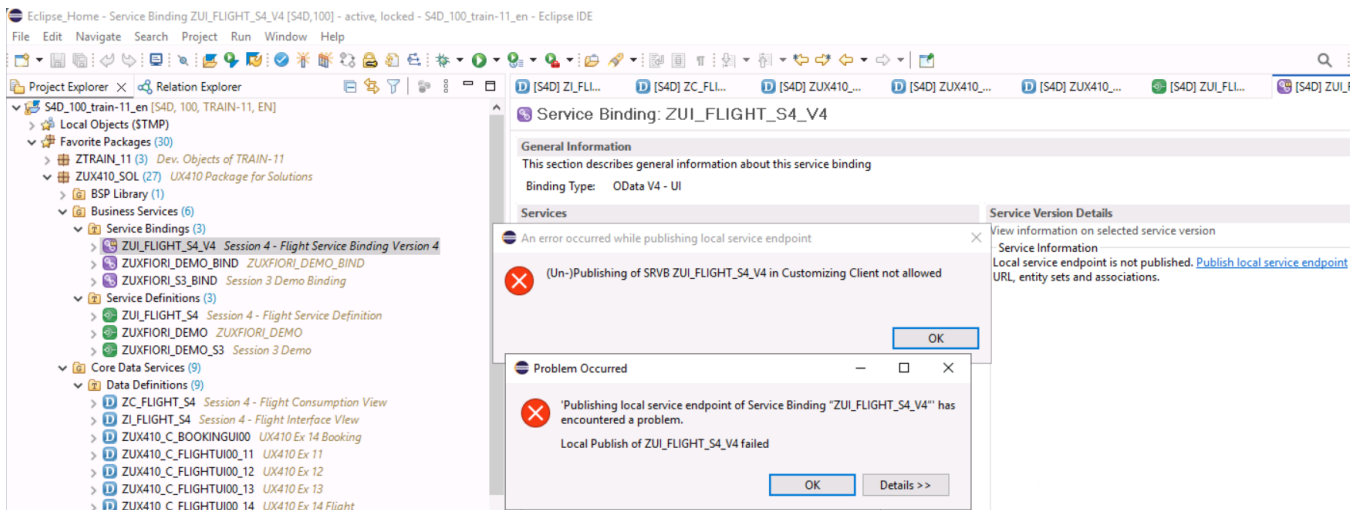
@EndUserText.label: 'Session 4 - Flight Service Definition'
define service ZUI_Flight_S4 {
  expose ZC_Flight_S4 as Flight;
}

```

Demonstration Notes

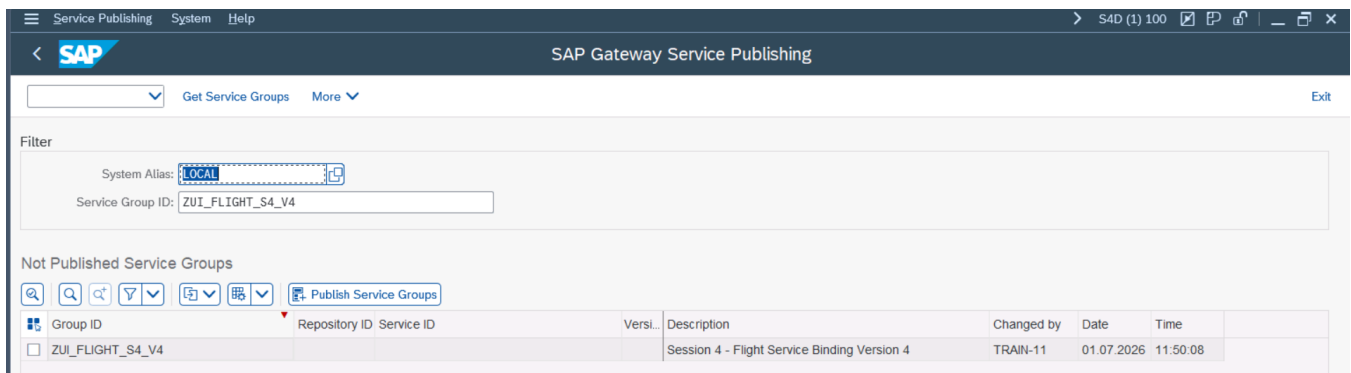
Screenshot Evidence — Backend: Activating the OData Service





Steps to resolve:

- ☐ Run transaction **/IWFND/V4_ADMIN**.
- ☐ Click **Publish Service Groups**.
- ☐ Enter your **System Alias** (usually LOCAL for an embedded/local RAP scenario).
- ☐ In the **Service Group ID** field, search for your service binding name (ZUI_FLIGHT_S4_V4).
- ☐ Select it and click **Publish Service Groups**.
- ☐ Go back to your Service Binding in Eclipse ADT and refresh — it should now show as published, and Preview will work normally.



Administration System Help

SAP Gateway Service Administration - ZUI_FLIGHT_S4_V4 -

Refresh Find Services Expand All Collapse All System Info Error Log Routing Configuration Publish Service Groups More

Service Group

- My Favorites
- Service Groups
 - /WBEP/SUPPORT
 - /WFND/CONFIG
 - /IWGW/NOTIFICATION
 - GW100_UI_SUPPLIERLIST_04
 - STMC
 - UX_UI_CUSTOMER_04
 - ZUI_FLIGHT_S4_V4
 - LOCAL

Service Group ZUI_FLIGHT_S4_V4 - System Alias LOCAL

Available Services

System Alias	Default	User Role	Host Name
LOCAL	☒		

Assigned Service Groups

Hide Service Groups

Service Group	Description
ZUI_FLIGHT_S4_V4	Session 4 - Flight Service Binding Version 4

ZUI_FLIGHT_S4_V4 - LOCAL - Available Services: 2

Add to Favorites Error Log Service Implementation Metadata Service Test

Line	Repository	Service ID	Version	Service Alias	Service	Description
1	DEFAULT	/WBEP/COMMON	1			V4 Common Service
2	SRVD	ZUI_FLIGHT_S4	1	ZUI_FLIGHT_S4	1	Session 4 - Flight Service Definition

Administration System Help

SAP Gateway Service Administration - ZUI_FLIGHT_S4_V4 -

Refresh Find Services Expand All Collapse All System Info Error Log Routing Configuration Publish Service Groups More

Service Group

- My Favorites
- Service Groups
 - /WBEP/SUPPORT
 - /WFND/CONFIG
 - /IWGW/NOTIFICATION
 - GW100_UI_SUPPLIERLIST_04
 - STMC
 - UX_UI_CUSTOMER_04
 - ZUI_FLIGHT_S4_V4
 - LOCAL

Service Group ZUI_FLIGHT_S4_V4 - System Alias LOCAL

Available Services

System Alias	Default	User Role	Host Name
LOCAL	☒		

Assigned Service Groups

Hide Service Groups

Service Group	Description
ZUI_FLIGHT_S4_V4	Session 4 - Flight Service Binding Version 4

ZUI_FLIGHT_S4_V4 - LOCAL - Available Services: 2

Add to Favorites Error Log Service Implementation Metadata Service Test

Line	Repository	Service ID	Version	Service Alias	Service	Description
1	DEFAULT	/WBEP/COMMON	1			V4 Common Service
2	SRVD	ZUI_FLIGHT_S4	1	ZUI_FLIGHT_S4	1	Session 4 - Flight Service Definition

Gateway Client

Test Cases

Web Browser

SAP Gateway Client Edit Goto Metadata System Help

SAP Gateway Client

Last Request Execute Next Request Select Service Administration Service Implementation EntitySets Add URI Option Switch User More

HTTP Method: ☒ GET ☐ POST ☐ PUT ☐ PATCH ☐ MERGE ☐ DELETE ☐ HEAD

Request URI: /sap/opu/odata4/sap/zui_flight_s4_v4/srvt/sap/zui_flight_s4/0001/\$metadata?sap-statistics=true

Protocol: ☒ HTTP ☐ HTTPS

Test Group: Test Case:

HTTP Request

Header Name Value

HTTP Response - Processing Time = 883 ms, HTTP Overhead = 35 ms

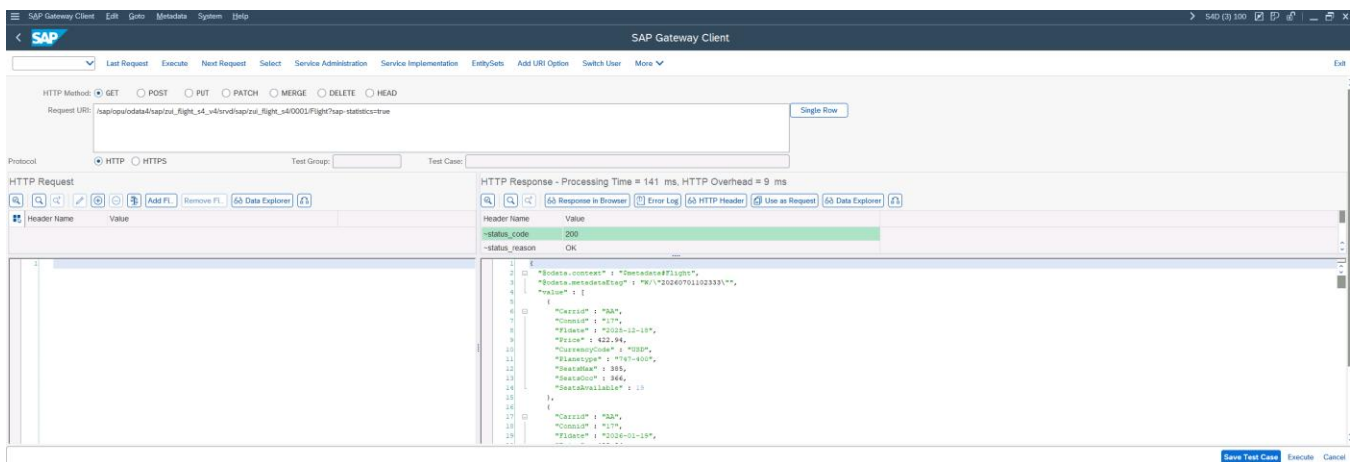
Header Name Value

Header Name	Value
~status_code	200
~status_reason	OK

This XML file does not appear to have any style information associated with it. The document tree is shown below.

```
<?xml version="1.0" encoding="UTF-8" standalone="yes" ?>
<edmx:Edmx xmlns:edmx="http://docs.oasis-open.org/odata/ns/edmx" xmlns="http://docs.oasis-open.org/odata/ns/edm" Version="4.0">
  <edmx:Reference
    Uri="/sap/opu/odata4/WFND/CATALOGSERVICE/v=2/Vocabularies(TechnicalName='%2FIWBEP%2FVOC_COMMUNICATION',Version='0001',SAP__Origin='LOCAL',SAP__Language='en')/$metadata"
    Alias="Communication" />
  <edmx:Include Namespace="com.sap.vocabularies.Communication.v1" Alias="Communication" />
</edmx:Edmx>
```

Save Test Case Execute Cancel



Fiori Front End Application

Paste your code here

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta http-equiv="X-UA-Compatible" content="IE=edge">
  <title>Session 4 Demo Flight Application</title>
  <style>
    html, body, body > div, #container, #container-uiarea {
      height: 100%;
    }
  </style>
  <script
    id="sap-ui-bootstrap"
    src="resources/sap-ui-core.js"
    data-sap-ui-theme="sap_fiori_3"
    data-sap-ui-resourceroots='{
      "zflights4.flights4": "/"
    }'
    data-sap-ui-oninit="module:sap/ui/core/ComponentSupport"
    data-sap-ui-compatVersion="edge"
    data-sap-ui-async="true"
    data-sap-ui-frameOptions="trusted"
  ></script>
</head>
<body class="sapUiBody sapUiSizeCompact" id="content">
  <div
    data-sap-ui-component
    data-name="zflights4.flights4"
    data-id="container"
    data-settings='{ "id" : "zflights4.flights4" }'
    data-handle-validation="true"
  ></div>
</body>
</html>
```

```
{
  "_version": "1.37.0",
  "sap.app": {
```

```

    "id": "zflights4.flights4",
    "type": "application",
    "i18n": "i18n/i18n.properties",
    "applicationVersion": {
      "version": "0.0.1"
    },
    "title": "{{appTitle}}",
    "description": "{{appDescription}}",
    "resources": "resources.json",
    "sourceTemplate": {
      "id": "@sap/generator-fiori:drop",
      "version": "1.27.0",
      "toolsId": "e5bca3c7-41d4-4059-b935-9acedabc856d"
    },
    "dataSources": {
      "annotation": {
        "type": "ODataAnnotation",
        "uri": "annotations/annotation.xml",
        "settings": {
          "localUri": "annotations/annotation.xml"
        }
      },
      "mainService": {
        "uri":
"/sap/opu/odata4/sap/zui_flight_s4_v4/srvd/sap/zui_flight_s4/0001/",
        "type": "OData",
        "settings": {
          "annotations": [
            "annotation"
          ],
          "localUri": "localService/mainService/metadata.xml",
          "odataVersion": "4.0"
        }
      }
    },
    "crossNavigation": {
      "inbounds": {
        "zfioridemos4-display": {
          "semanticObject": "zfioridemos4",
          "action": "display",
          "title": "{{zfioridemos4-display.flpTitle}}",
          "subTitle": "{{zfioridemos4-display.flpSubTitle}}",
          "signature": {
            "parameters": {},
            "additionalParameters": "allowed"
          }
        }
      }
    },
    "sap.ui": {
      "technology": "UI5",
      "icons": {
        "icon": "",
        "favIcon": "",
        "phone": "",
        "phone@2": "",
        "tablet": "",
        "tablet@2": ""
      },
      "deviceTypes": {
        "desktop": true,
        "tablet": true,
        "phone": true
      }
    },
  },

```

```

"sap.ui5": {
  "flexEnabled": true,
  "dependencies": {
    "minUI5Version": "1.96.33",
    "libs": {
      "sap.m": {},
      "sap.ui.core": {},
      "sap.fe.templates": {}
    }
  },
  "contentDensities": {
    "compact": true,
    "cozy": true
  },
  "models": {
    "i18n": {
      "type": "sap.ui.model.resource.ResourceModel",
      "settings": {
        "bundleName": "zflights4.flights4.i18n.i18n"
      }
    },
    "": {
      "dataSource": "mainService",
      "preload": true,
      "settings": {
        "synchronizationMode": "None",
        "operationMode": "Server",
        "autoExpandSelect": true,
        "earlyRequests": true
      }
    },
    "@i18n": {
      "type": "sap.ui.model.resource.ResourceModel",
      "uri": "i18n/i18n.properties"
    }
  },
  "resources": {
    "css": []
  },
  "routing": {
    "config": {},
    "routes": [
      {
        "pattern": "?:?query:",
        "name": "FlightList",
        "target": "FlightList"
      },
      {
        "pattern": "Flight({key}):?query:",
        "name": "FlightObjectPage",
        "target": "FlightObjectPage"
      }
    ]
  },
  "targets": {
    "FlightList": {
      "type": "Component",
      "id": "FlightList",
      "name": "sap.fe.templates.ListReport",
      "options": {
        "settings": {
          "contextPath": "/Flight",
          "variantManagement": "Page",
          "navigation": {
            "Flight": {
              "detail": {
                "route": "FlightObjectPage"
              }
            }
          }
        }
      }
    }
  }
}

```

```

        }
    },
    "controlConfiguration": {
        "@com.sap.vocabularies.UI.v1.LineItem": {
            "tableSettings": {
                "type": "ResponsiveTable"
            }
        }
    }
},
"FlightObjectPage": {
    "type": "Component",
    "id": "FlightObjectPage",
    "name": "sap.fe.templates.ObjectPage",
    "options": {
        "settings": {
            "editableHeaderContent": false,
            "contextPath": "/Flight",
            "content": {
                "body": {
                    "sections": {
                        "myCustomSection": {
                            "template":
"zflights4.flights4.ext.fragment.BookingStatus",
                            "title": "Booking Status",
                            "position": {
                                "placement": "After",
                                "anchor": "GeneratedFacet1"
                            }
                        }
                    }
                }
            }
        }
    }
},
"sap.fiori": {
    "registrationIds": [],
    "archetype": "transactional"
},
"sap.fe": {
    "app": {
        "enableLazyLoading": true
    }
}
}

```

```

<core:FragmentDefinition xmlns:core="sap.ui.core" xmlns="sap.m"
xmlns:macros="sap.fe.macros">
    <MessageStrip id="_IDGenMessageStrip"
        text="This flight is fully booked. Price is locked until seats become available."
        type="Warning"
        showIcon="true"
        visible="{= ${SeatsAvailable} === 0}" />

    <macros:Field
        id="seatsAvailableCustomField"
        metaPath="@com.sap.vocabularies.UI.v1.FieldGroup#FlightDetails/Data/0" />
</core:FragmentDefinition>

```

```

<edmx:Edmx xmlns:edmx="http://docs.oasis-open.org/odata/ns/edmx" Version="4.0">
  <edmx:Reference Uri="https://sap.github.io/odata-vocabularies/vocabularies/UI.xml">
    <edmx:Include Namespace="com.sap.vocabularies.UI.v1" Alias="UI"/>
  </edmx:Reference>
  <edmx:Reference
Uri="/sap/opu/odata4/sap/zui_flight_s4_v4/srvd/sap/zui_flight_s4/0001/$metadata">
    <edmx:Include Namespace="com.sap.gateway.srvd.zui_flight_s4.v0001"
Alias="SAP__self"/>
  </edmx:Reference>
  <edmx:DataServices>
    <Schema xmlns="http://docs.oasis-open.org/odata/ns/edm" Namespace="local">
      <Annotations Target="SAP__self.FlightType">
        <Annotation Term="UI.FieldGroup" Qualifier="GeneratedGroup">
          <Record Type="UI.FieldGroupType">
            <PropertyValue Property="Data">
              <Collection>
                <Record Type="UI.DataField">
                  <PropertyValue Property="Value" Path="Carrid"/>
                </Record>
                <Record Type="UI.DataField">
                  <PropertyValue Property="Value" Path="Connid"/>
                </Record>
                <Record Type="UI.DataField">
                  <PropertyValue Property="Value" Path="Fldate"/>
                </Record>
                <Record Type="UI.DataField">
                  <PropertyValue Property="Value" Path="Price"/>
                </Record>
                <Record Type="UI.DataField">
                  <PropertyValue Property="Value" Path="CurrencyCode"/>
                </Record>
                <Record Type="UI.DataField">
                  <PropertyValue Property="Value" Path="Planetype"/>
                </Record>
                <Record Type="UI.DataField">
                  <PropertyValue Property="Value" Path="SeatsMax"/>
                </Record>
                <Record Type="UI.DataField">
                  <PropertyValue Property="Value" Path="SeatsOcc"/>
                </Record>
                <Record Type="UI.DataField">
                  <PropertyValue Property="Label"
String="SeatsAvailable"/>
                  <PropertyValue Property="Value"
Path="SeatsAvailable"/>
                </Record>
              </Collection>
            </PropertyValue>
          </Record>
        </Annotation>
        <Annotation Term="UI.Facets">
          <Collection>
            <Record Type="UI.ReferenceFacet">
              <PropertyValue Property="ID" String="GeneratedFacet1"/>
              <PropertyValue Property="Label" String="General Information"/>
              <PropertyValue Property="Target"
AnnotationPath="@UI.FieldGroup#GeneratedGroup"/>
            </Record>
          </Collection>
        </Annotation>
      </Annotations>
    </Schema>
  </edmx:DataServices>

```

</edmx:Edmx>

```
<?xml version="1.0" encoding="utf-8"?>
<edmx:Edmx Version="4.0" xmlns:edmx="http://docs.oasis-open.org/odata/ns/edmx"
xmlns="http://docs.oasis-open.org/odata/ns/edm">
  <edmx:Reference
Uri="/sap/opu/odata/IWFND/CATALOGSERVICE;v=2/Vocabularies(TechnicalName='%2FIWBEP%2FVOC_COMMON',Version='0001',SAP__Origin='LOCAL')/$value">
    <edmx:Include Namespace="com.sap.vocabularies.Common.v1" Alias="SAP__common"/>
  </edmx:Reference>
  <edmx:Reference
Uri="/sap/opu/odata/IWFND/CATALOGSERVICE;v=2/Vocabularies(TechnicalName='%2FIWBEP%2FVOC_MEASURES',Version='0001',SAP__Origin='LOCAL')/$value">
    <edmx:Include Namespace="Org.OData.Measures.V1" Alias="SAP__measures"/>
  </edmx:Reference>
  <edmx:Reference
Uri="/sap/opu/odata/IWFND/CATALOGSERVICE;v=2/Vocabularies(TechnicalName='%2FIWBEP%2FVOC_CORE',Version='0001',SAP__Origin='LOCAL')/$value">
    <edmx:Include Namespace="Org.OData.Core.V1" Alias="SAP__core"/>
  </edmx:Reference>
  <edmx:Reference
Uri="/sap/opu/odata/IWFND/CATALOGSERVICE;v=2/Vocabularies(TechnicalName='%2FIWBEP%2FVOC_CAPABILITIES',Version='0001',SAP__Origin='LOCAL')/$value">
    <edmx:Include Namespace="Org.OData.Capabilities.V1" Alias="SAP__capabilities"/>
  </edmx:Reference>
  <edmx:Reference
Uri="/sap/opu/odata/IWFND/CATALOGSERVICE;v=2/Vocabularies(TechnicalName='%2FIWBEP%2FVOC_AGGREGATION',Version='0001',SAP__Origin='LOCAL')/$value">
    <edmx:Include Namespace="Org.OData.Aggregation.V1" Alias="SAP__aggregation"/>
  </edmx:Reference>
  <edmx:Reference
Uri="/sap/opu/odata/IWFND/CATALOGSERVICE;v=2/Vocabularies(TechnicalName='%2FIWBEP%2FVOC_VALIDATION',Version='0001',SAP__Origin='LOCAL')/$value">
    <edmx:Include Namespace="Org.OData.Validation.V1" Alias="SAP__validation"/>
  </edmx:Reference>
  <edmx:Reference
Uri="/sap/opu/odata/IWFND/CATALOGSERVICE;v=2/Vocabularies(TechnicalName='%2FIWBEP%2FVOC_CODELIST',Version='0001',SAP__Origin='LOCAL')/$value">
    <edmx:Include Namespace="com.sap.vocabularies.CodeList.v1" Alias="SAP__CodeList"/>
  </edmx:Reference>
  <edmx:Reference
Uri="/sap/opu/odata/IWFND/CATALOGSERVICE;v=2/Vocabularies(TechnicalName='%2FIWBEP%2FVOC_UI',Version='0001',SAP__Origin='LOCAL')/$value">
    <edmx:Include Namespace="com.sap.vocabularies.UI.v1" Alias="SAP__UI"/>
  </edmx:Reference>
  <edmx:Reference
Uri="/sap/opu/odata/IWFND/CATALOGSERVICE;v=2/Vocabularies(TechnicalName='%2FIWBEP%2FVOC_HTML5',Version='0001',SAP__Origin='LOCAL')/$value">
    <edmx:Include Namespace="com.sap.vocabularies.HTML5.v1" Alias="SAP__HTML5"/>
  </edmx:Reference>
  <edmx:DataServices>
    <Schema Namespace="com.sap.gateway.default.iwbep.common.v0001" Alias="SAP__self">
      <Annotation Term="SAP__core.SchemaVersion" String="1.0.0"/>
      <EntityType Name="Currency">
        <Key>
          <PropertyRef Name="CurrencyCode"/>
        </Key>
        <Property Name="CurrencyCode" Type="Edm.String" Nullable="false"
MaxLength="5"/>
        <Property Name="ISOCCode" Type="Edm.String" Nullable="false"
MaxLength="3"/>
        <Property Name="Text" Type="Edm.String" Nullable="false" MaxLength="15"/>
        <Property Name="DecimalPlaces" Type="Edm.Byte" Nullable="false"/>
      </EntityType>
      <EntityType Name="T100Longtext">
```

```

        <Key>
            <PropertyRef Name="MessageClass"/>
            <PropertyRef Name="MessageNumber"/>
            <PropertyRef Name="Variable1"/>
            <PropertyRef Name="Variable2"/>
            <PropertyRef Name="Variable3"/>
            <PropertyRef Name="Variable4"/>
        </Key>
        <Property Name="MessageClass" Type="Edm.String" Nullable="false"
MaxLength="20"/>
        <Property Name="MessageNumber" Type="Edm.String" Nullable="false"
MaxLength="3"/>
        <Property Name="Variable1" Type="Edm.String" Nullable="false"
MaxLength="50"/>
        <Property Name="Variable2" Type="Edm.String" Nullable="false"
MaxLength="50"/>
        <Property Name="Variable3" Type="Edm.String" Nullable="false"
MaxLength="50"/>
        <Property Name="Variable4" Type="Edm.String" Nullable="false"
MaxLength="50"/>
        <Property Name="Documentation" Type="Edm.Stream" Nullable="false"/>
    </EntityType>
    <EntityType Name="UnitOfMeasure">
        <Key>
            <PropertyRef Name="UnitCode"/>
        </Key>
        <Property Name="UnitCode" Type="Edm.String" Nullable="false"
MaxLength="3"/>
        <Property Name="ISOCCode" Type="Edm.String" Nullable="false"
MaxLength="3"/>
        <Property Name="ExternalCode" Type="Edm.String" Nullable="false"
MaxLength="3"/>
        <Property Name="Text" Type="Edm.String" Nullable="false" MaxLength="30"/>
        <Property Name="DecimalPlaces" Type="Edm.Int16"/>
    </EntityType>
    <EntityContainer Name="Container">
        <EntitySet Name="Currencies"
EntityType="com.sap.gateway.default.iwbep.common.v0001.Currency"/>
        <EntitySet Name="T100Longtexts"
EntityType="com.sap.gateway.default.iwbep.common.v0001.T100Longtext"/>
        <EntitySet Name="UnitsOfMeasure"
EntityType="com.sap.gateway.default.iwbep.common.v0001.UnitOfMeasure"/>
    </EntityContainer>
    <Annotations Target="SAP__self.UnitOfMeasure/UnitCode">
        <Annotation Term="SAP__common.Text" Path="Text"/>
        <Annotation Term="SAP__common.UnitSpecificScale" Path="DecimalPlaces"/>
        <Annotation Term="SAP__CodeList.StandardCode" Path="ISOCCode"/>
        <Annotation Term="SAP__CodeList.ExternalCode" Path="ExternalCode"/>
    </Annotations>
    <Annotations Target="SAP__self.UnitOfMeasure">
        <Annotation Term="SAP__core.AlternateKeys">
            <Collection>
                <Record>
                    <PropertyValue Property="Key">
                        <Collection>
                            <Record>
                                <PropertyValue Property="Name"
PropertyPath="ExternalCode"/>
                                <PropertyValue Property="Alias"
String="ExternalCode"/>
                            </Record>
                        </Collection>
                    </PropertyValue>
                </Record>
            </Collection>
        </Annotation>

```

```

</Annotations>
<Annotations Target="SAP__self.Currency/CurrencyCode">
  <Annotation Term="SAP__common.Text" Path="Text"/>
  <Annotation Term="SAP__common.UnitSpecificScale" Path="DecimalPlaces"/>
  <Annotation Term="SAP__CodeList.StandardCode" Path="ISOCode"/>
</Annotations>
<Annotations Target="SAP__self.Container">
  <Annotation Term="SAP__capabilities.FilterFunctions">
    <Collection>
      <String>eq</String>
      <String>ne</String>
      <String>gt</String>
      <String>ge</String>
      <String>lt</String>
      <String>le</String>
      <String>and</String>
      <String>or</String>
      <String>contains</String>
      <String>startswith</String>
      <String>endswith</String>
      <String>any</String>
      <String>all</String>
    </Collection>
  </Annotation>
</Annotations>
<Annotations Target="SAP__self.Container/Currencies">
  <Annotation Term="SAP__capabilities.UpdateRestrictions">
    <Record>
      <PropertyValue Property="QueryOptions">
        <Record>
          <PropertyValue Property="SelectSupported" Bool="true"/>
        </Record>
      </PropertyValue>
    </Record>
  </Annotation>
</Annotations>
<Annotations Target="SAP__self.Container/T100Longtexts">
  <Annotation Term="SAP__capabilities.UpdateRestrictions">
    <Record>
      <PropertyValue Property="QueryOptions">
        <Record>
          <PropertyValue Property="SelectSupported" Bool="true"/>
        </Record>
      </PropertyValue>
    </Record>
  </Annotation>
</Annotations>
<Annotations Target="SAP__self.Container/UnitsOfMeasure">
  <Annotation Term="SAP__capabilities.UpdateRestrictions">
    <Record>
      <PropertyValue Property="QueryOptions">
        <Record>
          <PropertyValue Property="SelectSupported" Bool="true"/>
        </Record>
      </PropertyValue>
    </Record>
  </Annotation>
</Annotations>
</Schema>
</edmx:DataServices>
</edmx:Edmx>

```

```

<?xml version="1.0" encoding="utf-8"?>
<edmx:Edmx Version="4.0" xmlns:edmx="http://docs.oasis-open.org/odata/ns/edmx"
xmlns="http://docs.oasis-open.org/odata/ns/edm">

```

```

    <edm:Reference
Uri="/sap/opu/odata/IWFND/CATALOGSERVICE;v=2/Vocabularies(TechnicalName='%2FIWBEP%2FVOC_COMMUNICATION',Version='0001',SAP__Origin='LOCAL')/$value">
    <edm:Include Namespace="com.sap.vocabularies.Communication.v1"
Alias="Communication"/>
    </edm:Reference>
    <edm:Reference
Uri="/sap/opu/odata/IWFND/CATALOGSERVICE;v=2/Vocabularies(TechnicalName='%2FIWBEP%2FVOC_PERSONALDATA',Version='0001',SAP__Origin='LOCAL')/$value">
    <edm:Include Namespace="com.sap.vocabularies.PersonalData.v1"
Alias="PersonalData"/>
    </edm:Reference>
    <edm:Reference
Uri="/sap/opu/odata/IWFND/CATALOGSERVICE;v=2/Vocabularies(TechnicalName='%2FIWBEP%2FVOC_ANALYTICS',Version='0001',SAP__Origin='LOCAL')/$value">
    <edm:Include Namespace="com.sap.vocabularies.Analytics.v1" Alias="Analytics"/>
    </edm:Reference>
    <edm:Reference
Uri="/sap/opu/odata/IWFND/CATALOGSERVICE;v=2/Vocabularies(TechnicalName='%2FIWBEP%2FVOC_COMMON',Version='0001',SAP__Origin='LOCAL')/$value">
    <edm:Include Namespace="com.sap.vocabularies.Common.v1" Alias="SAP__common"/>
    </edm:Reference>
    <edm:Reference
Uri="/sap/opu/odata/IWFND/CATALOGSERVICE;v=2/Vocabularies(TechnicalName='%2FIWBEP%2FVOC_MEASURES',Version='0001',SAP__Origin='LOCAL')/$value">
    <edm:Include Namespace="Org.OData.Measures.V1" Alias="SAP__measures"/>
    </edm:Reference>
    <edm:Reference
Uri="/sap/opu/odata/IWFND/CATALOGSERVICE;v=2/Vocabularies(TechnicalName='%2FIWBEP%2FVOC_CORE',Version='0001',SAP__Origin='LOCAL')/$value">
    <edm:Include Namespace="Org.OData.Core.V1" Alias="SAP__core"/>
    </edm:Reference>
    <edm:Reference
Uri="/sap/opu/odata/IWFND/CATALOGSERVICE;v=2/Vocabularies(TechnicalName='%2FIWBEP%2FVOC_CAPABILITIES',Version='0001',SAP__Origin='LOCAL')/$value">
    <edm:Include Namespace="Org.OData.Capabilities.V1" Alias="SAP__capabilities"/>
    </edm:Reference>
    <edm:Reference
Uri="/sap/opu/odata/IWFND/CATALOGSERVICE;v=2/Vocabularies(TechnicalName='%2FIWBEP%2FVOC_AGGREGATION',Version='0001',SAP__Origin='LOCAL')/$value">
    <edm:Include Namespace="Org.OData.Aggregation.V1" Alias="SAP__aggregation"/>
    </edm:Reference>
    <edm:Reference
Uri="/sap/opu/odata/IWFND/CATALOGSERVICE;v=2/Vocabularies(TechnicalName='%2FIWBEP%2FVOC_VALIDATION',Version='0001',SAP__Origin='LOCAL')/$value">
    <edm:Include Namespace="Org.OData.Validation.V1" Alias="SAP__validation"/>
    </edm:Reference>
    <edm:Reference
Uri="/sap/opu/odata/IWFND/CATALOGSERVICE;v=2/Vocabularies(TechnicalName='%2FIWBEP%2FVOC_CODELIST',Version='0001',SAP__Origin='LOCAL')/$value">
    <edm:Include Namespace="com.sap.vocabularies.CodeList.v1" Alias="SAP__CodeList"/>
    </edm:Reference>
    <edm:Reference
Uri="/sap/opu/odata/IWFND/CATALOGSERVICE;v=2/Vocabularies(TechnicalName='%2FIWBEP%2FVOC_UI',Version='0001',SAP__Origin='LOCAL')/$value">
    <edm:Include Namespace="com.sap.vocabularies.UI.v1" Alias="SAP__UI"/>
    </edm:Reference>
    <edm:Reference
Uri="/sap/opu/odata/IWFND/CATALOGSERVICE;v=2/Vocabularies(TechnicalName='%2FIWBEP%2FVOC_HTML5',Version='0001',SAP__Origin='LOCAL')/$value">
    <edm:Include Namespace="com.sap.vocabularies.HTML5.v1" Alias="SAP__HTML5"/>
    </edm:Reference>
    <edm:DataService>
        <Schema Namespace="com.sap.gateway.srvd.zui_flight_s4.v0001" Alias="SAP__self">
            <Annotation Term="SAP__core.SchemaVersion" String="1.0.0"/>
            <EntityType Name="FlightType">

```

```

        <Key>
            <PropertyRef Name="Carrid"/>
            <PropertyRef Name="Connid"/>
            <PropertyRef Name="Fldate"/>
        </Key>
        <Property Name="Carrid" Type="Edm.String" Nullable="false" MaxLength="3"/>
        <Property Name="Connid" Type="Edm.String" Nullable="false" MaxLength="4"/>
        <Property Name="Fldate" Type="Edm.Date" Nullable="false"/>
        <Property Name="Price" Type="Edm.Decimal" Nullable="false" Precision="15"
Scale="variable"/>
        <Property Name="CurrencyCode" Type="Edm.String" Nullable="false"
MaxLength="5"/>
        <Property Name="Planetype" Type="Edm.String" Nullable="false"
MaxLength="10"/>
        <Property Name="SeatsMax" Type="Edm.Int32" Nullable="false"/>
        <Property Name="SeatsOcc" Type="Edm.Int32" Nullable="false"/>
        <Property Name="SeatsAvailable" Type="Edm.Int32" Nullable="false"/>
    </EntityType>
    <EntityContainer Name="Container">
        <EntitySet Name="Flight"
EntityType="com.sap.gateway.srvd.zui_flight_s4.v0001.FlightType"/>
    </EntityContainer>
    <Annotations Target="SAP__self.FlightType/Carrid">
        <Annotation Term="SAP__common.IsUpperCase"/>
        <Annotation Term="SAP__common.Label" String="Airline"/>
        <Annotation Term="SAP__common.Heading" String="ID"/>
        <Annotation Term="SAP__common.QuickInfo" String="Airline Code"/>
    </Annotations>
    <Annotations Target="SAP__self.FlightType/Connid">
        <Annotation Term="SAP__common.IsDigitSequence"/>
        <Annotation Term="SAP__common.Label" String="Connection Number"/>
        <Annotation Term="SAP__common.Heading" String="No."/>
        <Annotation Term="SAP__common.QuickInfo" String="Flight Connection
Number"/>
    </Annotations>
    <Annotations Target="SAP__self.FlightType/Price">
        <Annotation Term="SAP__measures.ISOCurrency" Path="CurrencyCode"/>
        <Annotation Term="SAP__common.Label" String="Airfare"/>
    </Annotations>
    <Annotations Target="SAP__self.FlightType/Planetype">
        <Annotation Term="SAP__common.IsUpperCase"/>
        <Annotation Term="SAP__common.Label" String="Plane Type"/>
        <Annotation Term="SAP__common.Heading" String="Plane"/>
        <Annotation Term="SAP__common.QuickInfo" String="Aircraft Type"/>
    </Annotations>
    <Annotations Target="SAP__self.FlightType">
        <Annotation Term="SAP__common.Label" String="Session 4 - Flight
Consumption View"/>
    </Annotations>
    <Annotation Term="SAP__UI.LineItem">
        <Collection>
            <Record Type="SAP__UI.DataField">
                <PropertyValue Property="Label" String="Price"/>
                <PropertyValue Property="Value" Path="Price"/>
            </Record>
        </Collection>
    </Annotation>
    <Annotation Term="SAP__UI.SelectionFields">
        <Collection>
            <PropertyPath>Carrid</PropertyPath>
            <PropertyPath>Connid</PropertyPath>
        </Collection>
    </Annotation>
    </Annotations>
    <Annotations Target="SAP__self.Container/Flight">
        <Annotation Term="SAP__capabilities.SearchRestrictions">
            <Record>

```

```

        <PropertyValue Property="Searchable" Bool="false"/>
    </Record>
</Annotation>
<Annotation Term="SAP__capabilities.InsertRestrictions">
    <Record>
        <PropertyValue Property="Insertable" Bool="false"/>
    </Record>
</Annotation>
<Annotation Term="SAP__capabilities.DeleteRestrictions">
    <Record>
        <PropertyValue Property="Deletable" Bool="false"/>
    </Record>
</Annotation>
<Annotation Term="SAP__capabilities.UpdateRestrictions">
    <Record>
        <PropertyValue Property="Updatable" Bool="false"/>
        <PropertyValue Property="QueryOptions">
            <Record>
                <PropertyValue Property="SelectSupported" Bool="true"/>
            </Record>
        </PropertyValue>
    </Record>
</Annotation>
<Annotation Term="SAP__capabilities.FilterRestrictions">
    <Record>
        <PropertyValue Property="FilterExpressionRestrictions">
            <Collection>
                <Record>
                    <PropertyValue Property="Property"
PropertyPath="Carrid"/>
                    <PropertyValue Property="AllowedExpressions"
String="MultiValue"/>
                </Record>
                <Record>
                    <PropertyValue Property="Property"
PropertyPath="Price"/>
                    <PropertyValue Property="AllowedExpressions"
String="MultiValue"/>
                </Record>
                <Record>
                    <PropertyValue Property="Property"
PropertyPath="Planetype"/>
                    <PropertyValue Property="AllowedExpressions"
String="MultiValue"/>
                </Record>
            </Collection>
        </PropertyValue>
    </Record>
</Annotation>
</Annotations>
<Annotations Target="SAP__self.Container">
    <Annotation Term="SAP__CodeList.CurrencyCodes">
        <Record>
            <PropertyValue Property="Url"
String="../../../default/iwbep/common/0001/$metadata"/>
            <PropertyValue Property="CollectionPath" String="Currencies"/>
        </Record>
    </Annotation>
    <Annotation Term="SAP__aggregation.ApplySupported">
        <Record>
            <PropertyValue Property="Transformations">
                <Collection>
                    <String>aggregate</String>
                    <String>groupby</String>
                    <String>filter</String>
                </Collection>
            </PropertyValue>
        </Record>
    </Annotation>

```

```

        </PropertyValue>
        <PropertyValue Property="Rollup"
EnumMember="SAP__aggregation.RollupType/None"/>
    </Record>
</Annotation>
<Annotation
Term="SAP__common.ApplyMultiUnitBehaviorForSortingAndFiltering" Bool="true"/>
    <Annotation Term="SAP__capabilities.FilterFunctions">
        <Collection>
            <String>eq</String>
            <String>ne</String>
            <String>gt</String>
            <String>ge</String>
            <String>lt</String>
            <String>le</String>
            <String>and</String>
            <String>or</String>
            <String>contains</String>
            <String>startswith</String>
            <String>endswith</String>
            <String>any</String>
            <String>all</String>
        </Collection>
    </Annotation>
</Annotations>
<Annotations Target="SAP__self.FlightType/Fldate">
    <Annotation Term="SAP__common.Label" String="Flight Date"/>
    <Annotation Term="SAP__common.Heading" String="Date"/>
    <Annotation Term="SAP__common.QuickInfo" String="Flight date"/>
</Annotations>
<Annotations Target="SAP__self.FlightType/CurrencyCode">
    <Annotation Term="SAP__common.Label" String="Airline Currency"/>
    <Annotation Term="SAP__common.Heading" String="Curr."/>
    <Annotation Term="SAP__common.QuickInfo" String="Local currency of
airline"/>
</Annotations>
<Annotations Target="SAP__self.FlightType/SeatsMax">
    <Annotation Term="SAP__common.Label" String="Max. Capacity Econ."/>
    <Annotation Term="SAP__common.Heading" String="Capacity in Economy
Class"/>
    <Annotation Term="SAP__common.QuickInfo" String="Maximum Capacity in
Economy Class"/>
</Annotations>
<Annotations Target="SAP__self.FlightType/SeatsOcc">
    <Annotation Term="SAP__common.Label" String="Occupied econ."/>
    <Annotation Term="SAP__common.Heading" String="Occupied economy class"/>
    <Annotation Term="SAP__common.QuickInfo" String="Occupied seats in economy
class"/>
</Annotations>
</Schema>
</edmx:DataServices>
</edmx:Edmx>

```


BookingStatus.fragment.xml

{ } package.json

Guided Development - flights4 X

{ } manifest.json 1

Storyboard - flights4

Project

flights4 /home/user/pr...

Group By

Page Type

View

Project Guides

Search guides

Configure Variant Management for Controls and Pages

Set Default Filter Values for the Filter Bar

Object Page

Add a Chart Facet to an Object Page

Add a Custom Action to a Page Using Extensions

Add a Custom Section to an Object Page Using Extensions

Add a Field Group to an Object Page

Add a Header Facet to an Object Page Using DataPoints

Add a Micro Chart Column to a Table

Add a New Contact View Column to a Table

Add a New Section to an Object Page

Add a Progress Indicator Column to a Table

Add a Rating Indicator Column to a Table

Add a Custom Section to an Object Page Using Extensions

Guide Information

1. Create a Fragment or a View for the New Section

2. Add an Extension Definition to the manifest.json File

Description

Application-specific sections are defined using annotations. Creating a section that integrates components in the front end, such as charts or attachments, requires extension points. Use this guide to create a new section or subsection for an object page using extension points.

Note: Use app extensions with caution and only if you cannot produce the required behavior by other means, such as manifest settings or annotations. To correctly integrate your app extension coding with SAP Fiori elements, use only the `extensionAPI` of SAP Fiori elements.

Sample Preview

The sample preview shows a SAP Fiori application interface. At the top, there's a header bar with the SAP logo and 'Product'. Below it, a sub-header shows 'Notebook Basic 18' and 'W1-9992'. The main content area is divided into two parts. On the left, there's a donut chart with a red box highlighting the text 'Donut chart (Extension 1)'. On the right, there's a table with three columns: 'Phase', 'Progress', and 'Status'. The table has three rows: 'Implement Phase' (40.0%), 'Design Phase' (21.5%), and 'Test Phase' (38.5%).

Start Guide

Exit Guide

SESSION 5

Advanced Fiori Elements

UX406 · Fiori Elements & FPM (21–30%) · Annotations (11–20%)

• MATERIALS BUILT (PPTX + Study Guide)

Exam Points Raised

Session 5 flips the build from Session 4's RAP work to the CAP counterpart, on the same Flight scenario. Real exam points from the prepared materials:

- RAP vs CAP is a recognition question on the exam, not a preference — ABAP CDS + Eclipse ADT = RAP; Node.js + .cds files + cds watch = CAP
- Same annotation vocabulary either way: @UI.HeaderInfo, @UI.Facets, @UI.FieldGroup are identical terms in both backends
- Bound action = declared inside actions {} on the entity (CAP) or in the Behavior Definition (RAP) — needs a selected record either way
- Unbound action = declared at service level (CAP) or as a static action (RAP) — independent of any selection either way
- Criticality is always an integer (1/2/3), never a colour-name string, regardless of backend
- FPM (Building Blocks, custom sections) is 100% backend-agnostic — lives in webapp/ext + manifest.json either way
- Deployment genuinely differs: CAP uses cds build + mta.yaml + cf deploy; RAP uses transport requests + /IWFND/MAINT_SERVICE or V4_ADMIN — full depth in Session 6 (UX410)

Code Examples

Validated CAP CDS for the same Flight scenario built live in Session 4 (RAP) — compiled cleanly with the real CAP compiler, no syntax errors.

db/schema.cds — same Flight entity as the RAP build

```
namespace sap.fe.demo.flight;
using { cuid, managed } from '@sap/cds/common';

entity Flights : cuid, managed {
  carrierID      : String(3)      @title: 'Carrier';
  connectionID   : String(4)      @title: 'Connection';
  price          : Decimal(9,2)   @title: 'Price';
  bookedSeats    : Integer;
  plannedSeats   : Integer;

  priceCriticality : Integer = case
    when price <= 500 then 3 // green
    when price <= 1000 then 2 // amber
    else 1 // red
  end;
}
```

srv/flight-service.cds — OData V4 service, draft, actions

```

using { sap.fe.demo.flight as db } from '../db/schema';

@path: '/odata/v4/flight'
service FlightService {
  @odata.draft.enabled
  entity Flights as projection on db.Flights actions {
    action cancelFlight() returns Flights; // BOUND
  };
  action openBookingWindow(forDate: Date) returns String; // UNBOUND
}

```

Both files were run through the real CAP compiler (cds compile --to edmx / --to sql) during preparation for this session and produced valid OData V4 metadata with no warnings — see the CDS Code Examples companion document for the full annotated walkthrough.

Your Demonstration Notes

What was actually demonstrated in this session:
Created the Session 5 Study Guide (UX406) — RAP vs CAP architecture comparison, CAP annotation syntax for U3, U6–U9, mirrored against the Session 4 RAP build
Validated the db/schema.cds and srv/flight-service.cds shown above against the real CAP compiler (cds compile) — clean OData V4 metadata, no warnings
Generated a front end Fiori app using the CAP demo as the backend service.
Front End – Fiori Backend – CAP service
https://github.com/AngeliqueSabTec/ZUXFioriDemoS6

Your Code

Paste your code here

```

namespace sap.fe.demo.flight;

using { cuid, managed } from '@sap/cds/common';

entity Flights : cuid, managed {
  carrierID      : String(3)      @title: 'Carrier';
  connectionID   : String(4)      @title: 'Connection';
  flightDate     : Date           @title: 'Flight Date';
  price          : Decimal(9,2)   @title: 'Price';
  currency_code  : String(3)      @title: 'Currency' default 'USD';
  plannedSeats   : Integer        @title: 'Planned Seats';
  bookedSeats    : Integer        @title: 'Booked Seats';

  // Calculated element – CAP's equivalent of a CASE WHEN
  // column in an ABAP CDS interface view
  priceCriticality : Integer = case
    when price <= 500 then 3 // green
    when price <= 1000 then 2 // amber
    else 1 // red
  end
}

```

```

end;

// Field control for Price: 3 = Optional, 1 = ReadOnly
priceFieldControl : Integer = case
    when bookedSeats < plannedSeats then 3    // seats still available
    else 1                                     // fully booked, closed
end;

isFullyBooked : Boolean = bookedSeats >= plannedSeats;
}

ID,carrierID,connectionID,flightDate,price,currency_code,plannedSeats,bookedSeats
0a1b2c3d-0001-0000-0000-000000000001,AA,0017,2026-08-15,422.94,USD,385,368
0a1b2c3d-0002-0000-0000-000000000002,AA,0017,2026-09-20,422.94,USD,385,120
0a1b2c3d-0003-0000-0000-000000000003,LH,0400,2026-08-18,650.00,USD,300,299
0a1b2c3d-0004-0000-0000-000000000004,LH,0400,2026-10-01,650.00,USD,300,50

using { FlightService } from './flight-service';

annotate FlightService.Flights with @(
    UI.SelectionFields: [
        carrierID,
        connectionID
    ],
    Search.searchable: true
);

annotate FlightService.Flights with @(
    UI.LineItem: [
        { Value: carrierID },
        { $Type: 'UI.DataFieldWithIntentBasedNavigation',
          Value: connectionID,
          SemanticObject: 'Flight',
          Action: 'displayFactSheet' },
        { Value: flightDate },
        { Value: price, Criticality: priceCriticality },
        { Value: bookedSeats },
        { Value: plannedSeats },

        // BOUND action - needs a selected row
        { $Type: 'UI.DataFieldForAction',
          Action: 'FlightService.cancelFlight',
          Label: 'Cancel Flight' },

        // UNBOUND action - always available in the toolbar
        { $Type: 'UI.DataFieldForAction',
          Action: 'FlightService.openBookingWindow',
          Label: 'Open Booking Window' }
    ]
);

annotate FlightService.Flights with @(
    UI.HeaderInfo: {
        TypeName: 'Flight',
        TypeNamePlural: 'Flights',
        Title: { Value: connectionID },
        Description: { Value: carrierID }
    },
    UI.Facets: [{
        $Type: 'UI.ReferenceFacet',
        Label: 'Flight Details',
        Target: '@UI.FieldGroup#GeneralData'
    }],
    UI.FieldGroup #GeneralData: {
        Data: [

```

```

        { Value: flightDate },
        { Value: price, ![@Common.FieldControl]: priceFieldControl },
        { Value: currency_code },
        { Value: bookedSeats },
        { Value: plannedSeats }
    ]
}
);

annotate FlightService.Flights with @(
    UI.SelectionPresentationVariant #Open: {
        Text: 'Open',
        SelectionVariant: {
            SelectOptions: [{ PropertyName: isFullyBooked,
                Ranges: [{ Sign: #I, Option: #EQ, Low: false }] }]
        },
        PresentationVariant: { Visualizations: ['@UI.LineItem'] }
    },
    UI.SelectionPresentationVariant #Closed: {
        Text: 'Closed',
        SelectionVariant: {
            SelectOptions: [{ PropertyName: isFullyBooked,
                Ranges: [{ Sign: #I, Option: #EQ, Low: true }] }]
        },
        PresentationVariant: { Visualizations: ['@UI.LineItem'] }
    }
);

annotate FlightService.Flights with {
    priceCriticality @UI.Hidden: true;
    priceFieldControl @UI.Hidden: true;
    isFullyBooked @UI.Hidden: true;
};

```

```

const cds = require('@sap/cds');

module.exports = cds.service.impl(async function () {
    const { Flights } = this.entities;

    // Validation - same job as req.error() everywhere in CAP,
    // equivalent to RAP's IF ... IS INITIAL + NEW_MESSAGE_WITH_TEXT
    this.before(['CREATE', 'UPDATE'], Flights, (req) => {
        if (req.data.bookedSeats > req.data.plannedSeats) {
            req.error(400, 'Booked seats cannot exceed planned seats');
        }
    });

    // BOUND action - acts on the single selected Flights record
    this.on('cancelFlight', Flights, async (req) => {
        const flight = req.params[0];
        await UPDATE(Flights, flight.ID).with({ plannedSeats: flight.bookedSeats });
        return await SELECT.one.from(Flights, flight.ID);
    });

    // UNBOUND action - independent, no row selection
    this.on('openBookingWindow', async (req) => {
        const { forDate } = req.data;
        await UPDATE(Flights).set({ bookedSeats: 0 }).where({ flightDate: forDate });
        return `Booking window opened for flights on ${forDate}`;
    });
});

```

```

using { sap.fe.demo.flight as db } from '../db/schema';

```

```

@path: '/odata/v4/flight'
service FlightService {
  @odata.draft.enabled
  entity Flights as projection on db.Flights actions {
    action cancelFlight() returns Flights; // BOUND
  };
  action openBookingWindow(forDate: Date) returns String; // UNBOUND
}

//Can also have the annotations in here, but best practice is to have them in a
//separate file, so that the service definition is clean and easy to read.
// annotate FlightService.Flights with @(
//   UI.SelectionFields: [ carrierID, connectionID ],
//   Search.searchable: true
// );

```

```

_schema-version: 3.3.0
ID: flight-cap-demo
description: A simple CAP project.
version: 1.0.0
modules:
- name: flight-cap-demo-srv
  type: nodejs
  path: gen/srv
  provides:
  - name: srv-api
    properties:
      srv-url: ${default-url}
  parameters:
    buildpack: nodejs_buildpack
    instances: 1
  build-parameters:
    builder: npm-ci
- name: flight-cap-demo-app-content
  type: com.sap.application.content
  path: .
  requires:
  - name: flight-cap-demo-repo-host
    parameters:
      content-target: true
  build-parameters:
    build-result: resources
    requires:
  - artifacts:
    - sapfedemoflightsop.zip
      name: sapfedemoflightsop
      target-path: resources/
- name: sapfedemoflightsop
  type: html5
  path: app/flightsop
  build-parameters:
    build-result: dist
    builder: custom
    commands:
    - npm install
    - npm run build:cf
    supported-platforms: []
resources:
- name: flight-cap-demo-repo-host
  type: org.cloudfoundry.managed-service
  parameters:
    service: html5-apps-repo
    service-name: flight-cap-demo-html5-service
    service-plan: app-host

```

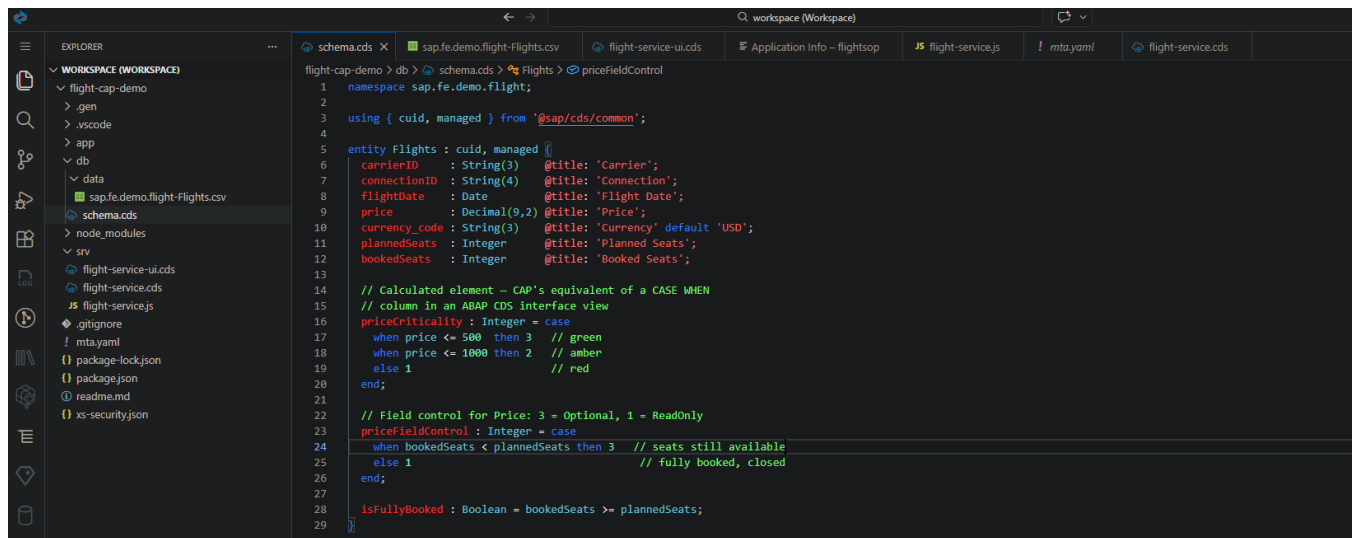
```

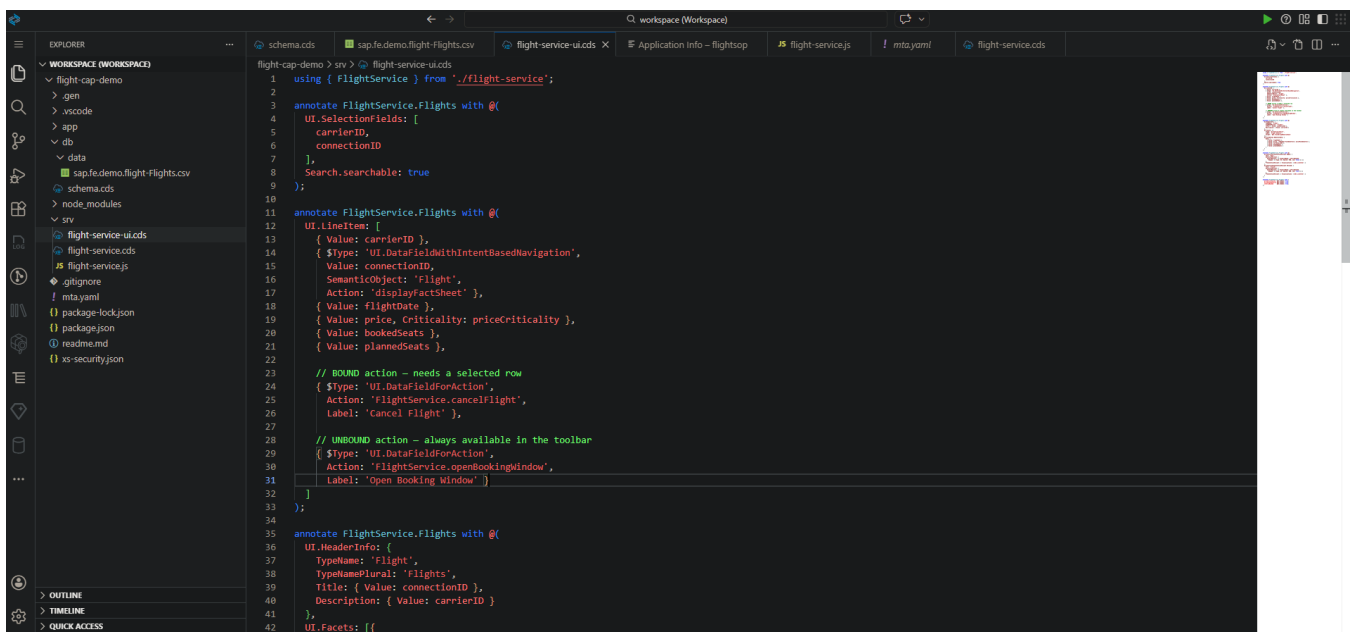
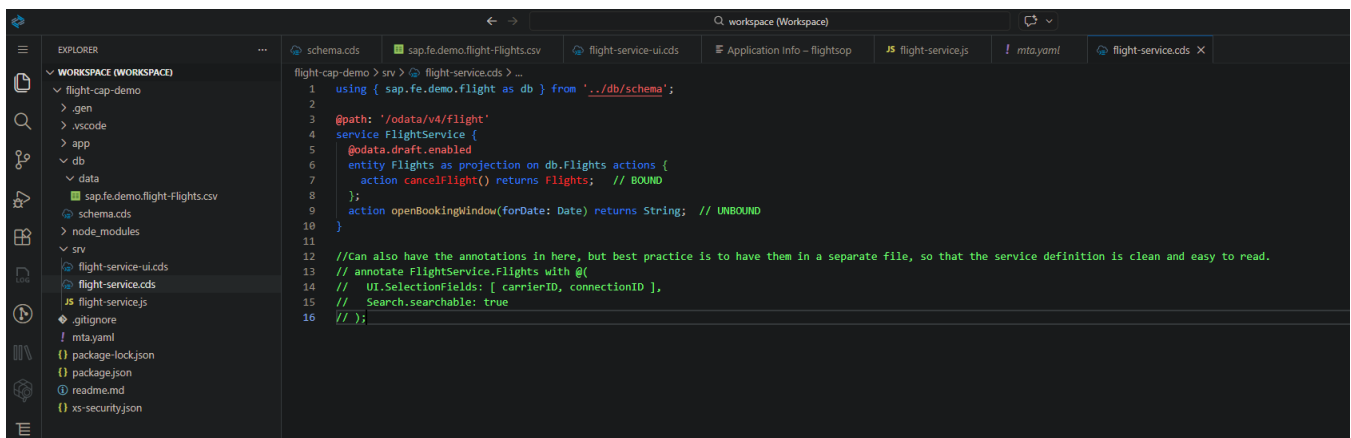
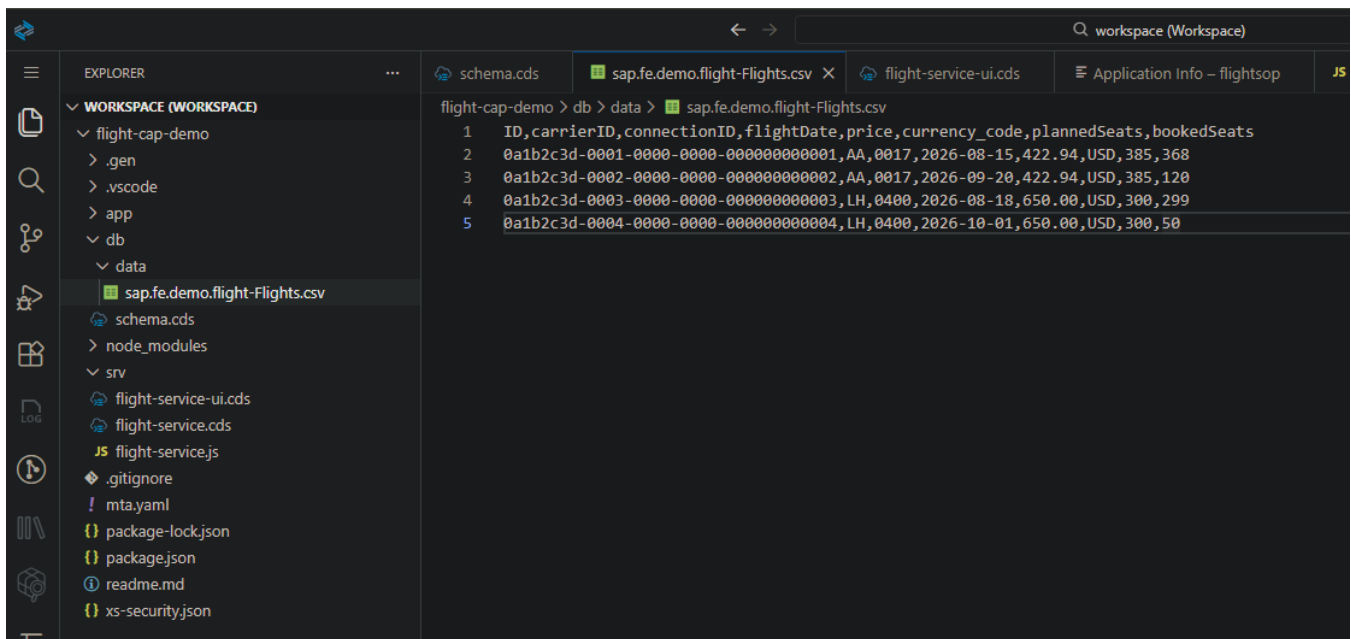
- name: flight-cap-demo-destination-service
  type: org.cloudfoundry.managed-service
  parameters:
    config:
      HTML5Runtime_enabled: false
      init_data:
        instance:
          destinations:
            - Authentication: NoAuthentication
              Name: ui5
              ProxyType: Internet
              Type: HTTP
              URL: https://ui5.sap.com
            existing_destinations_policy: update
          version: 1.0.0
      service: destination
      service-name: flight-cap-demo-destination-service
      service-plan: lite
  parameters:
    deploy_mode: html5-repo
    enable-parallel-deployments: true
  build-parameters:
    before-all:
      - builder: custom
      commands:
        - npm ci
        - npx cds build --production

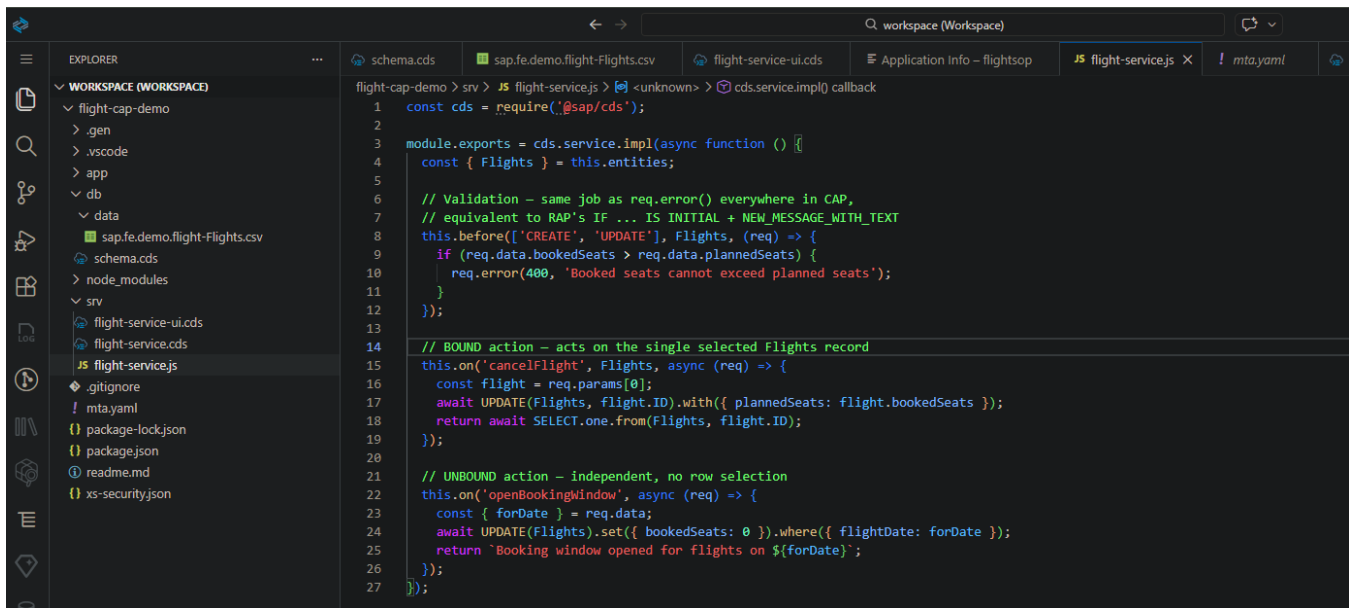
```

Demonstration Notes

Screenshot Evidence — Initial Application – Full stack CAP Application

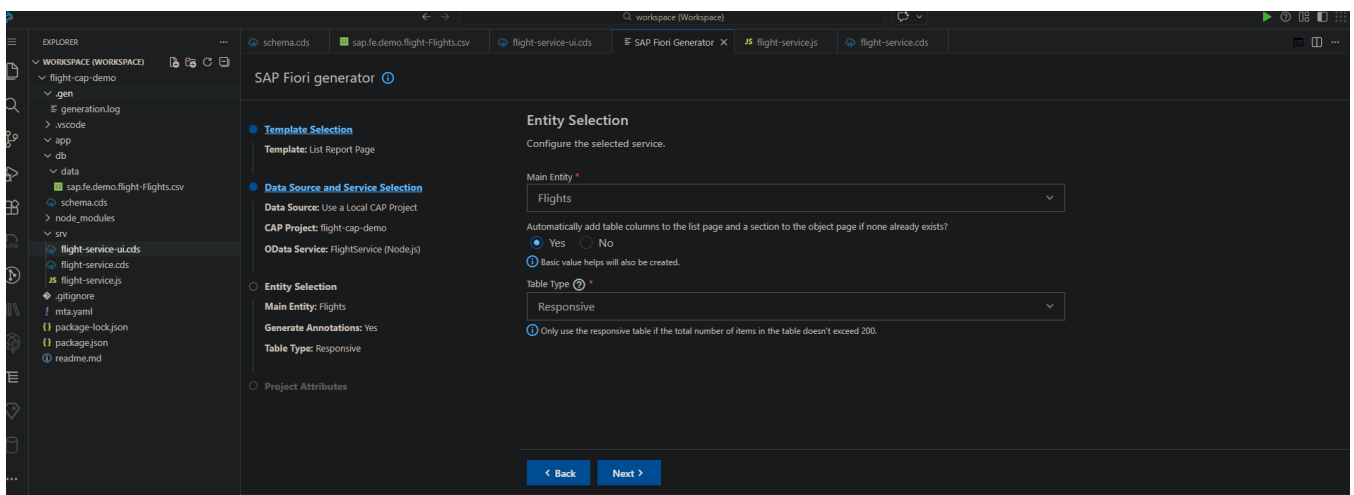
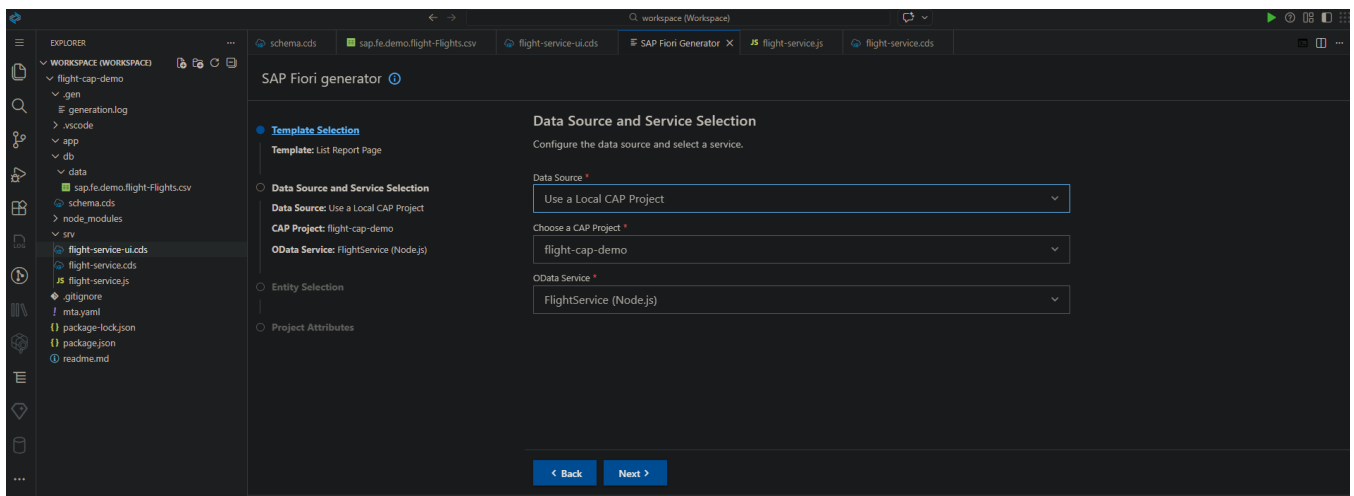
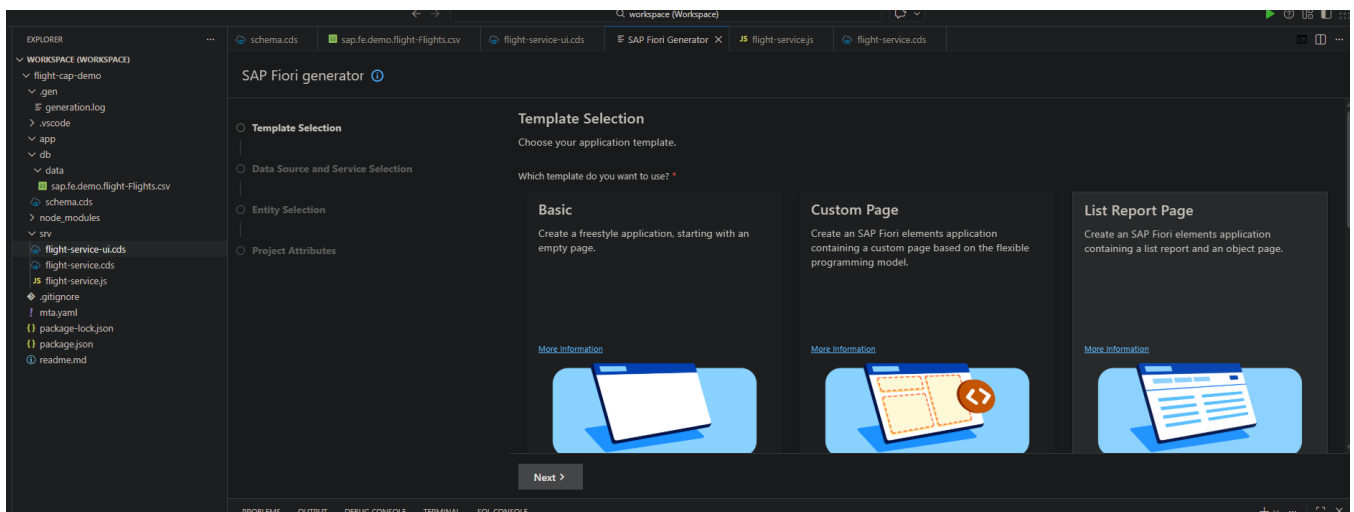


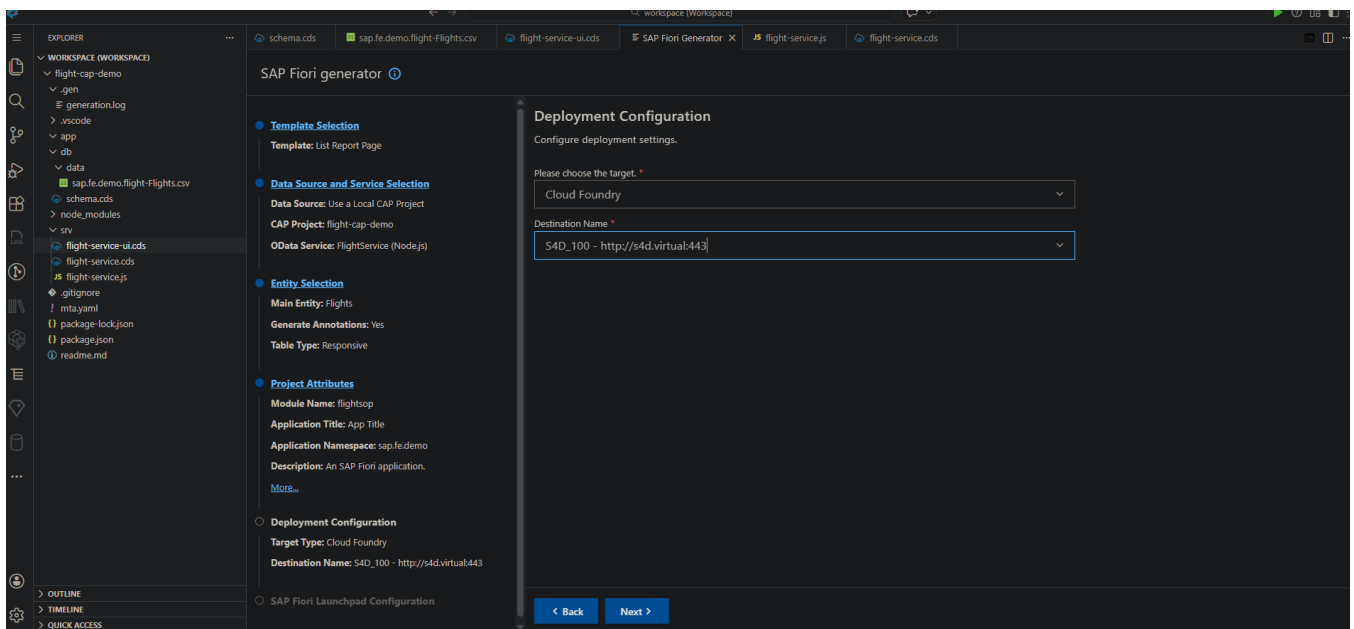
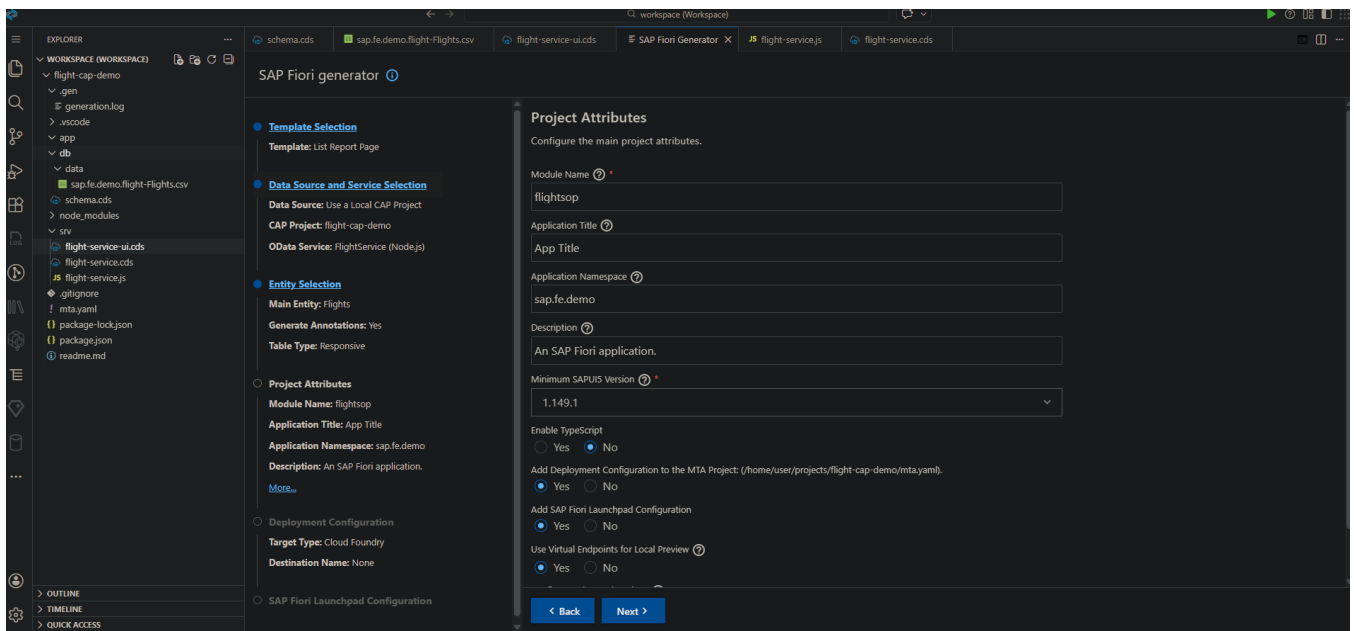


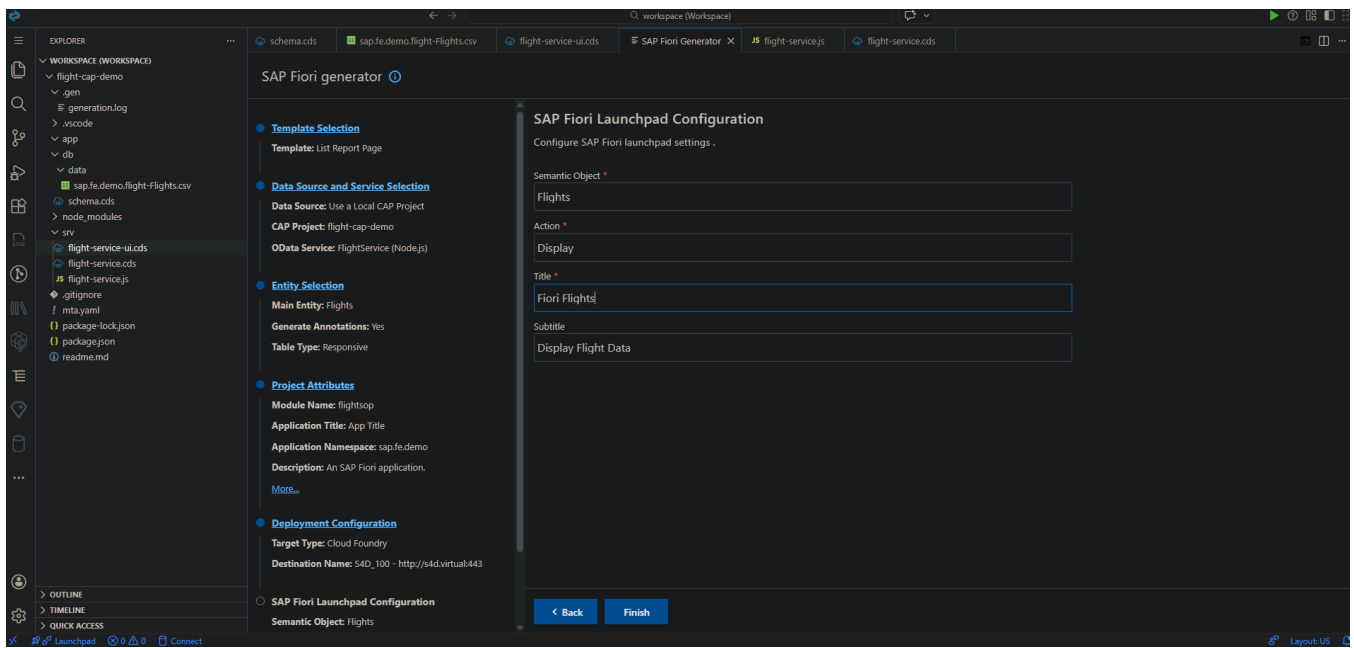


Step 13 — Application Generator

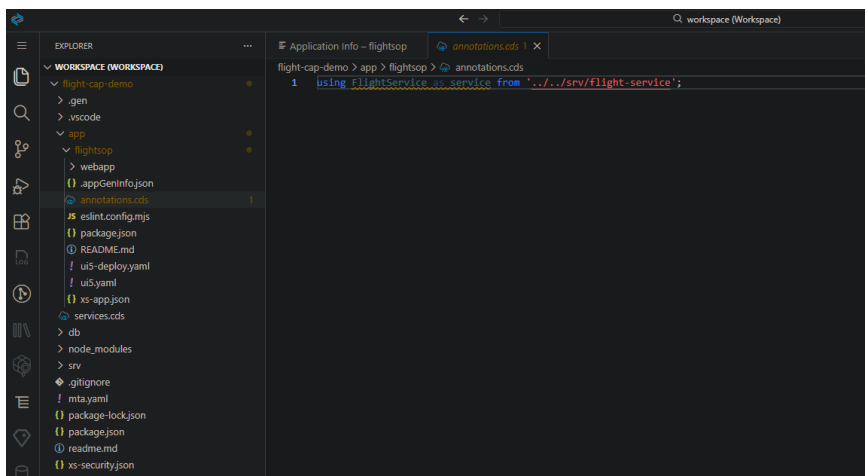
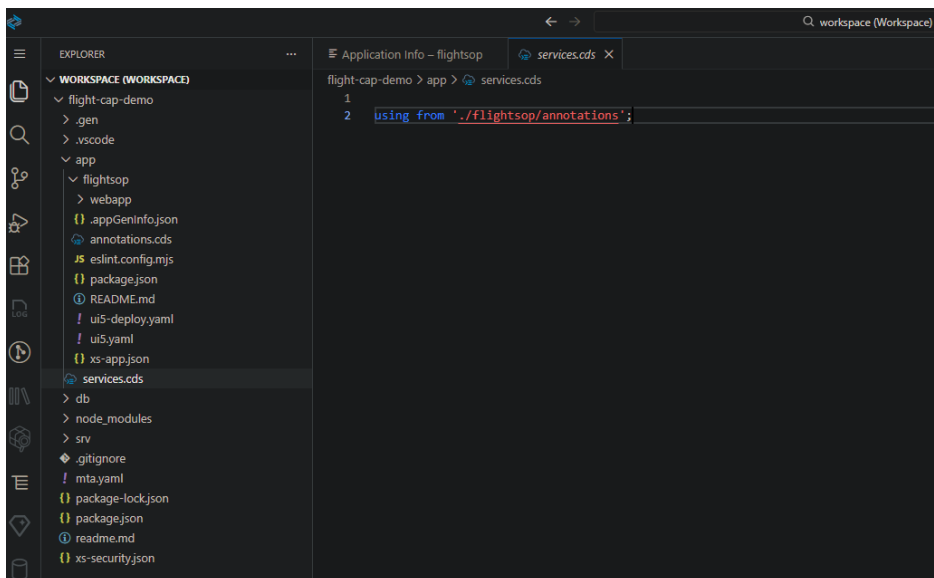
1. Open the **Command Palette** (Ctrl+Shift+P or Cmd+Shift+P) → run "**Fiori: Open Application Generator**".
2. **Floorplan:** List Report Object Page — matches everything you've already annotated.
3. **Data source:** this time choose "**Use a Local CAP Node.js Project**" (not "Connect to a System" — that's for RAP/remote services, since your CAP service already lives right here in this workspace).
4. **CAP project:** it should auto-detect flight-cap-demo — select it.
5. **OData service:** FlightService.
6. **Main entity:** Flights.
7. **Automatic annotation generation:** leave it enabled — the generator will pull in your existing flight-service-ui.cds annotations directly rather than writing separate ones.
8. **Add navigation? Draft enabled?** — draft support should auto-detect as already on from your @odata.draft.enabled; nothing to change there.
9. **Module name:** something like flightsop, namespace sap.fe.demo, leave the rest at defaults.
10. Click **Finish**. BAS scaffolds an app/flightsop/ folder inside your project this time — notice it's nested under the same CAP project, unlike the RAP build where the UI project was entirely separate from the ABAP system.

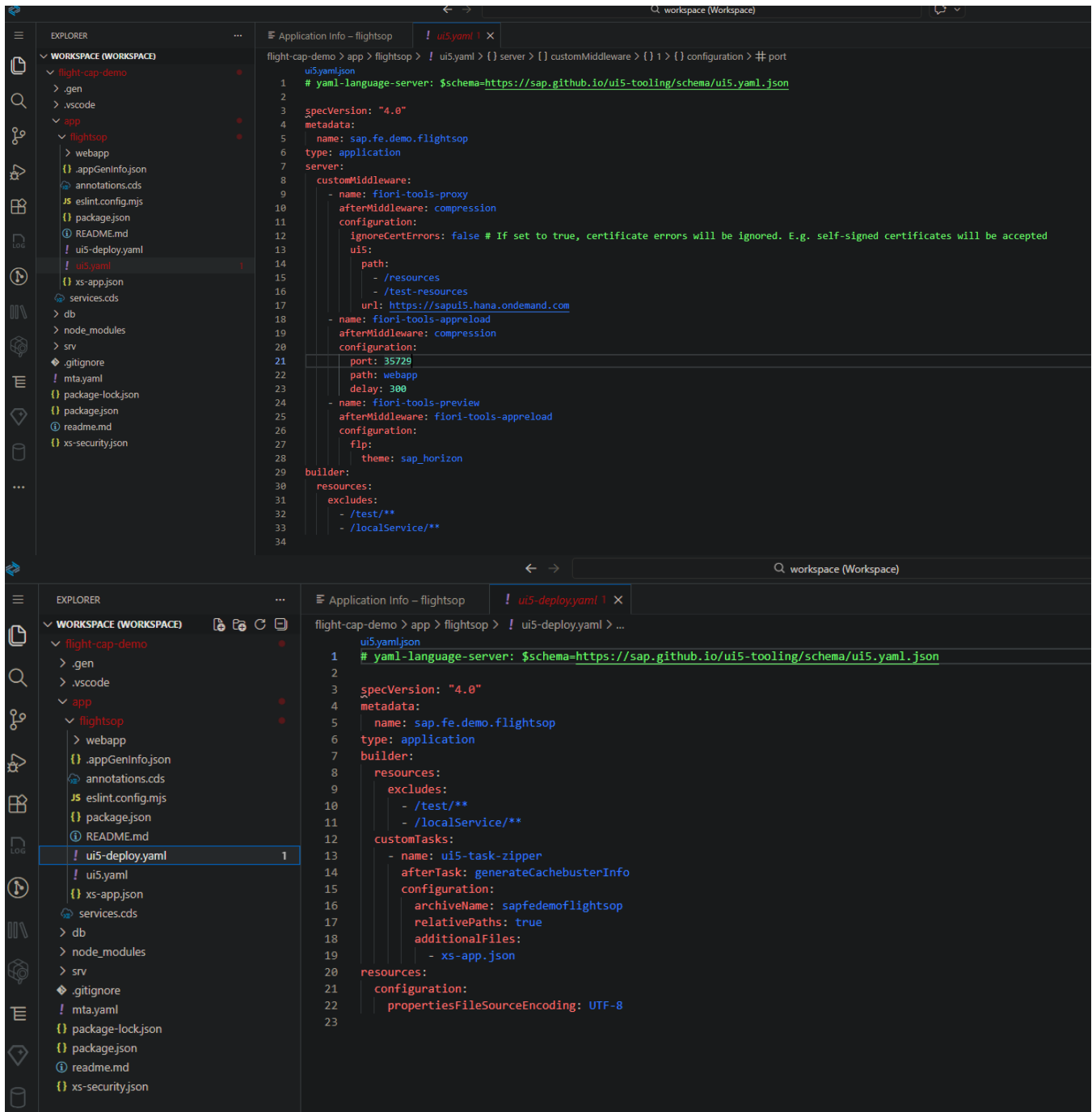






After app generation:





The screenshot shows the VS Code interface with the Explorer view on the left displaying the project structure. The file explorer shows a workspace named 'WORKSPACE (WORKSPACE)' containing a 'flight-cap-demo' folder. Inside 'flight-cap-demo', there are subfolders '.gen', '.vscode', and 'app'. The 'app' folder contains 'flight-sop', which in turn contains 'webapp'. The 'webapp' folder contains several files: '.appGenInfo.json', 'annotations.cds', 'eslint.config.mjs', 'package.json' (selected), 'README.md', 'ui5-deploy.yaml', 'ui5.yaml', 'xs-app.json', 'services.cds', 'db', 'node_modules', 'srv', '.gitignore', 'mta.yaml', 'package-lock.json', 'package.json', 'readme.md', and 'xs-security.json'. The main editor area shows the content of 'package.json' for the 'flight-cap-demo' project. The breadcrumb navigation at the top indicates the path: 'flight-cap-demo > app > flight-sop > package.json'. The code in 'package.json' is as follows:

```
1 {
2   "name": "flightsop",
3   "version": "0.0.1",
4   "description": "An SAP Fiori application.",
5   "keywords": [
6     "ui5",
7     "openui5",
8     "sapui5"
9   ],
10  "main": "webapp/index.html",
11  "dependencies": {},
12  "devDependencies": {
13    "@ui5/cli": "^4.0.33",
14    "@sap/ux-ui5-tooling": "1",
15    "@sap-ux/eslint-plugin-fiori-tools": "^10.0.0",
16    "eslint": "^10",
17    "ui5-task-zipper": "^3.4.x"
18  },
19  "scripts": {
20    "deploy-config": "npx -p @sap/ux-ui5-tooling fiori add deploy-config cf",
21    "build:cf": "ui5 build preload --clean-dist --config ui5-deploy.yaml --include-task-generateCachebusterInfo",
22    "build": "npm run build:cf"
23  }
24 }
```

The screenshot shows the VS Code interface with the Explorer view on the left displaying the project structure. The file explorer shows a workspace named 'WORKSPACE (WORKSPACE)' containing a 'flight-cap-demo' folder. Inside 'flight-cap-demo', there are subfolders '.gen', '.vscode', and 'app'. The 'app' folder contains 'flight-sop', which in turn contains 'webapp'. The 'webapp' folder contains several files: 'i18n', 'test', 'Component.js', 'index.html', 'manifest.json' (selected), 'appGenInfo.json', 'annotations.cds', 'eslint.config.mjs', 'package.json', 'README.md', 'ui5-deploy.yaml', 'ui5.yaml', 'xs-app.json', 'services.cds', 'db', 'node_modules', 'srv', '.gitignore', 'mta.yaml', 'package-lock.json', 'package.json', 'readme.md', and 'xs-security.json'. The main editor area shows the content of 'manifest.json' for the 'flight-cap-demo' project. The breadcrumb navigation at the top indicates the path: 'flight-cap-demo > app > flight-sop > webapp > manifest.json > sap.app'. The code in 'manifest.json' is as follows:

```
1 {
2   "_version": "1.86.0",
3   "sap.app": {
4     "id": "sap.fe.demo.flightsop",
5     "type": "application",
6     "i18n": "i18n/i18n.properties",
7     "applicationVersion": {
8       "version": "0.0.1"
9     },
10    "title": "{{appTitle}}",
11    "description": "{{appDescription}}",
12    "resources": "resources.json",
13    "sourceTemplate": {
14      "id": "@sap/generator-fiori:rop",
15      "version": "1.28.0",
16      "toolId": "705a35ff-6a30-43c4-b629-0172ad2ead47"
17    },
18    "dataSources": {
19      "mainService": {
20        "uri": "/odata/v4/flight/",
21        "type": "OData",
22        "settings": {
23          "annotations": [],
24          "odataVersion": "4.0"
25        }
26      }
27    },
28    "crossNavigation": {
29      "inbounds": {
30        "Flights-Display": {
31          "semanticObject": "Flights",
32          "action": "Display",
33          "title": "{{Flights-Display.flpTitle}}",
34          "subTitle": "{{Flights-Display.flpSubTitle}}",
35          "signature": {
36            "parameters": {},
37            "additionalParameters": "allowed"
38          }
39        }
40      }
41    }
42  },
```

The screenshot shows the VS Code editor interface. On the left is the Explorer sidebar showing the project structure: WORKSPACE (WORKSPACE), flight-cap-demo, .gen, .vscode, app, flightsop, webapp, i18n, test, Component.js, index.html, manifest.json, appGenInfo.json, annotations.cds, eslint.config.mjs, package.json, README.md, ui5-deploy.yaml, ui5.yaml, xs-app.json, services.cds, db, node_modules, srv, .gitignore, mta.yaml, package-lock.json, package.json, readme.md, and xs-security.json. The main editor area displays the content of index.html, which is an HTML file for a SAP Fiori application. The code includes a DOCTYPE declaration, meta tags for charset, viewport, and X-UA-Compatible, a title 'App Title', and a script block for SAP UI5. The body contains a div with class 'sapUiBody sapUiSizeCompact' and id 'content', which contains another div with data-sap-ui-component and data-name attributes.

The screenshot shows the SAP CDS Server welcome page. The page has a dark blue background with white text. At the top, it says 'Welcome to @sap/cds Server'. Below this, it says 'Serving flight-cap-demo 1.0.0'. Then, it says 'These are the paths currently served:'. Underneath, there is a section titled 'Web Applications:' with a list of paths: '/sap.fe.demo.flightsop/editor.html', '/sap.fe.demo.flightsop/index.html', '/sap.fe.demo.flightsop/preview.html', and '/sap.fe.demo.flightsop/test/f1p.html#app-preview'. Below this is a section titled 'Service Endpoints:' with a table showing the endpoint '/odata/v4/flight / \$metadata' and the service 'Flights'. At the bottom, it says 'This is an automatically generated page. You can replace it with a custom ./app/index.html.'

The screenshot shows the SAP Fiori 'Standard' view for the 'Flights' application. The view has a search bar at the top with the text 'Search' and a dropdown menu for 'Editing Status' set to 'All'. There are also fields for 'Carrier' and 'Connection'. Below the search bar is a table with the following columns: Carrier, Connection, Flight Date, Price, Booked Seats, Planned Seats, Changed By, Changed On, Created By, Created On, Currency, and a 'Go' button. The table contains four rows of flight data:

Carrier	Connection	Flight Date	Price	Booked Seats	Planned Seats	Changed By	Changed On	Created By	Created On	Currency
AA	0017	Aug 15, 2026	422.94	368	385	anonymous	Jul 8, 2026, 3:02:37 PM	anonymous	Jul 8, 2026, 3:02:37 PM	USD
AA	0017	Sep 20, 2026	422.94	120	385	anonymous	Jul 8, 2026, 3:02:37 PM	anonymous	Jul 8, 2026, 3:02:37 PM	USD
LH	0400	Aug 18, 2026	650.00	299	300	anonymous	Jul 8, 2026, 3:02:37 PM	anonymous	Jul 8, 2026, 3:02:37 PM	USD
LH	0400	Oct 1, 2026	650.00	50	300	anonymous	Jul 8, 2026, 3:02:37 PM	anonymous	Jul 8, 2026, 3:02:37 PM	USD

0017 AA Edit Delete   Ctrl+E				
Flight Details				
Flight Date: Aug 15, 2026	Price: 422.94	Currency: USD	Booked Seats: 368	Planned Seats: 385

SESSION 6

Deployment & Testing

UX410 · Deployment & Testing — ≤ 10% of C_FIORD

- LIVE EVIDENCE CAPTURED (see debugging log below)

Exam Points Raised

- CDS view activation order and dependency chains
- CDS view composition / layering — interface (base) views vs consumption views
- Classic @OData.publish: true (Gateway/OData V2) vs RAP Service Definition + Service Binding (OData V4)
- Service registration in /IWFND/MAINT_SERVICE for classic Gateway-published services
- Reading ADT activation logs to pinpoint exact syntax errors

Code Examples — Real Demo Views

These are the actual ZUX410 CDS views built and activated during this course, included here for reuse:

Interface/base view — does the arithmetic

```
define view ZUX410_I_FLIGHTUI00_14_BASE as select from sflight as sf
  association to scarr as sc on sf.carriid = sc.carriid
  association to spfli as sp on sf.carriid = sp.carriid
                        and sf.connid = sp.connid
{
  key sf.carriid, key sf.connid, key sf.fldate,
  sc.carrname, sf.price,
  division( sf.seatsocc * 100, sf.seatsmax, 0 ) as OccupancyPct
}
```

Consumption view — only plain comparisons in CASE WHEN

```
define view ZUX410_C_FLIGHTUI00_14_Enced as
  select from ZUX410_I_FLIGHTUI00_14_BASE as base
{
  @UI.hidden: true
  case
    when base.OccupancyPct >= 80 then 3    // green
    when base.OccupancyPct >= 50 then 2    // amber
    else                                1    // red
  end as OccupancyCriticality
}
```

✓ Real Evidence — Live Debugging

This pattern (and the three errors below) came from real ADT activation attempts during this course, not a textbook example. Capturing it here means the next time a CASE WHEN needs a calculated condition, the fix is already known rather than rediscovered.

Watch Out For — Live CDS Debugging Log

Three real activation errors were hit and fixed in sequence while building the Flight/Booking demo views. Each one is genuinely exam-relevant, not just a one-off typo.

Error 1 — "Unexpected word /"

Cause: A raw "/" division operator was used directly in a SELECT expression

Rule: ABAP CDS has no division operator at all — only the built-in division(dividend, divisor, decimal_shift) function

```
// WRONG
sf.seatsocc * 100 / sf.seatsmax >= 80

// RIGHT
division( sf.seatsocc * 100, sf.seatsmax, 0 ) >= 80
```

Error 2 — "Unexpected word :"

Cause: A @UI.dataPoint #qualifier: {...} hash-qualifier annotation was introduced — the only place in the file using that particular syntax pattern, and it wasn't wired into a facet via #DATAPOINT_REFERENCE

Fix: Removed the unused/unproven annotation block to isolate the real issue; confirmed everything else was structurally sound via automated brace/bracket balance checking

Error 3 — "User-defined functions are not supported in the SEARCHED CASE WHEN clause"

Cause: division() was used inside a CASE WHEN condition

Rule: CASE WHEN conditions only support plain field-to-literal comparisons. No arithmetic, no function calls, inside the condition itself

```
// WRONG - division() inside WHEN
when division( sf.seatsocc * 100, sf.seatsmax, 0 ) >= 80 then 3

// ALSO WRONG - raw arithmetic inside WHEN
when sf.seatsocc * 100 >= 80 * sf.seatsmax then 3    // still fails!

// RIGHT - split into two views.
// Base view computes OccupancyPct as a plain SELECT value.
// Consumption view does only: when base.OccupancyPct >= 80 then 3
```

⚠ Watch Out For

If a CASE WHEN needs to compare a calculated value, calculate it in a lower-level (interface/base) CDS view first, then reference the already-computed field by name in the consuming view's CASE WHEN. Never try to calculate AND compare in the same WHEN condition.

Service Exposure — Two Valid Paths

	Classic (@OData.publish)	RAP (Service Definition)
Extra CDS object needed	No — annotation on the view is enough	Yes — separate Service Definition + Service Binding

Activation step	/IWFND/MAINT_SERVICE → Add Service	Publish the Service Binding in ADT
Protocol	OData V2	OData V2 or V4
Used in this course	✓ Yes — all ZUX410 demo views	Optional reference only

CAP Deployment Path (Added from Session 5)

Session 5 built the CAP counterpart of the same Flight scenario. The classic/RAP table above covers two service-exposure paths; CAP is a genuinely different third path, not a variant of either.

Step 1-2 — Compile and scaffold the MTA descriptor

```
cds build --production
cds add mta
```

Step 3 — mta.yaml (generated skeleton)

```
ID: flight-demo
modules:
  - name: flight-demo-srv
    type: nodejs
    path: gen/srv
    requires: [ flight-demo-db ]
resources:
  - name: flight-demo-db
    type: com.sap.xs.hdi-container
```

Step 4 — Build and deploy to Cloud Foundry

```
mbt build
cf deploy mta_archives/flight-demo_1.0.0.mtar
```

💡 Key Point — Added from Session 5

Classic (@@odata.publish) and RAP (Service Definition/Binding) both stay inside the ABAP system and use a transport request. CAP deployment leaves the ABAP world entirely — it packages as a Multi-Target Application and deploys to Cloud Foundry or Kyma on SAP BTP. Same end goal (a running OData V4 service), three genuinely different mechanical paths.

Your Demonstration Notes

What was actually demonstrated in this session:

RAP is deployed as a BSP application to the backend. Always check the sicf node, the object navigator for the bsp app, and the manifest.json inside the BSP app.

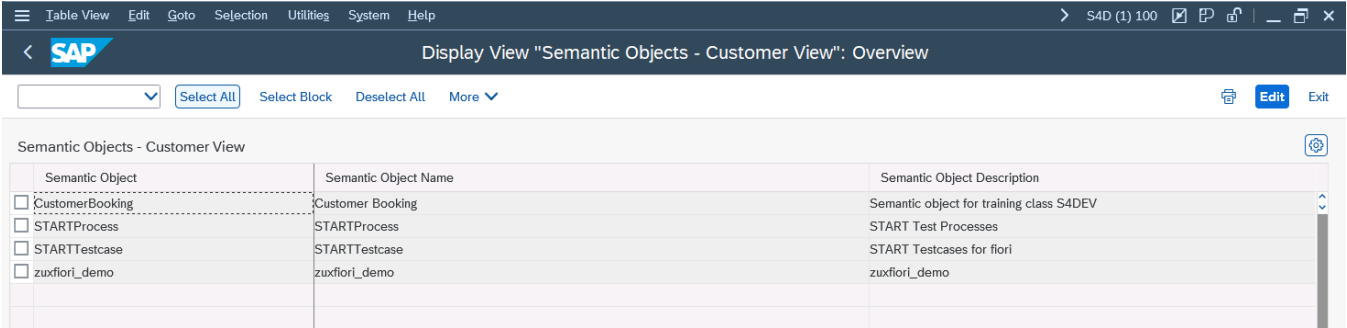
CAP is deployed to the CLOUD Foundry environment on BTP itself. This is where you can click on links and run the app from there.

Your Code

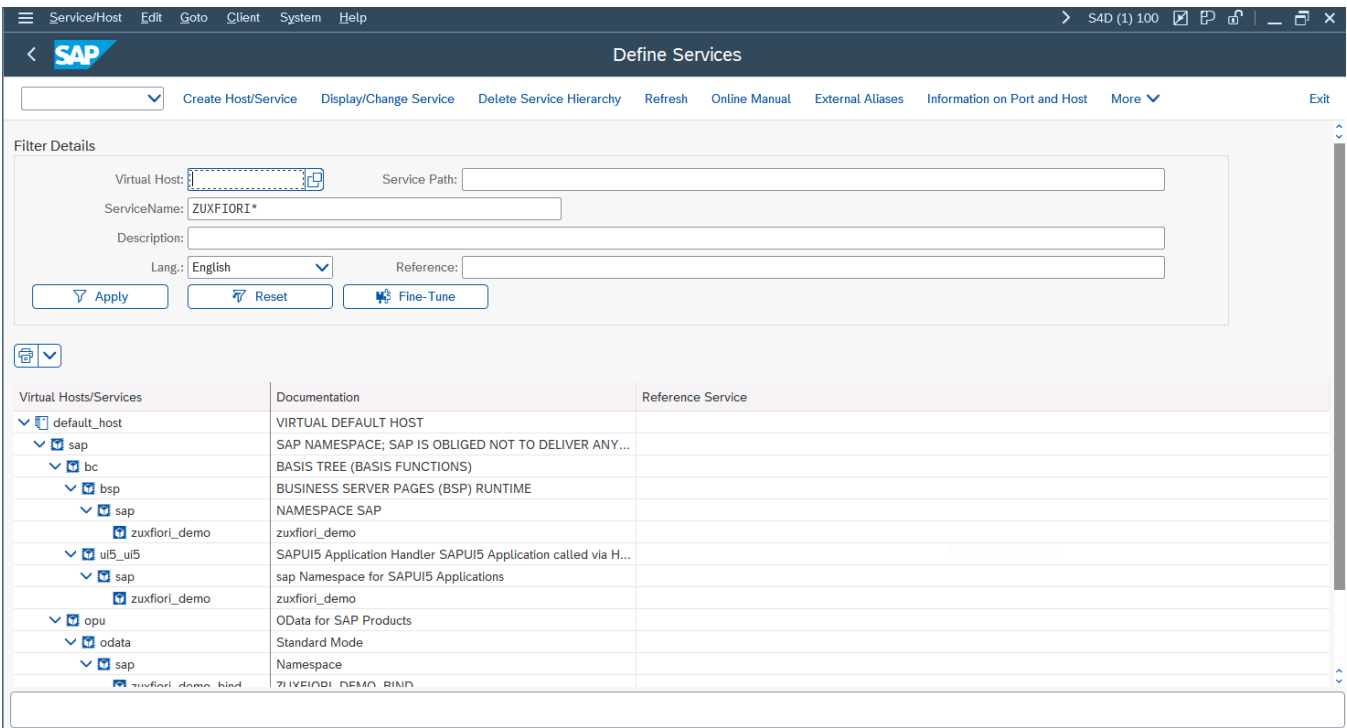
Paste your code here

None

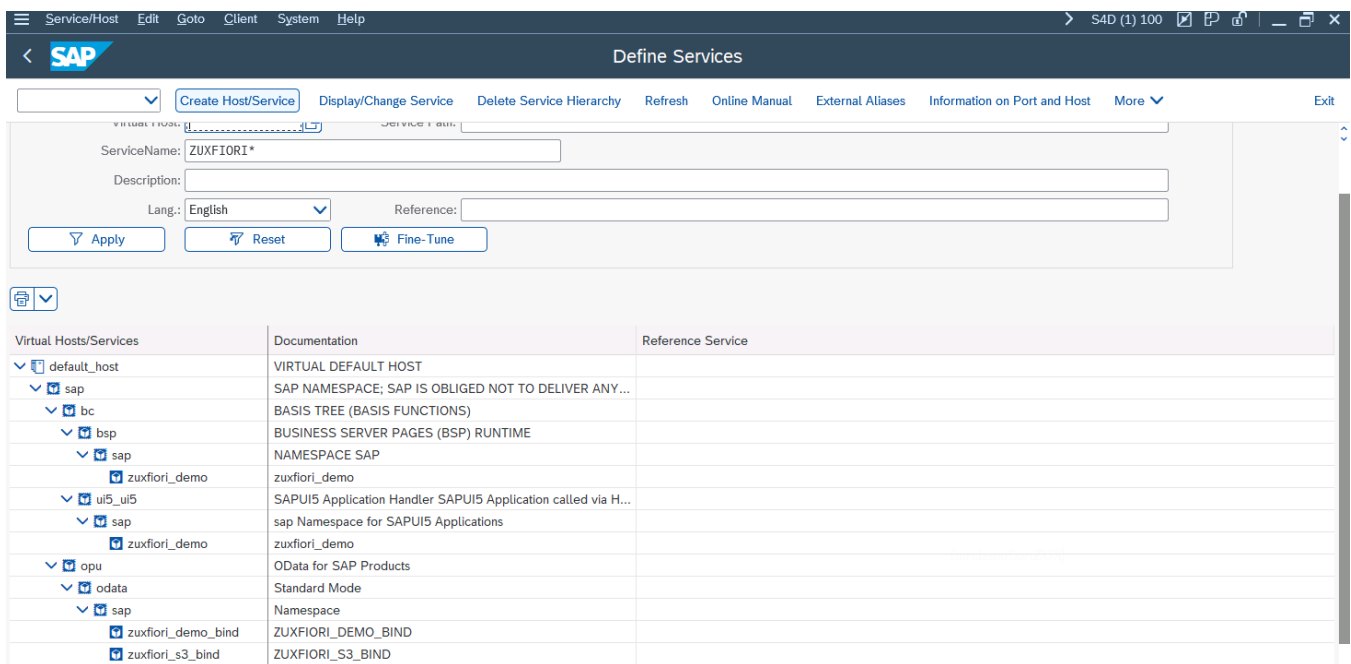
Screenshot Evidence — Semantic Object & Service Path Registration



SM30 — Semantic Objects (Customer View) table showing zuxfiori_demo registered as a semantic object.

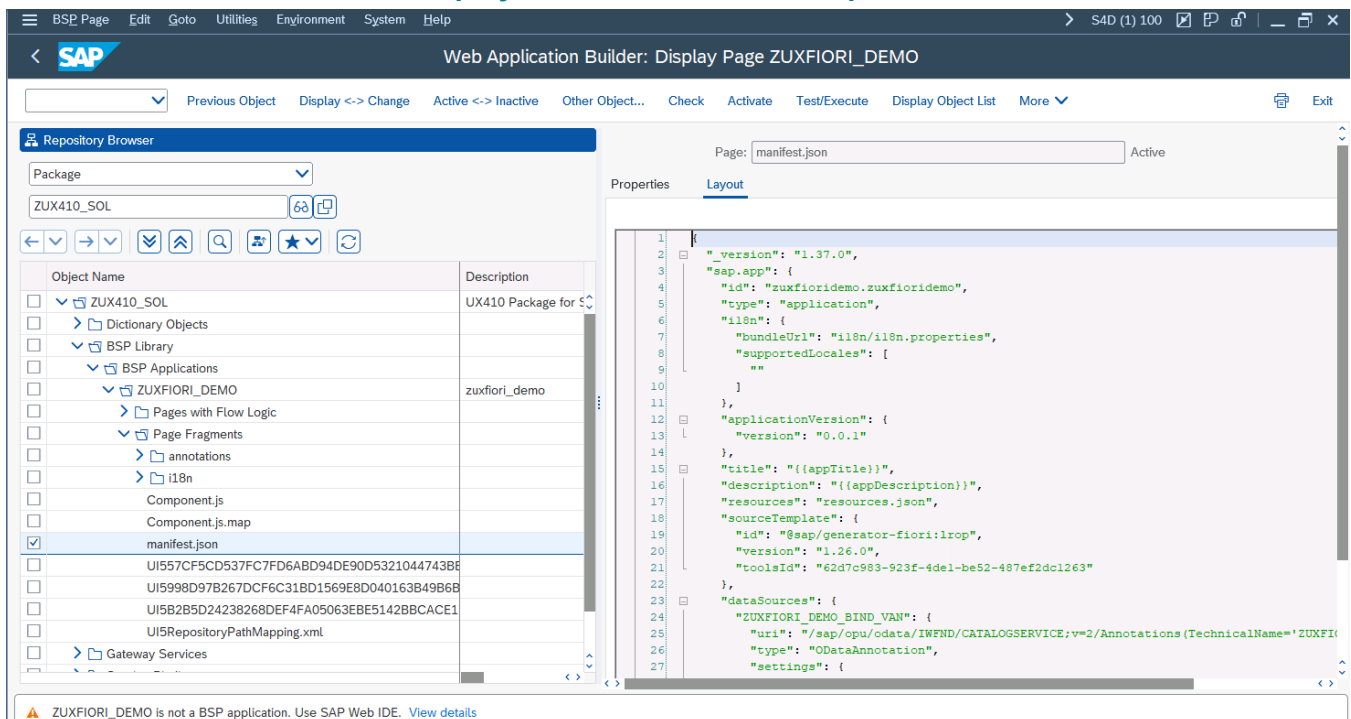


SICF (Define Services) — the BSP/UI5 application node tree for zuxfiori_demo, plus the OData node ZUXFIORI_DEMO_BIND.



SICF tree expanded further — showing both zuxfiori_demo_bind and zuxfiori_s3_bind OData service nodes registered side by side.

Screenshot Evidence — BSP Deployment & PFCG Role Setup



SE80 Web Application Builder — the deployed BSP application ZUXFIORI_DEMO, with manifest.json open. Note the warning that classic SE80 isn't the recommended tool for SAPUI5 apps — BAS/Fiori Tools is.

Screenshot Evidence — CF Deployment example

Subaccount

Overview

Services

Service Marketplace

Instances and Subscriptions

Cloud Foundry

Spaces

Space Quotas

Org Members

HTML5

Connectivity

Security

Entitlements

Usage Analytics

Space

Applications

Services

Service Marketplace

Instances

Routes

Security Groups

Space Events

Space Members

Application

Application Overview

Service Bindings

Security

Roles

Scopes

Role Templates

User-Provided Variables

Environment Variables

Application Events

Logs

Subaccount: [redacted] - DEV - Spaces

Take a walkthrough to learn the key concepts in Cloud Foundry through the UIs. [Go to Tutorial](#)

Spaces (1)

DEV

3 Started | 0 Stopped Applications

Org Quota

Space Memory 3GB of 200GB

Service Instances 4 of 20000

< 1 >

Space: DEV - Applications

Deploy Application Restage Application

Available Memory: 197GB

Auto Refresh: [refresh icon]

Applications (3)

All Applications Search by application name

Name	Requested State	Instances	Disk per Instance	Memory per Instance	All Instances Memory	Actions
[redacted]-app	Started	1/1	2GB	1GB	1GB	[play] [stop] [trash]
[redacted]-ui	Started	1/1	2GB	1GB	1GB	[play] [stop] [trash]
[redacted]-srv	Started	1/1	1GB	1GB	1GB	[play] [stop] [trash]

< 1 >

Application: [redacted] app - Application Overview

Requested State: Started

Start Stop Restart

Auto Refresh: [refresh icon]

General Information

Application Details

Application Name: [redacted]

Lifecycle: buildpack

Buildpack: nodejs_buildpack

Last Updated: July 7, 2025, 4:05:37 PM GMT+02:00

SSH Enabled: false

Stack: cftrunfs4

Change Stack

Mapped Routes (1)

https://[redacted]

Map Route Unmap Route

Instances Information

Instances Configuration

App Instances: 1

Disk per Instance: 2GB

Memory per Instance: 1GB

Application Autoscaler Instance: Bind Application Autoscaler

Change Configuration

Current Application Memory Usage

1GB

Available Space Memory

197GB

Add Instance Remove Instance

App Instances (1)

#	Status	Running Since	CPU Entitlement	Memory	Disk
0	Running	July 7, 2025, 4:05:47 PM GMT+02:00	7%	0.24GB / 1GB	0.24GB / 2GB

Appendix — Full Activation Order Reference

For the live Flight/Booking demo built across this course:

- 1.: ZUX410_C_BOOKINGUI00 — no dependencies on other custom views
- 2.: ZUX410_I_FLIGHTUI00_14_BASE — interface view, references sflight/scarr/spfli and ZUX410_C_BOOKINGUI00
- 3.: ZUX410_C_FLIGHTUI00_14_Enced — consumption view, references ZUX410_I_FLIGHTUI00_14_BASE
- 4.: Register in /IWFND/MAINT_SERVICE (classic) OR create Service Definition + Service Binding (RAP)

Appendix — Document Revision Log

Add a line each time this document is updated with new session evidence, so it stays a clear audit trail.

Date	Session	What was added
01 Jul 2026	Session 4	Added the full RAP build for the Flight app (actions, criticality, Object Page facets, field control, tabs, intent-based navigation, FPM custom section) and the Live RAP Debugging Log covering 10 real activation/runtime errors.
07 Jul 2026	Session 6	Added the CAP Deployment Path (cds build, cds add mta, mta.yaml, mbt build + cf deploy) from Session 5 alongside the existing classic/RAP service exposure comparison.